Semiconductor Electronics: A Lecture Course

ECE 340, Spring 2026, ZJUI Institute (Prof. Said Mikki)

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1 Introduction - Why Semiconductors Matter

Welcome to ECE 340: Semiconductor Electronics. This course is built around a single unifying theme: the development of a rigorous, physically-based understanding of the **drift-diffusion model**—the fundamental set of equations that govern all semiconductor devices. Our journey will systematically derive the drift-diffusion-Poisson equations from first principles, then apply them to understand and analyze real devices.

Course Roadmap: Building the Drift-Diffusion Model (24 Lectures)

- **Lecture 1**: Introduction: Why semiconductors matter and course philosophy.
- **Lectures 2-3**: Band Theory Foundation: Dispersion relations $E(k)$, the concept of holes, effective mass m^*, and the distinction between conduction and valence bands.
- **Lectures 4-5**: Quantum Statistics and Density of States: Fermi-Dirac distribution, DoS, and the foundation for counting carriers.
- **Lectures 6-7**: Equilibrium Carrier Concentrations: Derivation of n and p in thermal equilibrium, the $np = n_i^2$ relation, and the position of the Fermi level.
- **Lecture 8**: Doping and Extrinsic Semiconductors: Donors, acceptors, n-type and p-type materials, and carrier concentrations in doped semiconductors.
- **Lectures 9-10**: Non-Equilibrium Distributions: Excess carriers, generation, recombination mechanisms, and the dynamics of charge relaxation.
- **Lecture 11**: Drift and Diffusion Currents: Derivation of drift current ($\mathbf{J} = qn\mu\mathbf{E}$), diffusion current ($\mathbf{J} = qD\nabla n$), and the Einstein relation $D/\mu = kT/q$.
- **Lectures 12-13**: The Drift-Diffusion-Poisson Model I: Continuity equations and current density equations — derivation and physical interpretation.
- **Lectures 14-15**: The Drift-Diffusion-Poisson Model II: Poisson's equation, coupling with continuity equations, and the complete semiconductor transport model.
- **Lectures 16-17**: p-n Junction in Equilibrium: Built-in potential, space charge region, and applying the drift-diffusion-Poisson model.
- **Lectures 18-19**: p-n Junction Under Bias: Derivation of the ideal diode equation $I = I_S(e^{qV/kT} - 1)$, junction capacitance, and breakdown.
- **Lectures 20-21**: Optical Devices: Photodiodes, solar cells, photoconductors, LEDs, and laser diodes — analyzed using the drift-diffusion framework with generation terms.
- **Lectures 22-23**: Bipolar Junction Transistors (BJTs): Structure, operating principles, current gain, Ebers-Moll model, and frequency response.
- **Lecture 24**: MOSFETs: Field-effect principle, threshold voltage, current-voltage characteristics, and body effect.

1.1 Course Philosophy: From First Principles to Device Operation

This course is structured around a deliberate, progressive derivation of the drift-diffusion-Poisson equations because this model represents the intellectual core of semiconductor device physics. Unlike a purely descriptive approach, our derivation-based methodology ensures that:

- Every equation we use has a clear physical origin—whether from quantum mechanics, statistical me-

chanics, or electromagnetism.

- The assumptions and approximations inherent in device models are explicitly identified and understood.
- Students gain the ability to derive results rather than merely memorize them.
- The unity of device physics becomes apparent: the same drift-diffusion equations, with appropriate boundary conditions, describe diodes, transistors, and optical devices.

1.2 The Central Goal: Mastering the Drift-Diffusion-Poisson Model

By the end of this course, you will have derived and understood the complete set of equations that form the foundation of all semiconductor device analysis:

$$\textbf{Current density equations:} \quad \mathbf{J}_n = q\mu_n n\mathbf{E} + qD_n\nabla n \tag{1}$$

$$\mathbf{J}_p = q\mu_p p\mathbf{E} - qD_p\nabla p \tag{2}$$

$$\textbf{Continuity equations:} \quad \frac{\partial n}{\partial t} = \frac{1}{q}\nabla\cdot\mathbf{J}_n + G - R \tag{3}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\nabla\cdot\mathbf{J}_p + G - R \tag{4}$$

$$\textbf{Poisson's equation:} \quad \nabla^2\phi = -\frac{\rho}{\epsilon} = -\frac{q}{\epsilon}(p - n + N_D^+ - N_A^-) \tag{5}$$

These five equations, properly understood and applied, constitute the complete description of semiconductor device operation. Every device we study—diodes, solar cells, BJTs, MOSFETs—represents a particular solution to this system under specific boundary conditions and operating regimes.

1.3 The Drift-Diffusion Model as a Unifying Framework

What makes the drift-diffusion-Poisson model so powerful is its universality. Whether analyzing a simple p-n junction or a complex MOSFET, the same physical principles apply:

- **Drift** describes carrier motion under electric fields—governed by mobility μ and effective mass m^*.
- **Diffusion** describes carrier motion from concentration gradients—governed by the diffusion coefficient D and the Einstein relation $D/\mu = kT/q$.
- **Generation and recombination** describe how carriers are created and destroyed—linking to optical absorption, thermal processes, and device lifetime.
- **Poisson's equation** couples the electrostatic potential to the charge distribution—closing the system and enabling self-consistent solutions.

The Unified Vision

By the conclusion of this course, you will recognize that:

- The **diode** is the simplest solution to the drift-diffusion equations in a p-n structure.
- The **BJT** represents coupled solutions in a three-terminal, two-junction structure.
- The **MOSFET** introduces field-effect control through the electrostatic boundary conditions.
- **Optical devices** add generation terms G to the continuity equations.

All emerge naturally from the same foundational equations—this is the elegance and power of the drift-diffusion model.

1.4 Prerequisites and Mathematical Background

To succeed in this course, you should be comfortable with:

- **Calculus**: Differentiation, integration, and basic differential equations.
- **Physics**: Basic electromagnetism (electric fields, potentials, Gauss's law).
- **Quantum mechanics** (optional but helpful): Basic concepts of energy levels and wave-particle duality will be reviewed as needed.
- **Statistical mechanics** (optional): The Fermi-Dirac distribution will be developed from first principles.

All necessary quantum and statistical concepts will be introduced as we build the drift-diffusion model from the ground up. No prior exposure to semiconductor physics is assumed.

What You Will Gain

Upon completing this course, you will be able to:

1. Derive the drift-diffusion-Poisson equations from fundamental physical principles.
2. Calculate equilibrium and non-equilibrium carrier concentrations in doped semiconductors.
3. Analyze p-n junctions, including the ideal diode equation, breakdown, and capacitance.
4. Design and analyze bipolar and field-effect transistors.
5. Understand the physics of optoelectronic devices including solar cells and photodetectors.
6. Apply the drift-diffusion framework to emerging device technologies.

1.5 The Engine of the Information Age

The device you are using to read this document—whether it's a laptop, a tablet, or a smartphone—is a testament to the triumph of semiconductor electronics. At the heart of every computer, server, and mobile device lies the integrated circuit (IC), a complex network of billions of tiny transistors fabricated on a single piece of semiconductor material, usually silicon.

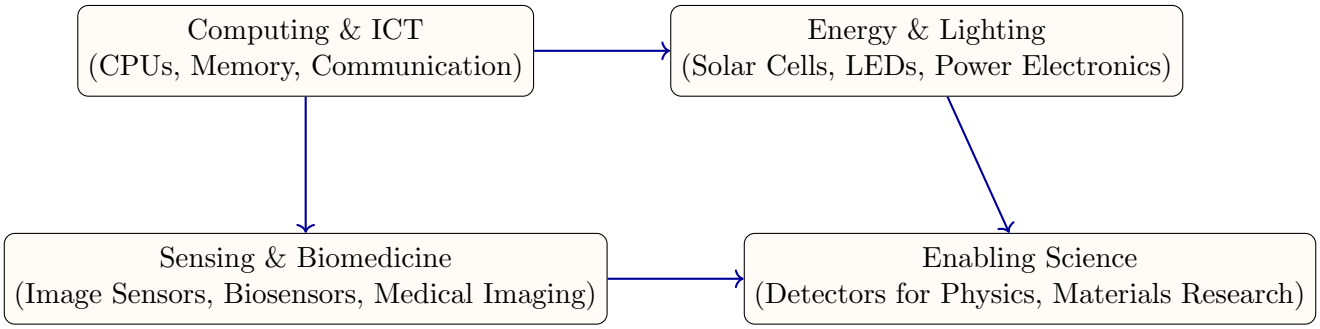
The semiconductor industry has relentlessly followed Moore's Law for decades, doubling the number of transistors on a chip approximately every two years. This has been achieved by continuously shrinking the size of transistors, which requires an ever-deeper understanding of quantum mechanics and materials science at the nanoscale—topics central to this course.

The Centrality of Semiconductors in Computing

- **The Physics of Computation**: All modern computing, from the simplest logic gate to the most complex AI processor, relies on the controlled flow of electrical charge through semiconductors. Understanding semiconductor physics means understanding the very **physical foundation of information processing**. You learn not just *how* to use a transistor, but *why* it works.
- **The Language of Integrated Circuits**: The entire ecosystem of microelectronics—including digital logic, memory (DRAM, SRAM, Flash), and processors (CPUs, GPUs)—is built upon semiconductor technology. A solid grasp of this field is essential for anyone aspiring to work in chip design, hardware engineering, or any related discipline.
- **The Unrivaled Technology**: For over half a century, semiconductor-based computing devices have been the only technology to achieve mass production and economic viability for general-purpose computing. While other approaches (like vacuum tubes, magnetic logic, or optical computing) have been explored, none have matured to challenge the scalability, cost-effectiveness, and energy efficiency of semiconductors. This course explains the unique material properties that make this dominance possible.

1.6 Beyond Computing: A Plethora of Applications

While computing is a giant of the field, the impact of semiconductors extends far beyond, enabling solutions in energy, health, and fundamental science.



- **Renewable Energy and Power Electronics**: The transition to a sustainable energy future is heavily reliant on semiconductors.
  - **Photovoltaic Solar Cells**: These devices are essentially large-area semiconductor diodes that convert light directly into electricity. The physics of the p-n junction, which we will study in detail, is the very principle that allows solar panels to generate power.
  - **Power Electronics**: Efficiently managing and converting electrical power from sources like solar panels and batteries requires power transistors (e.g., IGBTs, MOSFETs) made from semiconductors like silicon and, increasingly, wide-bandgap materials like silicon carbide (SiC) and gallium nitride (GaN).
- **Materials Science and Nanotechnology**: Semiconductors are a playground for materials science. By precisely engineering materials at the atomic scale—through techniques like doping, creating alloys (e.g., SiGe), and building nanostructures (quantum wells, wires, and dots)—we can design materials with entirely new electronic and optical properties. This course provides the foundational knowledge to understand these advanced material systems.

- **Sensing and the Internet of Things (IoT)**: Semiconductor sensors are everywhere. The CMOS image sensor in your phone's camera, the MEMS (Micro-Electro-Mechanical Systems) accelerometer that detects your screen's orientation, and chemical sensors for environmental monitoring are all semiconductor-based. Their small size, low cost, and low power consumption, all derived from semiconductor physics, make the IoT revolution possible.
- **Fundamental Physics and Scientific Discovery**: Semiconductor devices are indispensable tools in experimental physics. High-energy physics experiments, like those at CERN's Large Hadron Collider, use sophisticated silicon pixel detectors to track the paths of subatomic particles after collisions. Astrophysicists use semiconductor-based CCDs and infrared detectors in telescopes to study distant galaxies. Understanding these devices allows engineers to build better instruments for scientific discovery.

Why This Matters for You

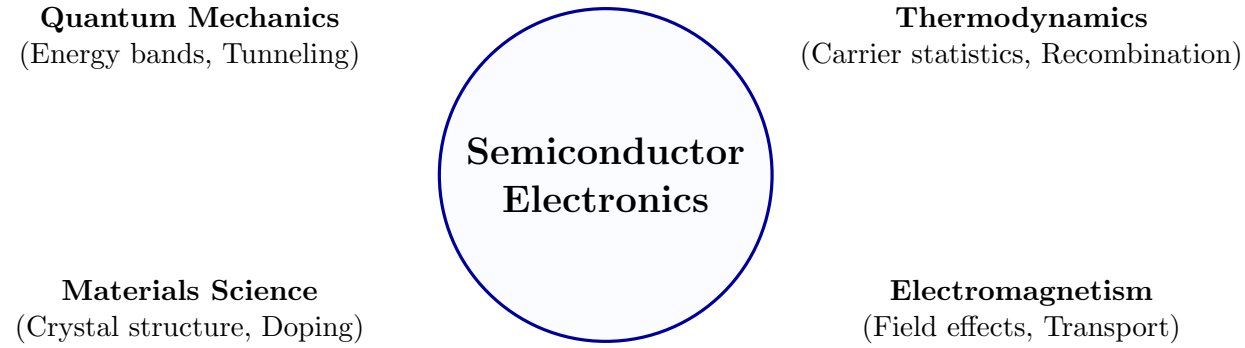
As an undergraduate in engineering or science, this course will equip you with a fundamental understanding of the materials that underpin your discipline.

- **For Electrical and Computer Engineers**: It is the core science of your profession. It connects the atomic-level properties of silicon to the macroscopic I-V curves of a transistor you will use in circuit design.
- **For Computer Scientists**: It demystifies the hardware. You will understand the physical "brick" upon which all algorithms and software are built, providing insight into the ultimate limits of computing performance.
- **For Mechanical Engineers**: It opens doors to the world of micro- and nano-systems (MEMS/NEMS), where mechanical structures are integrated with electronics, and to the growing field of smart materials and sensors.
- **For Physicists and Materials Scientists**: It is the premier applied example of quantum mechanics and solid-state physics, showing how fundamental principles lead to world-changing technologies.

This course will provide you with the essential toolkit to understand, innovate, and work with the technologies that define our modern world.

1.7 The Interdisciplinary Nature of Semiconductor Engineering

Success in semiconductor engineering requires a blend of physics, chemistry, and engineering principles. Throughout this course, we will draw on concepts from quantum mechanics, statistical thermodynamics, and electromagnetism to build a complete picture of how semiconductor devices operate.



This interdisciplinary nature is what makes the field so rich and impactful. As we proceed through the course, we will see how these different areas of knowledge converge to explain and enable the behavior of

real-world devices. The journey begins with understanding the fundamental quantum mechanics of electrons in solids, which is the focus of our next lecture.

2 Crystal Structures: The Foundation of Semiconductor Devices

Semiconductors are not amorphous materials like glass; they are **crystals**. To understand how a transistor or a diode works, we must first understand the ordered, periodic arrangement of atoms that gives semiconductors their unique electronic properties. This section provides the minimal essential knowledge of crystal structures and bonding as a foundation for the upcoming discussion on energy bands.

2.1 What is a Crystal?

Definition 1 (Crystal)**.** A crystal is a solid material whose constituents (atoms, molecules, or ions) are arranged in a highly ordered, repeating pattern that extends in all three spatial dimensions. This periodic arrangement is known as the **crystal lattice**.

Think of a crystal as a three-dimensional wallpaper pattern. Just as a wallpaper pattern is created by repeating a single tile, a crystal is built by repeating a fundamental unit called the **unit cell**.

- **Lattice**: A regular, periodic array of points in space.
- **Basis**: An atom or group of atoms attached to each lattice point.
- **Crystal Structure = Lattice + Basis**.

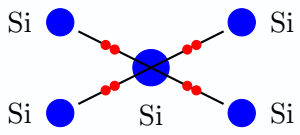
For semiconductors like silicon (Si) and germanium (Ge), the crystal structure is the **diamond cubic** structure. Compound semiconductors like gallium arsenide (GaAs) have a similar but slightly different structure called the **zincblende** structure. Both are characterized by each atom being covalently bonded to four nearest neighbors in a tetrahedral configuration.

2.2 Atomic Bonding in Semiconductors

The type of chemical bonding between atoms is crucial for determining whether a material is a semiconductor, a conductor, or an insulator.

Covalent Bonding in Silicon

Silicon, the workhorse of the semiconductor industry, has four valence electrons. In a silicon crystal, each atom shares its four valence electrons with its four nearest neighbors, forming **covalent bonds**. This sharing creates a stable configuration where each atom effectively has eight electrons in its outer shell (the "octet rule").



Shared valence electrons form covalent bonds

At absolute zero temperature ($T = 0$ K), all valence electrons are locked in these covalent bonds, and the material behaves as an insulator. As temperature increases, some electrons may gain enough thermal energy to break free from their bonds, becoming "free" to conduct electricity. This temperature-dependent conductivity is a hallmark of semiconductors.

2.3 Essential Quantum Mechanics Review

Before we can understand how electrons move through a crystal, we need a few fundamental concepts from quantum mechanics.

Definition 2 (Planck's Relation)**.** Energy and frequency are proportional. A photon (or any quantum particle) with frequency f has an energy E given by:

$$\boxed{E = hf = \hbar\omega}$$

where $h = 6.626 \times 10^{-34}$ J·s is Planck's constant, $\hbar = h/2\pi$, and $\omega = 2\pi f$ is the angular frequency.

Definition 3 (de Broglie Relation)**.** A particle with momentum p has an associated wavelength λ, called the de Broglie wavelength:

$$\boxed{\lambda = \frac{h}{p}} \quad \text{or} \quad \boxed{p = \hbar k}$$

where $k = 2\pi/\lambda$ is the wave number.

Definition 4 (Schrödinger Equation)**.** The behavior of a quantum particle (like an electron) is described by its wave function $\Psi(x, t)$, which satisfies the time-dependent Schrödinger equation:

$$\boxed{i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi}$$

For a single particle in one dimension with potential energy $V(x)$, the Hamiltonian $\hat{H}$ is:

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

The wave function itself is not directly measurable, but its square magnitude $|\Psi(x, t)|^2$ gives the probability of finding the particle at position x at time t.

Definition 5 (Electron Volt (eV))**.** The electron volt is a convenient unit of energy in semiconductor physics. It is defined as the kinetic energy gained by an electron accelerated through a potential difference of 1 volt:

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

Typical semiconductor bandgap energies are on the order of 1 eV (e.g., silicon: 1.12 eV, GaAs: 1.42 eV).

3 Energy Bands: From Discrete Atoms to Crystals

When atoms are brought together to form a crystal, their discrete energy levels evolve into continuous bands of allowed energies, separated by forbidden regions called **bandgaps**. This band structure determines whether a material is a conductor, semiconductor, or insulator.

3.1 Electron Motion in a Periodic Potential

A key question: How can an electron move freely through a densely packed crystal structure without constantly colliding with atoms? The answer lies in quantum mechanics.

Consider an electron entering a crystal. The atoms in the crystal create a periodic potential $V(x)$ that repeats with the lattice spacing a:

$$V(x+a) = V(x)$$

Bloch's Theorem

For an electron in a periodic potential, the Schrödinger equation has solutions of the form:

$$\boxed{\psi_k(x) = u_k(x)e^{ikx}}$$

where $u_k(x)$ is a function that has the same periodicity as the lattice: $u_k(x+a) = u_k(x)$. These are called **Bloch waves**.

The Bloch wave is a plane wave (e^{ikx}) modulated by a periodic function $u_k(x)$. This means:

- The electron is not localized; its wave function extends throughout the entire crystal.
- The probability density $|\psi_k(x)|^2 = |u_k(x)|^2$ is periodic, meaning the electron is equally likely to be found near any atom.
- Unlike a bound state that decays exponentially away from a nucleus (like in a hydrogen atom), the Bloch wave does **not decay**. It propagates through the crystal like a wave through a transmission line.

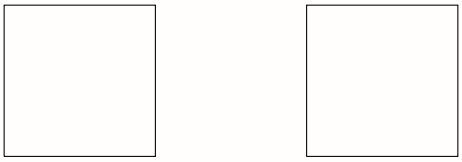
This is the quantum mechanical explanation for electron mobility in crystals: the periodic structure allows oscillatory, propagating solutions to the Schrödinger equation, not just decaying ones.

3.2 Energy Splitting: The Resonance Analogy

When atoms are far apart, they have discrete, sharp energy levels. As they come together to form a crystal, these levels split into bands. This phenomenon can be understood through a classical analogy.

Classical Analogy: Coupled Resonant Circuits

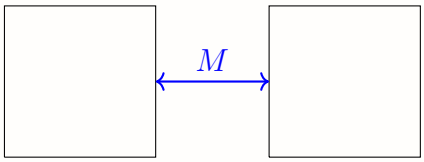
Consider two identical LC circuits, each with its own resonant frequency $f_0 = 1/(2\pi\sqrt{LC})$. When the circuits are isolated, they each oscillate at f_0.



Circuit 1: f_0 Circuit 2: f_0

Now, bring them close together so their inductors are coupled magnetically (mutual inductance M). The system now has two **normal modes** of oscillation:

- A **symmetric mode** where currents flow in the same direction: frequency $f_1 < f_0$
- An **antisymmetric mode** where currents flow in opposite directions: frequency $f_2 > f_0$



Result:
$f_1 < f_0$ (symmetric)
$f_2 > f_0$ (antisymmetric)

Key insight: The interaction (coupling) between two identical resonant systems causes the single resonant frequency to **split** into two distinct frequencies.

Now apply Planck's relation $E = hf$:

$$f_1 < f_0 < f_2 \quad \Rightarrow \quad E_1 < E_0 < E_2$$

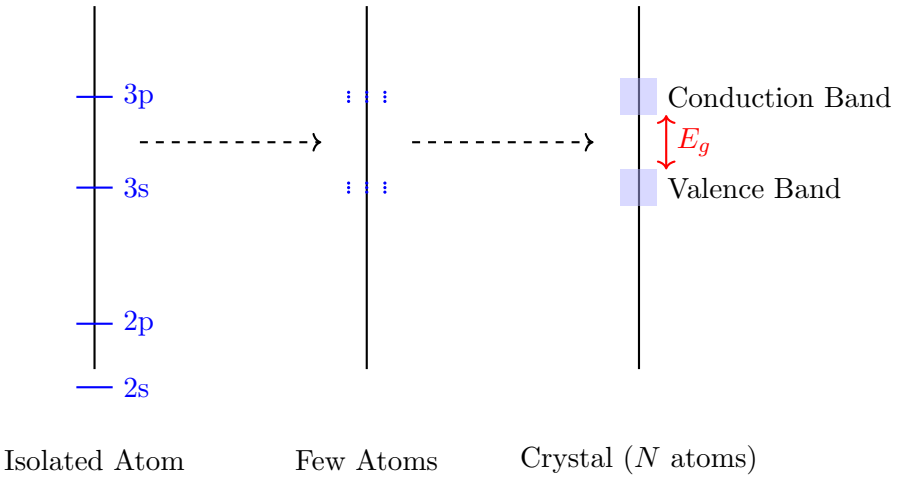
The single energy level E_0 of an isolated system splits into two distinct energy levels E_1 and E_2 when two identical systems are coupled.

3.3 From Discrete Levels to Energy Bands

Now generalize this to a crystal with N atoms:

- **Isolated atom**: Each electron orbital has a discrete energy level (e.g., 1s, 2s, 2p).
- **Two atoms**: When two atoms bond, each energy level splits into two closely spaced levels (one lower, one higher), corresponding to bonding and antibonding states.
- **Many atoms (N very large)**: When N atoms come together to form a crystal, each original atomic level splits into N discrete but very closely spaced energy levels.

Since N is enormous (typically $\sim 10^{22}$ atoms/cm^3), these N levels are so close together that they form a quasi-continuous **energy band**.



3.4 The Bandgap and Semiconductor Classification

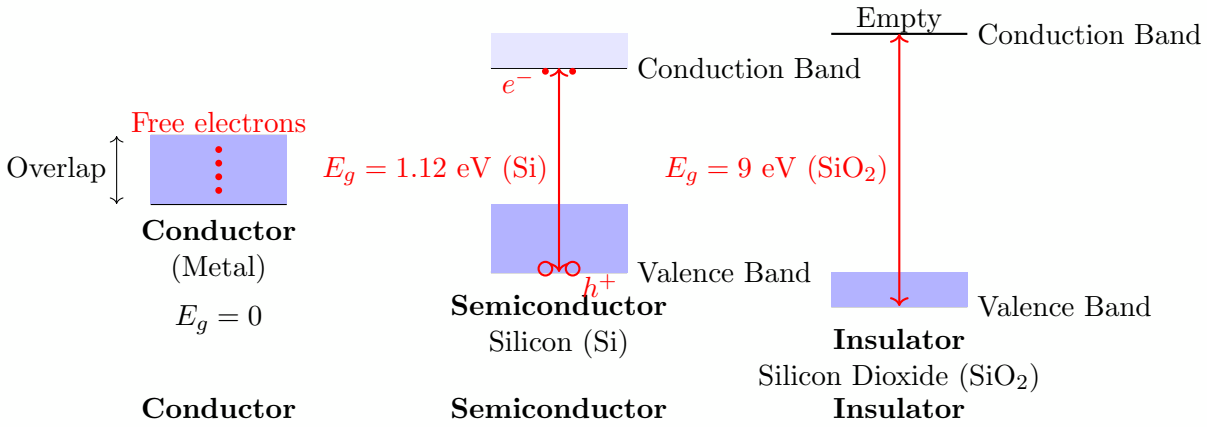
The splitting creates two important bands:

- **Valence Band**: The lower band, formed from the splitting of the outermost (valence) electron orbitals. At low temperatures, this band is fully occupied by electrons.
- **Conduction Band**: The upper band, formed from the splitting of the next-higher energy orbitals (the ones that would be empty in isolated atoms). This band is empty at low temperatures.
- **Bandgap (E_g)**: The forbidden energy region between the top of the valence band and the bottom of the conduction band. No electron states exist here.

Bandgap Energies for Common Semiconductors

Material	Symbol	Bandgap E_g at 300 K (eV)
Germanium	Ge	0.67
Silicon	Si	1.12
Gallium Arsenide	GaAs	1.42
Gallium Nitride	GaN	3.4
Silicon Carbide	SiC	3.3

Visual representation of the energy band diagram:



Note: Bandgap values are material-dependent
Si: 1.12 eV, Ge: 0.67 eV, GaAs: 1.42 eV, SiO_2: ∼9 eV

3.5 Why Semiconductors are Special

The key to semiconductor behavior lies in the bandgap size:

- **Conductors (Metals)**: The valence and conduction bands overlap ($E_g = 0$). Electrons can move freely without needing extra energy.
- **Insulators**: The bandgap is very large ($E_g > 3-4$ eV). At room temperature, essentially no electrons have enough thermal energy to jump from the valence band to the conduction band.
- **Semiconductors**: The bandgap is intermediate (~ 1 eV). At room temperature, a small but significant number of electrons gain enough thermal energy ($k_BT \approx 0.026$ eV at 300 K) to be excited across the gap, leaving behind "holes" in the valence band and enabling conduction.

This ability to control conductivity through temperature, doping, and electric fields is what makes semiconductors the foundation of modern electronics. In the next sections, we will explore how we can engineer these properties to create useful devices.

4 Dispersion Relations: Connecting Energy and Momentum

The single most important input from quantum theory to semiconductor physics is the **dispersion relation**. This mathematical relationship between the energy E and momentum p (or wave number k) of an electron determines virtually all of its dynamic properties—how it moves, how it responds to electric fields, and how it interacts with its environment.

4.1 The Free Electron Dispersion Relation

Let us begin with the simplest possible case: a completely free electron in empty space, with no forces acting on it.

Free Electron Schrödinger Equation

For a free electron, the potential energy $V(x) = 0$. The Hamiltonian consists only of kinetic energy:

$$\hat{H} = \frac{\hat{p}^2}{2m_e} = -\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2}$$

The time-independent Schrödinger equation is:

$$-\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2} = E\psi$$

The solution to this equation is the plane wave:

$$\boxed{\psi(x) = Ae^{ikx} + Be^{-ikx}}$$

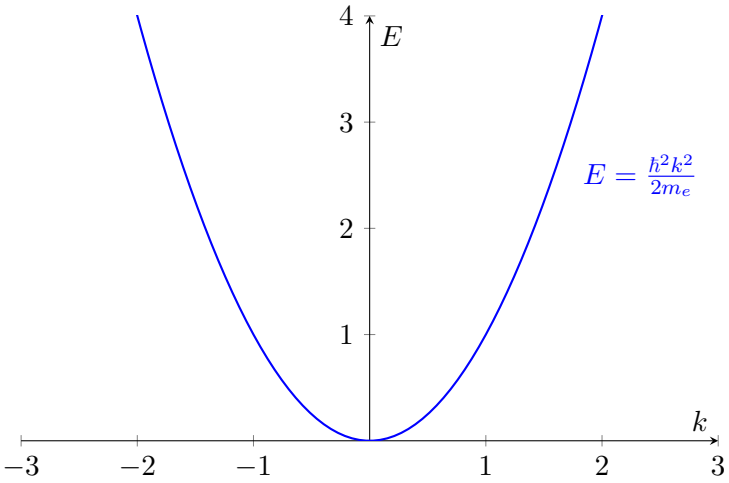
Recall from de Broglie's relation that the wave number k is related to momentum by $p = \hbar k$. Substituting the plane wave solution $\psi(x) = e^{ikx}$ into the Schrödinger equation:

$$-\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2}e^{ikx} = -\frac{\hbar^2}{2m_e}(ik)^2e^{ikx} = \frac{\hbar^2k^2}{2m_e}e^{ikx} = Ee^{ikx}$$

This yields the **free electron dispersion relation**:

$$\boxed{E = \frac{\hbar^2k^2}{2m_e}}$$

Free Electron Parabolic E-k Relation



This parabolic relationship is fundamental. It tells us that:

- Energy increases quadratically with wave number (momentum).
- For a given $|k|$, there are two states: one with positive momentum ($+k$) and one with negative momentum ($-k$).
- The curvature of the parabola is determined by the electron mass m_e.

While this derivation assumed a completely free electron, and electrons in solids are clearly *not* free, this parabolic relation serves as the conceptual foundation for all that follows. The modifications introduced by the crystal lattice will preserve this basic quadratic form but with crucial adjustments.

4.2 Electrons in a Periodic Potential: The Effective Mass Approximation

In a real crystal, the electron moves through a complex potential created by the positively charged atomic nuclei and all the other electrons. This potential $V(\mathbf{r})$ is periodic with the lattice spacing:

$$V(\mathbf{r}+\mathbf{R}) = V(\mathbf{r})$$

where $\mathbf{R}$ is any lattice translation vector. The full Schrödinger equation becomes extremely complicated—far too complex to solve exactly for a real crystal with $\sim 10^{23}$ interacting particles.

The Independent Electron Approximation

To make progress, we use the **independent electron approximation**. The key idea is this: we assume that each electron propagates independently through the crystal, but with all the effects of the other electrons and the nuclei absorbed into an **effective background potential** $V_{\text{eff}}(\mathbf{r})$.

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V_{\text{eff}}(\mathbf{r})\right]\psi(\mathbf{r}) = E\psi(\mathbf{r})$$

This effective potential is still periodic, but it incorporates:

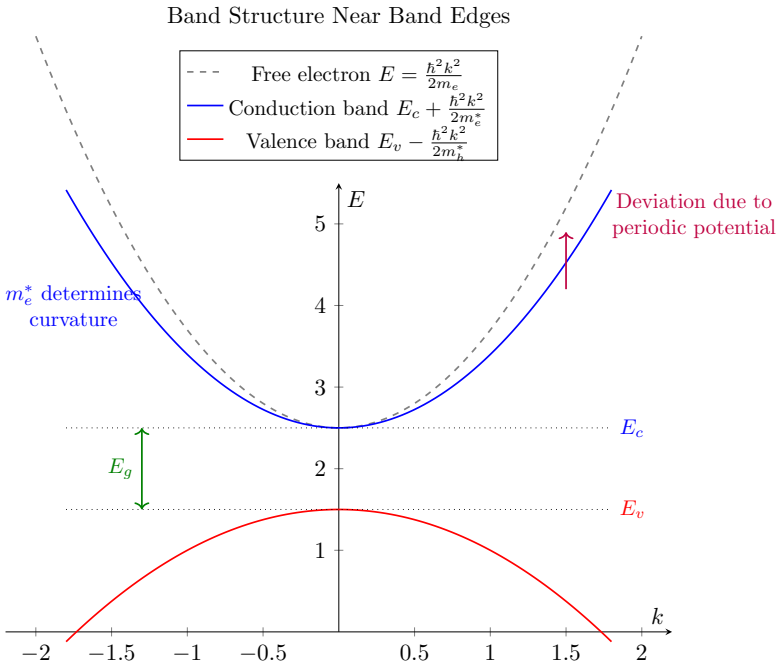
- The average repulsive interaction with all other electrons
- The attractive potential from the nuclei
- Exchange and correlation effects (quantum mechanical interactions between electrons)

Solving this equation (using methods from solid state physics like the nearly-free electron model or tight-binding model) yields the actual band structure $E(k)$ for the crystal. Remarkably, near the band edges (the bottom of the conduction band or the top of the valence band), this $E(k)$ relationship can often be approximated as parabolic, but with a modified curvature.

$$\boxed{E(k) \approx E_c + \frac{\hbar^2 k^2}{2m_e^*}} \quad \text{(near conduction band minimum)}$$

$$\boxed{E(k) \approx E_v - \frac{\hbar^2 k^2}{2m_h^*}} \quad \text{(near valence band maximum)}$$

Here, m_e^* and m_h^* are the **effective masses** for electrons and holes (missing electrons in the valence band), respectively. E_c is the conduction band edge energy, and E_v is the valence band edge energy.



4.3 Effective Mass: Curvature is Mass

The effective mass is not just a fitting parameter—it has a rigorous and profound definition rooted in the band structure itself.

Effective Mass Definition

Start with the free electron dispersion relation:

$$E = \frac{\hbar^2 k^2}{2m_e}$$

Differentiate once with respect to k:

$$\frac{dE}{dk} = \frac{\hbar^2 k}{m_e} = \frac{\hbar p}{m_e} = \hbar v_g$$

This gives the **group velocity** $v_g = (1/\hbar)(dE/dk)$.
Differentiate a second time:

$$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m_e}$$

Solving for mass:

$$\boxed{m_e = \frac{\hbar^2}{d^2E/dk^2}}$$

For electrons in a crystal, we *define* the effective mass by this same relationship:

$$\boxed{m^* = \frac{\hbar^2}{d^2E/dk^2}}$$

The effective mass is therefore a measure of the **curvature** of the $E(k)$ dispersion relation:

- **Large curvature** (d^2E/dk^2 large) $\Rightarrow$ **small effective mass** — electron responds easily to forces

- **Small curvature** (d^2E/dk^2 small) $\Rightarrow$ **large effective mass** — electron is "heavy" and sluggish
- **Negative curvature** ($d^2E/dk^2 < 0$) $\Rightarrow$ **negative effective mass** — occurs near the top of the valence band and gives rise to holes (positively charged quasi-particles)

Example 1 (Electron and Hole Effective Masses in Common Semiconductors)**.**

Material	**Electron effective mass** (m_e^*/m_0)	**Hole effective mass** (m_h^*/m_0)
Silicon (Si)	0.98 (longitudinal) 0.19 (transverse)	0.49 (heavy) 0.16 (light)
Germanium (Ge)	1.64 (longitudinal) 0.082 (transverse)	0.28 (heavy) 0.044 (light)
Gallium Arsenide (GaAs)	0.067	0.45 (heavy) 0.082 (light)

Note: $m_0 = 9.11 \times 10^{-31}$ kg is the free electron rest mass.

4.4 Where Do Dispersion Relations Come From?

In practice, the detailed $E(k)$ dispersion relations for real materials come from two sources:

1. **Ab initio numerical calculations**: Using powerful computers and density functional theory (DFT) or other first-principles methods, solid state physicists can solve the Schrödinger equation for the periodic potential to high accuracy. These calculations yield the full band structure $E(k)$ for all k in the Brillouin zone.
2. **Experimental measurement**: Techniques like angle-resolved photoemission spectroscopy (ARPES) can directly measure the $E(k)$ relationship by shining X-rays on a material and analyzing the energy and angle of emitted electrons.

From these dispersion relations, we estimate effective masses by numerically computing the second derivative d^2E/dk^2 near the band edges. These effective masses then become essential parameters for device modeling, allowing us to treat electrons in semiconductors using modified versions of classical equations, but with m_e replaced by m^*.

Why Effective Mass Matters: A glimpse into future topics

The effective mass appears in virtually every important equation in semiconductor device physics:

- **Density of states**: $g_c(E) = \frac{(2m_e^*)^{3/2}}{2\pi^2\hbar^3}\sqrt{E - E_c}$
- **Mobility**: $\mu = \frac{q\tau}{m^*}$ (where τ is the scattering time)
- **Drift velocity**: $\mathbf{v}_d = \mu\mathbf{E} = \frac{q\tau}{m^*}\mathbf{E}$
- **Thermal velocity**: $v_{th} = \sqrt{\frac{3k_BT}{m^*}}$

A smaller effective mass means higher mobility, faster devices, but also lighter, more quantum-mechanically delocalized electrons. This trade-off is central to semiconductor materials engineering.

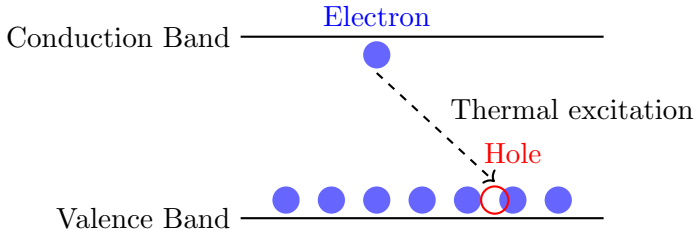
In the next sections, we will use these dispersion relations and effective masses to understand how electrons populate energy bands, how they respond to electric fields, and ultimately how they enable the operation of diodes and transistors.

5 Holes: The Missing Electrons?

One of the most elegant and practically important concepts in semiconductor physics is the **hole**. Holes are not merely an abstract convenience—they are essential for understanding current flow in semiconductors and form the basis for entire classes of devices (p-type materials, p-n junctions, bipolar transistors). In this section, we will develop the concept of holes systematically, from intuitive picture to rigorous quantum mechanical foundation.

5.1 The Intuitive Picture: A Missing Electron

Imagine a nearly full valence band, with most electron states occupied but a few left empty. This situation arises naturally in semiconductors at room temperature, where thermal energy excites some electrons from the valence band to the conduction band.



Now consider what happens when we apply an electric field **E** to this system:

1. An electron from a neighboring site can hop into the empty state (the hole).
2. This leaves behind a new empty state at the electron's original location.
3. The empty state (hole) appears to move in the direction **opposite** to electron motion.



The hole behaves as if it were a positively charged particle moving in the direction of the electric field (opposite to electron flow). This intuitive picture is powerful, but it raises a fundamental question: **Why are holes prominent in semiconductors but not in metals?**

5.2 Why Holes Matter in Semiconductors, Not in Metals

The answer lies in the band structure and the position of the Fermi level (assumed as know in the following box).

Holes in Metals vs. Semiconductors

- **In metals**: The Fermi level lies inside a band. There are plenty of empty states *just above* the Fermi level and occupied states *just below*. When an electric field is applied, electrons near the Fermi level can easily move into nearby empty states. Current is carried primarily by electrons, and the concept of "holes" is not particularly useful because there are so many empty states available everywhere.
- **In semiconductors**: At low temperatures, the valence band is completely full and the conduction band is completely empty. No current can flow. At finite temperatures, a small number of electrons are excited to the conduction band, leaving behind an equal number of empty states (holes) in the valence band. These holes are **the only available empty states** for the remaining valence electrons to move into. Consequently, hole current dominates in the valence band.

The key insight: holes become prominent when the valence band is **nearly full** and empty states are scarce. This is precisely the situation in semiconductors, where the bandgap limits the number of thermally excited electrons.

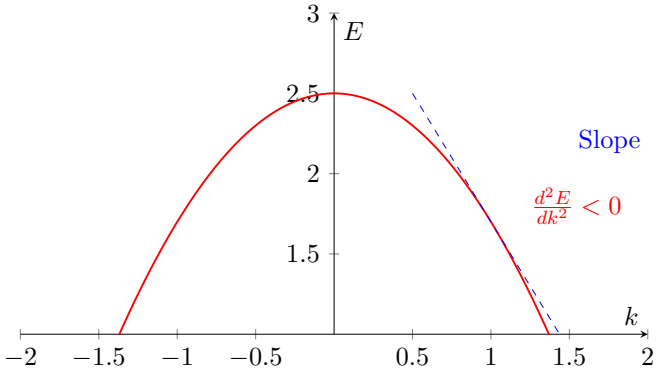
5.3 Quantum Mechanical Foundation: Negative Effective Mass

To understand holes rigorously, we must analyze the dynamics of electrons near the **top of the valence band**. Consider the dispersion relation $E(k)$ near the valence band maximum. For a simple parabolic band model:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2|m^*|}$$

where E_v is the energy at the valence band top, and we explicitly write $|m^*|$ to indicate the magnitude of the effective mass. Note the **negative sign** in front of the quadratic term.

Valence Band Top - Negative Curvature



Now compute the effective mass using our definition from Section 4:

$$m^* = \frac{\hbar^2}{d^2E/dk^2}$$

Since $d^2E/dk^2 < 0$ at the valence band top, we obtain:

$$\boxed{m^* < 0}$$

This is a profound result: **Electrons near the top of the valence band have negative effective mass.**

What Does Negative Mass Mean?

Consider Newton's second law for an electron in an external electric field $\mathbf{E}$:

$$\mathbf{F} = -e\mathbf{E} = m^*\mathbf{a}$$

If m^* is negative, then the acceleration $\mathbf{a}$ is in the direction *opposite* to what we would normally expect:

$$\mathbf{a} = \frac{\mathbf{F}}{m^*} = \frac{-e\mathbf{E}}{m^*}$$

Since $m^* < 0$, the denominator is negative, making $\mathbf{a}$ positive when $\mathbf{E}$ is positive. The electron accelerates in the *same direction* as the electric field—opposite to the force on it!

This counterintuitive behavior arises entirely from the quantum mechanical $E(k)$ relation. However, dealing with negative mass particles is mathematically inconvenient. We can instead redefine our description.

5.4 The Hole as a Positive Charge Carrier

Let us transform our description by defining a new particle—the hole—with the following properties:

Definition 6 (Hole)**.** A hole is a fictitious particle that represents the absence of an electron in an otherwise full valence band. Its properties are defined as:

$$\begin{aligned} \text{Charge:} \quad & q_h = +e \\ \text{Effective mass:} \quad & m_h^* = -m_e^* > 0 \\ \text{Energy:} \quad & E_h = -E_e \quad \text{(increases downward in band diagram)} \end{aligned}$$

To see why this works, consider the dynamics. For an electron with negative mass $m_e^* < 0$:

$$\mathbf{a} = \frac{-e\mathbf{E}}{m_e^*}$$

Since m_e^* is negative, let us write $m_e^* = -m_h^*$ where $m_h^* > 0$. Then:

$$\mathbf{a} = \frac{-e\mathbf{E}}{-m_h^*} = \frac{e\mathbf{E}}{m_h^*}$$

The Hole Transformation Summary

Property	Electron at valence band top	Hole
Charge	$-e$	$+e$
Effective mass	$m_e^* < 0$	$m_h^* = -m_e^* > 0$
Velocity	$v_e = \frac{1}{\hbar}\frac{dE_e}{dk}$	$v_h = v_e$ (same)
Wave vector	$\mathbf{k}_e$	$\mathbf{k}_h = -\mathbf{k}_e$
Energy	$E_e(k)$ (max at $k = 0$)	$E_h(k) = -E_e(k)$ (min at $k = 0$)

Example 2 (Hole Motion in an Electric Field)**.** Consider an electron near the valence band top with $m_e^* = -0.5m_0$. An electric field $\mathbf{E} = 10^4$ V/m is applied in the $+x$ direction.

Electron perspective: Force on electron: $\mathbf{F} = -e\mathbf{E} = -1.6 \times 10^{-15}$ N in $+x$ direction. Acceleration: $\mathbf{a} = \mathbf{F}/m_e^* = (-1.6 \times 10^{-15})/(-0.5 \times 9.11 \times 10^{-31}) = +3.51 \times 10^{15}$ m/s^2 in $+x$ direction.

Hole perspective: A hole with $m_h^* = +0.5m_0$ experiences force $\mathbf{F} = +e\mathbf{E} = +1.6 \times 10^{-15}$ N. Acceleration: $\mathbf{a} = \mathbf{F}/m_h^* = (1.6 \times 10^{-15})/(0.5 \times 9.11 \times 10^{-31}) = +3.51 \times 10^{15}$ m/s^2 in $+x$ direction.

Both descriptions give identical physical behavior, but the hole picture uses familiar positive mass and positive charge.

5.5 Alternative Justification of the Hole Concept: Current Argument

We can also understand the hole concept by examining how current is carried in a nearly full valence band. This approach avoids the negative effective mass formulation and focuses directly on current contributions.

Current Density in Semiconductors

The current density $\mathbf{J}$ (units: A/cm^2 or A/m^2) due to a collection of charged particles is given by:

$$\mathbf{J} = q\, n\, \mathbf{v}_d$$

where:

- q is the charge per particle (coulombs)
- n is the particle concentration (cm^{-3} or m^{-3})
- $\mathbf{v}_d$ is the average drift velocity (cm/s or m/s)

In three dimensions, for a distribution of electrons with different velocities, we must sum over all electrons. The total current (units: amperes) is obtained by integrating the current density over a cross-sectional area. For our purposes, we can work with the total current contribution from a collection of electrons:

$$\mathbf{I}_{\text{total}} = -e \sum_i \mathbf{v}_i$$

where the sum runs over all electrons in the band, and $\mathbf{v}_i$ is the velocity of the i-th electron. The negative sign reflects the electron's negative charge $-e$.

Current in a Nearly Full Valence Band

Consider a valence band that is completely full except for **one missing electron** at some state. The total current carried by all electrons in this band can be written as:

$$\mathbf{I}_{\text{total}} = \mathbf{I}_{\text{full band}} - \mathbf{I}_{\text{missing electron}}$$

where:

- $\mathbf{I}_{\text{full band}} = -e \sum_{\text{all states}} \mathbf{v}_i$ is the current if the band were completely full
- $\mathbf{I}_{\text{missing electron}} = -e\, \mathbf{v}_j$ is the current that would be contributed by the missing electron if it were present (with $\mathbf{v}_j$ being its velocity)

Why a Full Band Carries No Net Current

In a perfectly periodic crystal, the electron velocity is given by the quantum mechanical **group velocity**:

$$\mathbf{v}(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E(\mathbf{k})$$

For the valence band, the dispersion relation $E(\mathbf{k})$ has a important symmetry property:

$$E(-\mathbf{k}) = E(\mathbf{k})$$

This symmetry implies that for every electron state with wavevector $\mathbf{k}$ and velocity $\mathbf{v}(\mathbf{k})$, there is a corresponding state with wavevector $-\mathbf{k}$ and velocity $\mathbf{v}(-\mathbf{k}) = -\mathbf{v}(\mathbf{k})$.

Therefore, when we sum over **all** states in a completely full valence band:

$$\mathbf{I}_{\text{full band}} = -e \sum_{\text{all } \mathbf{k}} \mathbf{v}(\mathbf{k}) = -e \left[\sum_{\mathbf{k}>0} \mathbf{v}(\mathbf{k}) + \sum_{\mathbf{k}<0} \mathbf{v}(\mathbf{k}) \right] = -e \left[\sum_{\mathbf{k}>0} \mathbf{v}(\mathbf{k}) + \sum_{\mathbf{k}>0} (-\mathbf{v}(\mathbf{k})) \right] = 0$$

Each velocity contribution is exactly canceled by its opposite counterpart. Hence, a completely full band carries **no net current**.

Current from a Missing Electron

Returning to our expression for the nearly full band:

$$\mathbf{I}_{\text{total}} = \mathbf{I}_{\text{full band}} - \mathbf{I}_{\text{missing electron}} = 0 - (-e\,\mathbf{v}_j) = +e\,\mathbf{v}_j$$

The missing electron (the empty state) thus contributes a current equivalent to a particle with:

- **Positive charge** $+e$ (instead of the electron's $-e$)
- The same velocity $\mathbf{v}_j$ as the missing electron would have had

Physical Interpretation

This derivation reveals that:

1. The current in a nearly full valence band can be completely described by considering only the **empty states** (holes).
2. Each empty state behaves as if it were a positively charged particle with charge $+e$ moving with velocity $\mathbf{v}(\mathbf{k})$.
3. We never needed to invoke negative effective mass—the hole emerges naturally from the cancellation of currents in the full band and the symmetry of the valence band $E(\mathbf{k})$ relation.
4. The hole's dynamics under external fields will follow from how $\mathbf{v}(\mathbf{k})$ changes, which is governed by the band structure.

This current-based argument provides a clean, mathematically rigorous justification for treating holes as independent positive charge carriers, complementing the negative effective mass picture developed earlier.

5.6 Experimental Consequences

The existence of holes is not merely theoretical—it has direct experimental consequences:

1. **Hall effect**: In p-type semiconductors, the Hall coefficient is positive, directly confirming that the majority carriers are positively charged.
2. **Cyclotron resonance**: Measurements of cyclotron frequency yield positive effective masses for holes, matching theoretical predictions.
3. **Thermoelectric effects**: The Seebeck coefficient changes sign between n-type and p-type materials, consistent with hole conduction.

The concept of holes thus provides a complete and consistent description of current flow in the valence band, allowing us to treat both electrons in the conduction band and holes in the valence band as ordinary positively-charged particles with positive effective masses. This duality forms the foundation for understanding p-n junctions, bipolar transistors, and virtually all semiconductor devices.

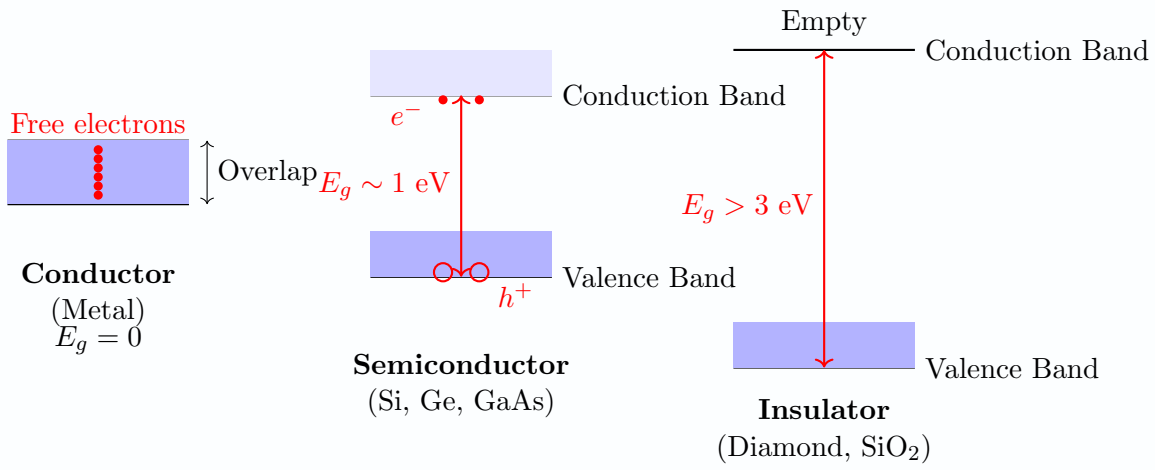
6 Classification of Solids: Insulators, Conductors, and Semiconductors

With our understanding of energy bands, effective mass, and holes, we can now systematically classify materials based on their electronic properties. This classification explains the vast differences in conductivity between materials like copper, silicon, and diamond, and it sets the stage for engineering semiconductors through doping.

6.1 Band Structure Classification

The key factor determining a material's electrical behavior is the arrangement of its energy bands—specifically, the valence band, the conduction band, and the bandgap between them.

The Three Classes of Solids



Property	Conductor	Semiconductor	Insulator
Bandgap E_g	0 (overlap)	$\sim 0.5 - 3$ eV	> 3 eV
Conductivity σ (S/m)	$10^7 - 10^8$	$10^{-6} - 10^4$	$< 10^{-10}$
Carrier concentration	$\sim 10^{22}$ cm^{-3}	$10^{10} - 10^{19}$ cm^{-3}	~ 0
Temperature dependence	σ decreases with T	σ increases with T	Insulating at all T

6.2 Detailed Analysis of Each Class

Conductors (Metals)

In conductors, either:

- The valence and conduction bands **overlap** (e.g., in alkali metals like Na, K), or
- The conduction band is **partially filled** (e.g., in noble metals like Cu, Ag, Au)

Since there is no bandgap, electrons have access to a continuum of empty states immediately above the Fermi level. Under an applied electric field, electrons can easily gain energy and move to these nearby empty states, resulting in very high conductivity.

Insulators

Insulators have a **large bandgap** ($E_g > 3$ eV). At room temperature, thermal energy $k_B T \approx 0.026$ eV is far too small to excite electrons across this gap. Consequently:

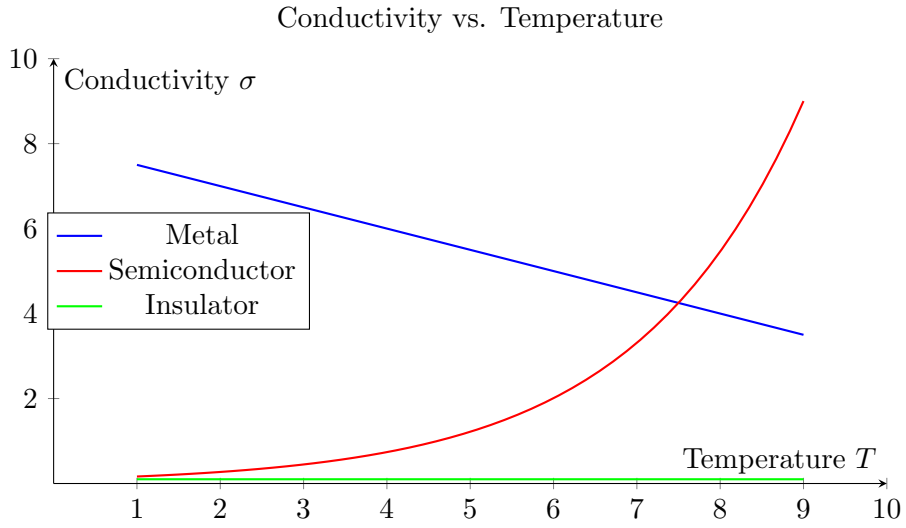
- The valence band remains completely full

- The conduction band remains completely empty
- No carriers are available for conduction

Even at elevated temperatures, the number of thermally excited electrons remains negligible. For example, diamond ($E_g = 5.5$ eV) remains an excellent insulator up to very high temperatures.

Semiconductors

Semiconductors have an **intermediate bandgap** ($\sim 0.5 - 3$ eV). At $T = 0$ K, they behave as perfect insulators—the valence band is full and the conduction band is empty. However, at finite temperatures, a significant number of electrons can be thermally excited across the gap.



6.3 Intrinsic Semiconductors

Definition 7 (Intrinsic Semiconductor)**.** An **intrinsic semiconductor** is a pure semiconductor material with no intentionally added impurities. In an intrinsic semiconductor, the number of electrons in the conduction band equals the number of holes in the valence band:

$$n = p = n_i$$

where n_i is the **intrinsic carrier concentration**.

The enormous range in n_i (seven orders of magnitude!) reflects the strong exponential dependence on bandgap. GaAs, with its larger bandgap, has far fewer intrinsic carriers than silicon, which is why it can be made semi-insulating and is useful for high-frequency devices.

Key Properties of Intrinsic Semiconductors

- **Fermi level**: Located approximately at the middle of the bandgap

$$E_F \approx \frac{E_c + E_v}{2} = E_v + \frac{E_g}{2}$$

- **Carrier concentration**: Given by the mass action law

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)$$

  where N_c and N_v are the effective density of states in the conduction and valence bands.

- **Material examples**: Pure silicon (Si), germanium (Ge), gallium arsenide (GaAs) at room temperature

Material	n_i at 300 K (cm^{-3})
Germanium (Ge)	2.4×10^{13}
Silicon (Si)	1.5×10^{10}
Gallium Arsenide (GaAs)	1.8×10^{6}

6.4 Extrinsic Semiconductors and Doping

The real power of semiconductor technology comes from our ability to **engineer** the carrier concentrations by introducing controlled amounts of impurities—a process called **doping**.

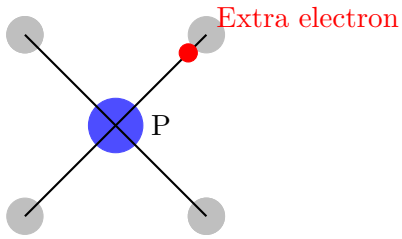
The Doping Revolution

Pure, intrinsic silicon has a carrier concentration of only $\sim 10^{10}$ cm^{-3}. By adding impurities at parts-per-million levels, we can increase this to $10^{15} - 10^{19}$ cm^{-3}—a factor of up to one billion! This ability to control conductivity over such a vast range is what makes semiconductor devices possible.

Donors and n-Type Semiconductors

Definition 8 (Donor Impurity)**.** A **donor** is an impurity atom that has more valence electrons than the host semiconductor atom it replaces. It "donates" an electron to the conduction band.

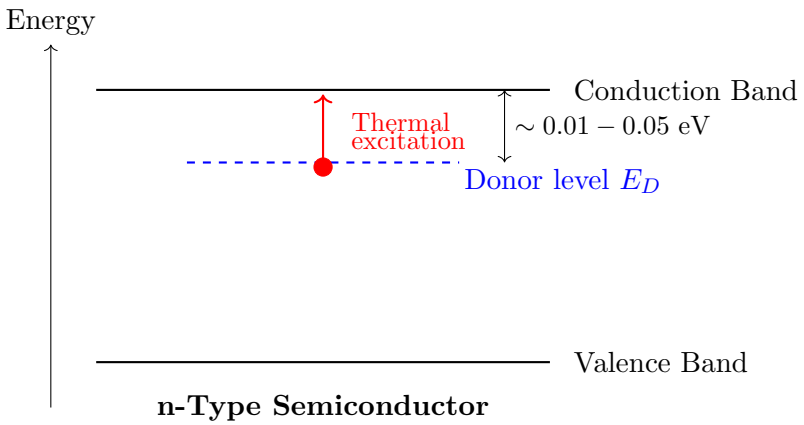
Consider silicon (Group IV, 4 valence electrons). If we replace a silicon atom with a phosphorus atom (Group V, 5 valence electrons):



Phosphorus donor in silicon

Four of phosphorus's five valence electrons form covalent bonds with neighboring silicon atoms. The fifth electron is loosely bound and can be easily excited into the conduction band at room temperature.

The energy level of this donor electron lies just below the conduction band edge—typically only $\sim 0.01 - 0.05$ eV for common donors like P, As, or Sb in Si.



Since each donor contributes one electron to the conduction band, and the donor concentration N_D is typically much larger than n_i, we have:

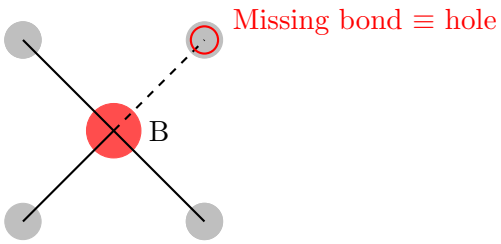
$$n \approx N_D, \quad p = \frac{n_i^2}{n} \approx \frac{n_i^2}{N_D} \ll n$$

Such a material is called **n-type** because the majority carriers are negatively charged electrons.

Acceptors and p-Type Semiconductors

Definition 9 (Acceptor Impurity)**.** An **acceptor** is an impurity atom that has fewer valence electrons than the host semiconductor atom. It "accepts" an electron from the valence band, creating a hole.

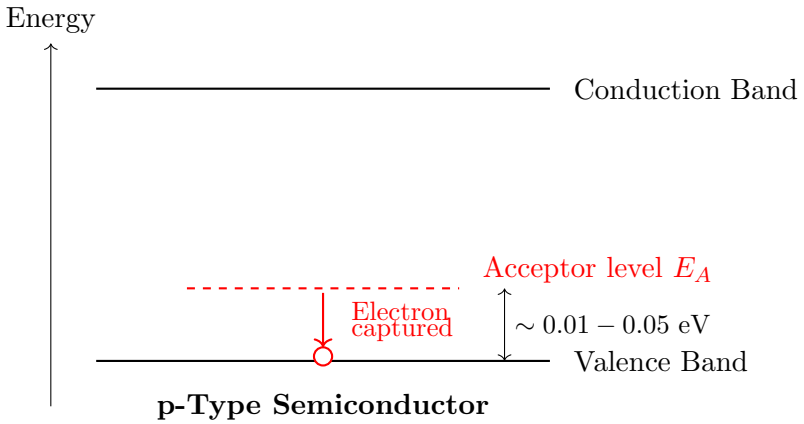
Now consider replacing a silicon atom with boron (Group III, 3 valence electrons):



Boron acceptor in silicon

Boron has only three valence electrons, so it can only form three complete covalent bonds. The fourth bond site is missing an electron—this is effectively a hole. Boron "accepts" an electron from the neighboring silicon to complete the bond, creating a hole in the valence band.

The acceptor energy level lies just above the valence band edge, typically $\sim 0.01 - 0.05$ eV for common acceptors like B, Al, or Ga in Si.



For acceptor concentration $N_A \gg n_i$, we have:

$$p \approx N_A, \quad n = \frac{n_i^2}{p} \approx \frac{n_i^2}{N_A} \ll p$$

This is **p-type** material, where the majority carriers are positively charged holes.

6.5 Compensation and Carrier Statistics

When both donors and acceptors are present, they **compensate** each other—donors provide electrons that fall into acceptor levels. The net carrier concentration is determined by the difference:

$$n \approx N_D - N_A \quad (\text{if } N_D > N_A, \text{ n-type})$$
$$p \approx N_A - N_D \quad (\text{if } N_A > N_D, \text{ p-type})$$

Doping Summary

Property	n-Type	p-Type
Dopant type	Donor	Acceptor
Dopant examples	P, As, Sb (Group V)	B, Al, Ga (Group III)
Majority carriers	Electrons ($n \approx N_D$)	Holes ($p \approx N_A$)
Minority carriers	Holes ($p = n_i^2/N_D$)	Electrons ($n = n_i^2/N_A$)
Fermi level	Near conduction band	Near valence band

- **Typical doping levels**: $10^{14} - 10^{20}$ cm^{-3}
- **Intrinsic concentration at 300K for Si**: 1.5×10^{10} cm^{-3}
- Doping can change carrier concentration by over 10 orders of magnitude!

The ability to create n-type and p-type regions in the same crystal, and to control the carrier concentrations precisely, is what enables all modern semiconductor devices—from simple diodes to complex integrated circuits. In the next sections, we will explore how these doped materials behave when brought together to form p-n junctions, the fundamental building blocks of semiconductor electronics.

7 n-Type and p-Type Semiconductors

With our understanding of donors and acceptors from the previous section, we can now define the two types of extrinsic semiconductors that form the basis of all semiconductor devices. At this stage, we will keep the discussion conceptual, focusing on what these materials are and why they matter, without delving into the statistical mechanics of carrier distributions.

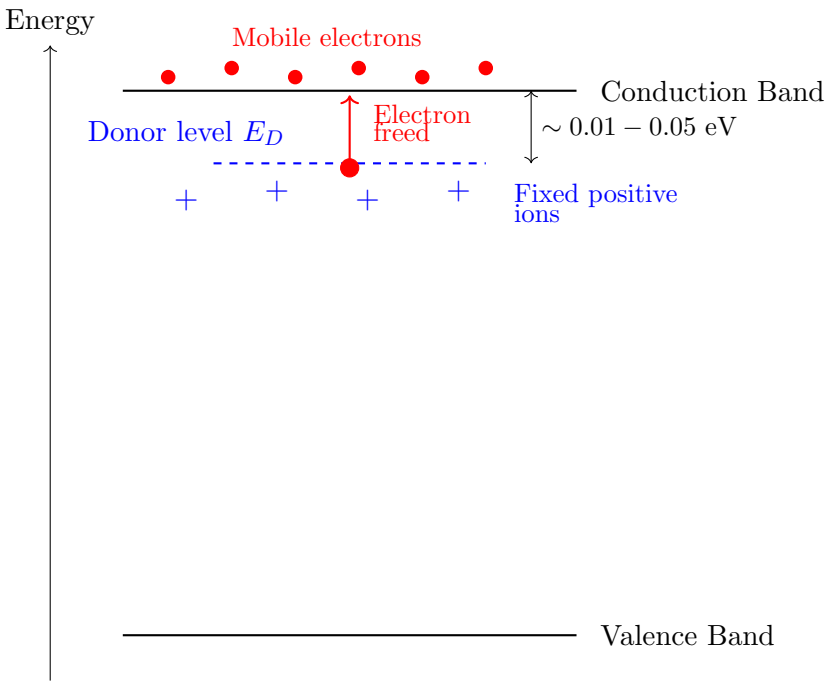
7.1 n-Type Semiconductors

Definition 10 (n-Type Semiconductor - Conceptual)**.** An **n-type semiconductor** is a semiconductor material that has been doped with **donor impurities**—atoms that have more valence electrons than the host semiconductor atoms.

The name "n-type" comes from the **negative** charge of the majority carriers (electrons).

What Makes a Material n-Type?

- **How it's made**: By adding donor impurities (Group V elements like phosphorus, arsenic, or antimony) to a pure semiconductor like silicon.
- **What happens**: Each donor atom has five valence electrons, but only four are needed to form covalent bonds with neighboring silicon atoms. The fifth electron is loosely bound and can easily break free and move around the crystal.
- **The result**: After losing its extra electron, the donor atom becomes a **positively charged ion** fixed in the crystal lattice. The freed electron becomes a mobile negative charge carrier.
- **Why "n-type"**: Because the material now has an excess of mobile **negative** charges (electrons) that can conduct electricity.



n-Type: Extra electrons, fixed positive ions

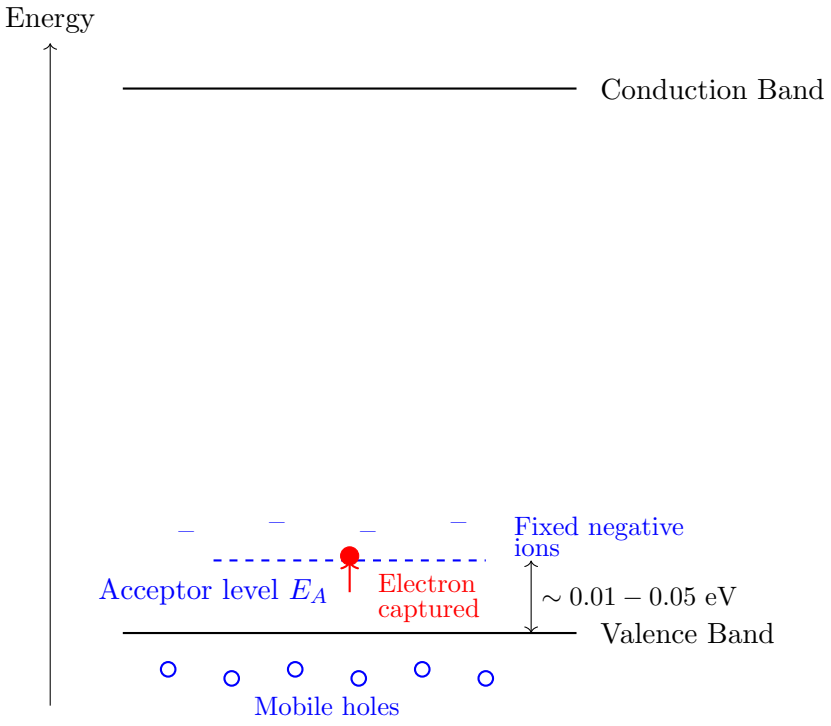
7.2 p-Type Semiconductors

Definition 11 (p-Type Semiconductor - Conceptual)**.** A **p-type semiconductor** is a semiconductor material that has been doped with **acceptor impurities**—atoms that have fewer valence electrons than the host semiconductor atoms.

The name "p-type" comes from the **positive** charge of the majority carriers (holes).

What Makes a Material p-Type?

- **How it's made**: By adding acceptor impurities (Group III elements like boron, aluminum, or gallium) to a pure semiconductor like silicon.
- **What happens**: Each acceptor atom has only three valence electrons, but needs four to complete all its covalent bonds. It "accepts" an electron from a neighboring silicon atom to complete the fourth bond.
- **The result**: After accepting an electron, the acceptor atom becomes a **negatively charged ion** fixed in the crystal lattice. The missing electron in the neighboring bond creates a **hole**—a mobile positive charge carrier.
- **Why "p-type"**: Because the material now has an excess of mobile **positive** charges (holes) that can conduct electricity.



p-Type: Extra holes, fixed negative ions

7.3 Intuitive Comparison

n-Type vs. p-Type: The Big Picture

Think of it this way:

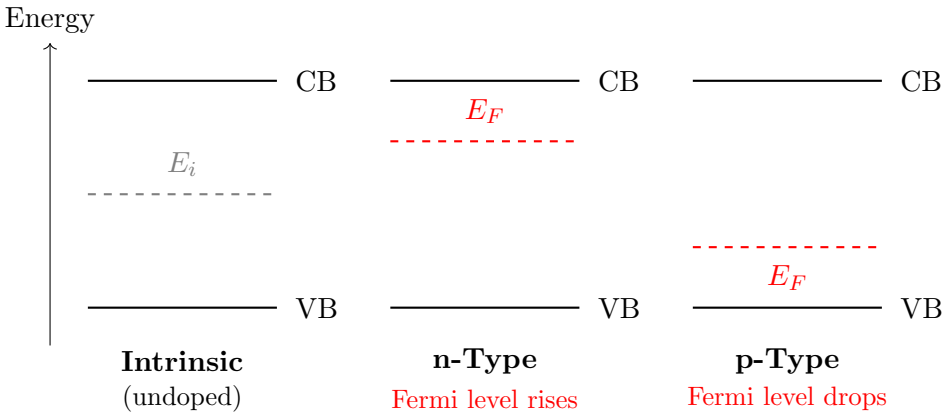
	n-Type	**p-Type**
Dopant example	Phosphorus (5 valence electrons)	Boron (3 valence electrons)
What the dopant does	"Donates" an extra electron	"Accepts" an electron, creating a hole
Mobile carriers	Electrons (negative charge)	Holes (positive charge)
Fixed ions	Positively charged donors	Negatively charged acceptors
Mnemonic	**n** for **n**egative carriers	**p** for **p**ositive carriers

- In **n-type** material, think: "Extra electrons = Negative carriers"
- In **p-type** material, think: "Missing electrons = Positive holes"

7.4 What About the Fermi Level?

Without diving into statistics, we can understand the Fermi level conceptually in doped materials:

- In **n-type** material, the extra electrons "push" the Fermi level **up** toward the conduction band. Think of it like adding more water to a reservoir—the water level rises.
- In **p-type** material, the missing electrons (holes) "pull" the Fermi level **down** toward the valence band. Think of it like removing water from a reservoir—the water level drops.
- In **intrinsic** (undoped) material, the Fermi level sits approximately in the middle of the bandgap.



Fermi Level Position in Semiconductors

7.5 Why This Matters

The creation of n-type and p-type regions in the same crystal is the fundamental building block of all semiconductor devices:

- **p-n junctions**: When n-type and p-type materials are brought together, the junction between them forms a diode—the simplest semiconductor device that allows current to flow in only one direction.

- **Transistors**: Both bipolar junction transistors (npn or pnp) and field-effect transistors rely on alternating n-type and p-type regions to provide amplification and switching.
- **Integrated circuits**: Modern chips contain billions of tiny n-type and p-type regions arranged to create complex logic circuits.

The Central Insight

The ability to create n-type and p-type materials in the same crystal, and to control where each type is located, is what enables all of modern electronics. Without this capability, we would have no diodes, no transistors, no integrated circuits—and no digital age.
In the next section, we will explore what happens when we bring n-type and p-type materials together to form a p-n junction, and discover how this simple structure can rectify current and form the basis of countless electronic devices.

8 The Fermi Energy: The "Fill Line" for Electrons

Before we can understand how electrons distribute themselves among available energy states, and how devices like p-n junctions work, we need to introduce one of the most important concepts in semiconductor physics: the **Fermi energy**.

At this point, we will avoid the mathematical complexity of Fermi-Dirac statistics. Instead, we will develop an intuitive, physical picture that captures the essence of what the Fermi energy represents and why it matters.

8.1 The Classical vs. Quantum Expectation

The Classical Expectation

To appreciate the strange behavior of electrons, let us first consider a classical system. Imagine a large classroom filled with students. Each student possesses a certain amount of thermal energy (body heat).

- **At room temperature**: The students move around with varying amounts of energy. Most will cluster around a typical average value, say E_{avg}, following a standard bell-shaped distribution.
- **As we lower the temperature**: The room gets colder. Students lose thermal energy and slow down.
- **At absolute zero ($T = 0$ K)**: In the classical world, **all students would eventually lose all their thermal energy**. They would all cluster into the absolute lowest possible energy state (imagine them all freezing perfectly still on the floor). The energy distribution would collapse into a single spike at zero.

This is how classical objects behave: given enough cold, everything settles into the absolute lowest energy state available.

The Quantum Reality: The "Strictly Assigned Seats" Rule

Now, replace the students with **electrons**. We perform the exact same experiment: we start at room temperature and gradually cool the crystal to absolute zero. What happens to their energy distribution?

The Pauli Exclusion Principle: No Sharing Allowed

Electrons are fermions, and they obey the **Pauli Exclusion Principle**. In the context of energy states, this principle acts like a strict seating chart in an auditorium:

> **Each quantum energy state can hold a maximum of two electrons**
> (one spin up, one spin down).
> No two electrons can occupy the exact same quantum state simultaneously.

This rule fundamentally changes how electrons respond to cold. They **cannot** all collapse into the lowest energy state because there is only room for two!

Because of this restriction, as we cool the system to $T = 0$ K, electrons must **stack up** in energy:

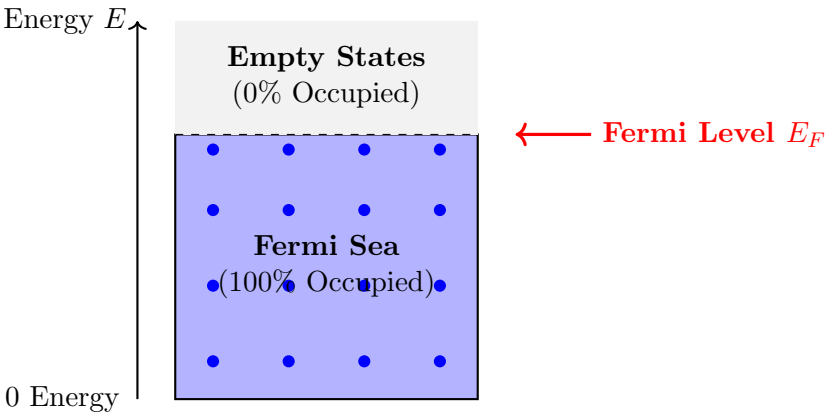
1. The lowest energy state gets filled (2 electrons).
2. The next lowest available state gets filled.
3. The next gets filled, and so on.

They continue filling state-by-state from the bottom up until every electron in the crystal has found a place.

8.2 Defining the Fermi Level

The Fermi Sea at Absolute Zero ($T = 0$ K)

This stacking creates a striking quantum phenomenon. At absolute zero, the electrons form a "Fermi sea"—a dense collection of occupied states rising from zero energy up to a maximum level, then abruptly stopping.



Electrons stack up to E_F at $T = 0$ K

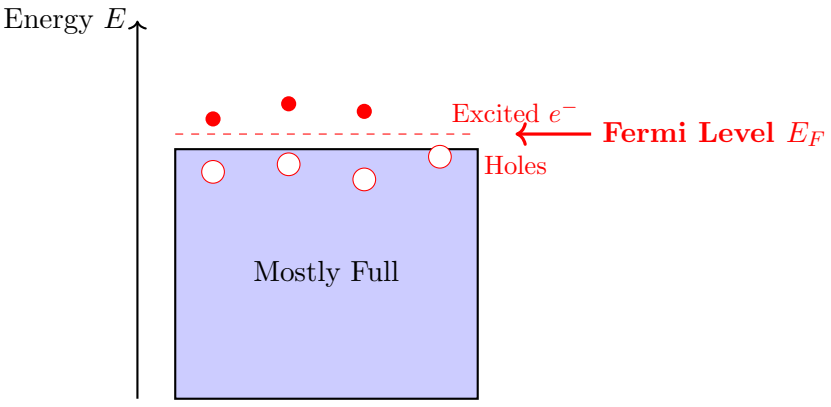
The highest energy level that contains electrons at $T = 0$ K is called the **Fermi energy** (E_F).

- **Below** E_F: Every single state is occupied.
- **Above** E_F: Every single state is empty.
- **At** E_F: The transition from "completely full" to "completely empty" is perfectly sharp.

This is the **Fermi-Dirac distribution at absolute zero**: a perfect step function. Electrons **refuse to freeze** at the bottom; they maintain a wide energy distribution up to E_F solely because they are forbidden from sharing states.

Thermal Smearing at Room Temperature ($T > 0$ K)

What happens when we warm the crystal back to room temperature? Thermal energy gives some electrons near the top of the "Fermi sea" a small kick. A tiny fraction jump above E_F, leaving behind empty spots (holes) just below it.



At $T > 0$ K, a few electrons jump above E_F

Intuitive Definition of Fermi Energy

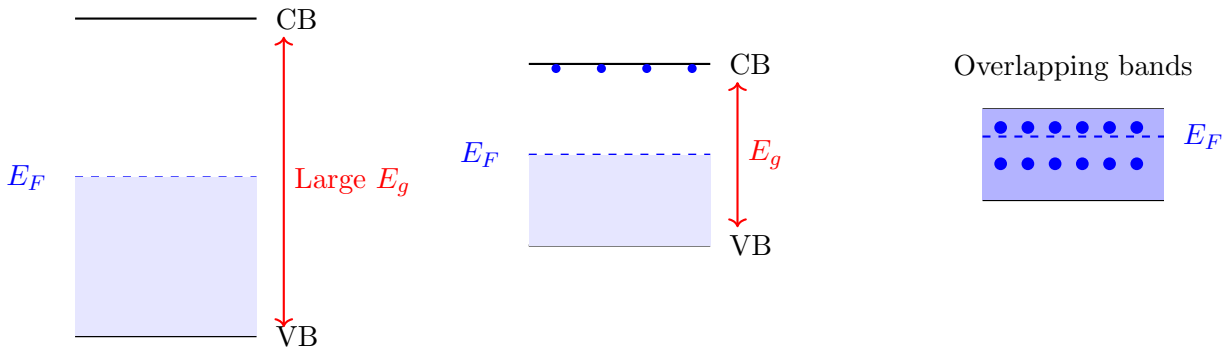
The **Fermi energy** E_F is the "fill line" or chemical potential for electrons. It acts as a rigid reference level:

- Energy states **well below** E_F are essentially **100% full**.
- Energy states **well above** E_F are essentially **100% empty**.
- The probability of finding an electron exactly at E_F is **50%** (at any $T > 0$ K).

In semiconductors, E_F typically lies inside the forbidden bandgap. Even though no actual electron states exist *at* E_F, it serves as the crucial reference point determining how many electrons are thermally excited into the conduction band.

8.3 Visualizing E_F Across Different Materials

Let us see how this "fill line" behaves in our three classes of materials:



Insulator	**Semiconductor**	**Conductor (Metal)**
E_F in gap	E_F near midgap	E_F inside band
No free electrons	Few electrons in CB	Many free electrons

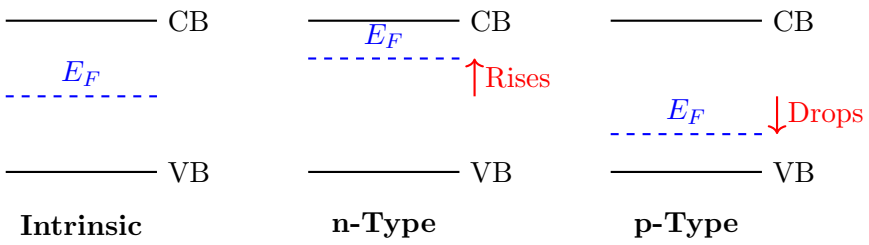
Energy increases upward ↑

8.4 What Determines E_F and Why It Matters

The position of the Fermi energy is not arbitrary; it acts as the "chemical potential" that balances the competition between quantum statistics and thermal agitation.

Factors Determining the Fermi Level

- **Doping**: Adding impurities dramatically shifts E_F. In **n-type** material (extra electrons), E_F rises toward the conduction band. In **p-type** material (extra holes), E_F drops toward the valence band.
- **Temperature**: As T increases, the sharp "fill line" softens. E_F shifts slightly to maintain charge neutrality as more carriers are thermally generated.
- **Bandgap**: Materials with larger E_g keep E_F farther from both bands, making it harder for carriers to populate the conduction/valence bands.



The Fermi energy is not just an abstract reference—it has profound practical implications:

1. **Junction formation**: When two materials with different E_F are brought together (e.g., p-type and n-type), electrons flow until their Fermi levels align. This equilibration creates the built-in electric fields essential for diodes and transistors.
2. **Carrier concentrations**: The distance between E_F and the band edges directly determines the number of free carriers:
   - E_F close to conduction band $\Rightarrow$ many electrons, few holes (n-type)
   - E_F close to valence band $\Rightarrow$ many holes, few electrons (p-type)
   - E_F far from both bands $\Rightarrow$ very few carriers (insulator/intrinsic)
3. **Device behavior**: Temperature shifts in E_F explain why semiconductor circuits behave differently in extreme environments compared to metals.

The Fermi Energy as a "Tug-of-War" Reference

Think of E_F as the rope in a tug-of-war between electrons trying to occupy low-energy states (valence band) and thermal energy trying to lift them to high-energy states (conduction band). The position of E_F tells us who is winning:

- E_F near conduction band $\Rightarrow$ electrons are winning (n-type)
- E_F near valence band $\Rightarrow$ holes are winning (p-type)
- E_F in the middle $\Rightarrow$ a draw (intrinsic)

8.5 A Simple Rule of Thumb

For most practical purposes in this course, remember:

The Fermi energy is the "fill line" for the electron sea.
States below it are generally full; states above it are generally empty.
Its distance from the band edges tells you how many carriers you have.

Example 3 (Quick Intuitive Check)**.** Consider three silicon samples at room temperature:

- Sample A: E_F is 0.05 eV below the conduction band edge.
- Sample B: E_F is exactly at midgap (0.56 eV from both bands).
- Sample C: E_F is 0.05 eV above the valence band edge.

Without calculations, we immediately identify:

- Sample A is **n-type** (fill line pushed up $\Rightarrow$ lots of electrons)
- Sample B is **intrinsic** (fill line centered $\Rightarrow$ balanced e^- and h^+)

- Sample C is **p-type** (fill line pulled down ⇒ lots of holes)

This intuitive understanding is sufficient for qualitative reasoning in device physics. The precise mathematics of the Fermi-Dirac distribution will be introduced when we need to calculate exact carrier concentrations. For now, E_F as the "maximum energy at $T = 0$ K" and the "reference fill line" will guide our analysis of junctions and devices.

9 Density of States: Counting Available Quantum States

So far, we have discussed energy bands qualitatively—we know that electrons in a semiconductor can only occupy certain energy ranges (the conduction and valence bands), and that there is a forbidden bandgap where no electron states exist. But to understand quantitatively how many electrons actually occupy these bands at a given temperature, we need to answer two questions:

1. **How many quantum states are available** at each energy? (This is the **density of states**.)
2. **What is the probability** that each available state is occupied by an electron? (This is the **Fermi-Dirac distribution**, which we will cover in the next section.)

In this section, we focus on the first question: counting the available states.

9.1 The Concept of Density of States

Definition 12 (Density of States)**.** The **density of states** $g(E)$ is a function that tells us the number of available quantum states per unit energy interval per unit volume of the crystal.

$$g(E)\,dE = \text{number of states with energy between } E \text{ and } E + dE, \text{ per unit volume}$$

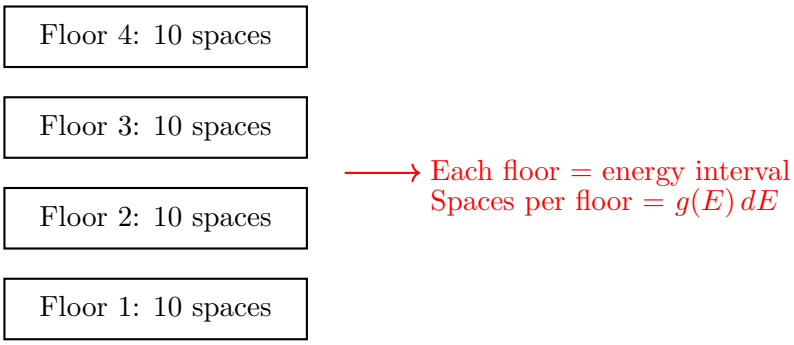
Units: $\text{cm}^{-3} \cdot \text{eV}^{-1}$ or $\text{m}^{-3} \cdot \text{J}^{-1}$.

Think of it this way: if you could look at the energy band and slice it into thin energy intervals of width dE, $g(E)\,dE$ would be the count of how many different quantum states (each capable of holding one electron, subject to the Pauli exclusion principle) exist in that slice.

9.2 Analogy: Parking Spaces in a Parking Garage

Imagine a multi-story parking garage:

- Each floor has a certain number of parking spaces.
- The "density of spaces" tells you how many parking spots are available on each floor.
- Higher floors might have the same or different numbers of spaces.
- The total number of cars that can be parked is the sum over all floors.



In our semiconductor:

- The "floors" are small energy intervals dE.
- The "parking spaces" are quantum states for electrons.
- $g(E)$ tells us how many spaces are available at each energy level.

9.3 Density of States for Free Electrons

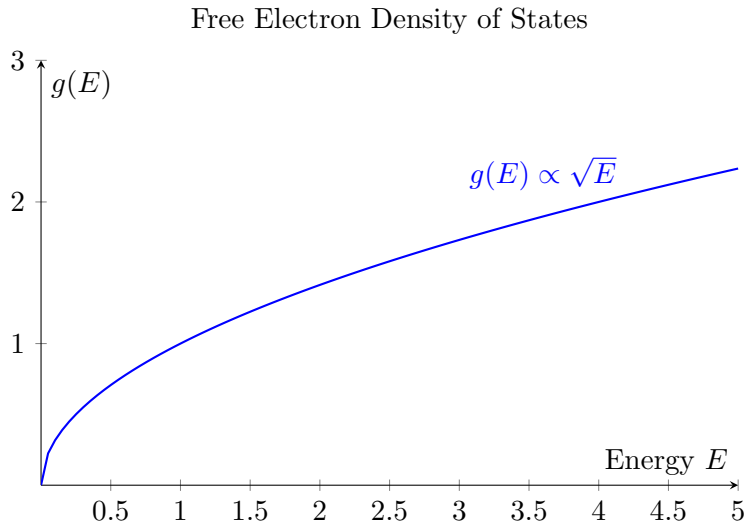
Let us begin with the simplest case: completely free electrons in a three-dimensional box. This is the situation we considered when deriving the parabolic dispersion relation $E = \hbar^2 k^2 / 2m_e$.

Density of States for Free Electrons (3D)

For free electrons in three dimensions, the density of states is:

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2} \right)^{3/2} \sqrt{E}$$

- This formula counts **both spin states** (spin up and spin down) for each k-state.
- It increases as $\sqrt{E}$—there are more available states at higher energies.
- The derivation comes from counting allowed k-states in k-space and converting to energy space using the dispersion relation.



9.4 Adapting the Formula for Semiconductors

In a semiconductor, electrons in the conduction band are **not completely free**—they experience the periodic potential of the crystal lattice. However, as we learned in Section 4, near the bottom of the conduction band, the dispersion relation can be approximated as parabolic:

$$E(k) = E_c + \frac{\hbar^2 k^2}{2m_e^*}$$

where E_c is the energy at the conduction band minimum, and m_e^* is the electron effective mass. This is mathematically identical to the free electron dispersion relation, except:

- The energy is shifted by E_c (the bottom of the conduction band).
- The electron mass m_e is replaced by the effective mass m_e^*.

Therefore, we can immediately write the density of states for the conduction band by making these substitutions in the free electron formula.

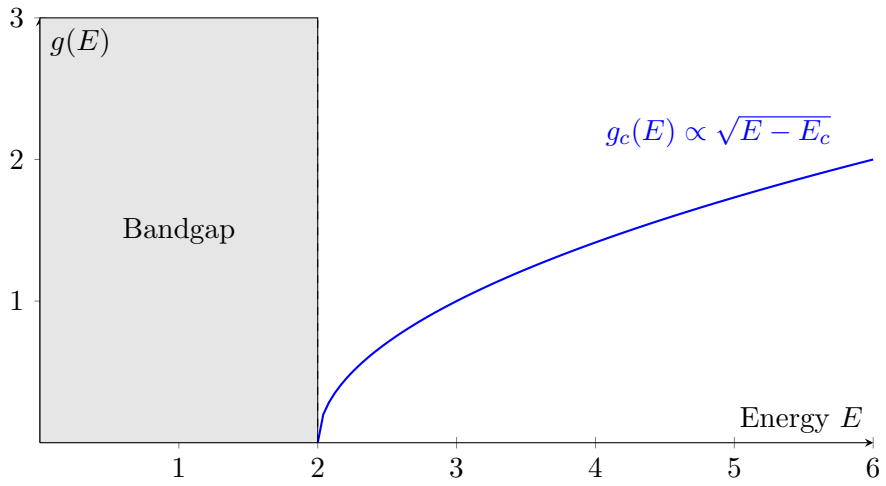
Conduction Band Density of States

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c} \quad \text{for } E \geq E_c$$

For energies below E_c (inside the bandgap), $g_c(E) = 0$—there are no conduction band states in the gap.
Key features:

- The density of states is **zero** below E_c (the bandgap).
- It increases as $\sqrt{E - E_c}$ above E_c.
- The effective mass m_e^* determines the curvature—a larger effective mass means a higher density of states (more states available at a given energy above the band edge).

Conduction Band Density of States



9.5 Valence Band Density of States and Holes

For the valence band, we have a similar situation but with a crucial difference. Near the top of the valence band, the dispersion relation is:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*}$$

where E_v is the energy at the valence band maximum, and m_h^* is the **hole effective mass** (positive). Notice the minus sign—energy decreases as k increases.

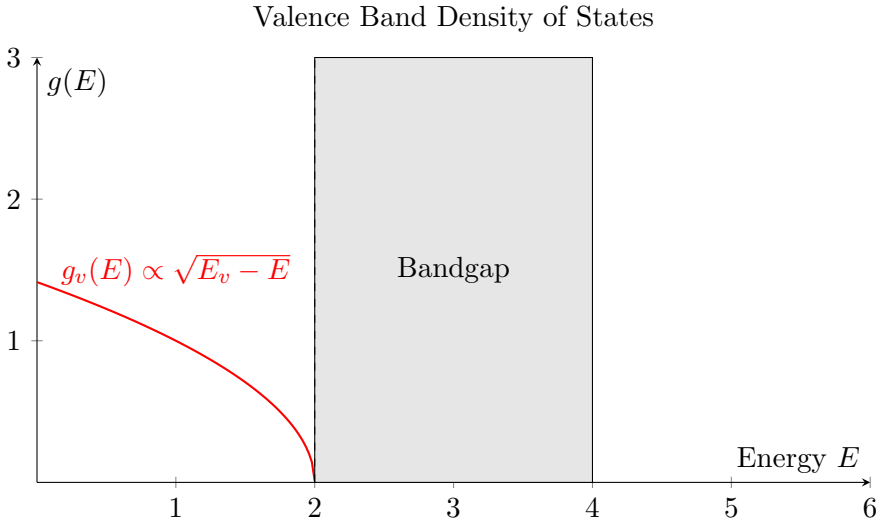
When dealing with holes, it is often more convenient to think in terms of **hole energy** increasing downward from the valence band edge. The density of states for holes in the valence band is:

Valence Band Density of States (for Holes)

$$g_v(E) = \frac{1}{2\pi^2}\left(\frac{2m_h^*}{\hbar^2}\right)^{3/2}\sqrt{E_v - E} \quad \text{for } E \leq E_v$$

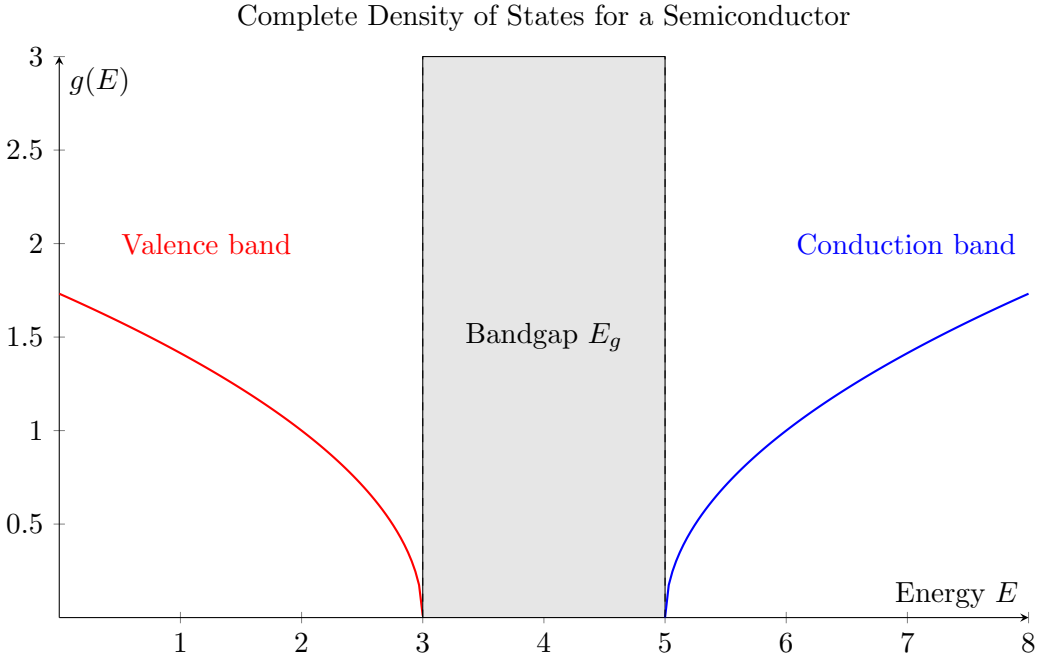
For energies above E_v, $g_v(E) = 0$.
Key features:

- The density of states increases as $\sqrt{E_v - E}$ as we go **down** into the valence band.
- The hole effective mass m_h^* plays the same role as m_e^* does for electrons.
- This formula counts states for **holes**—the absence of electrons.



9.6 Complete Picture: Conduction and Valence Bands Together

Putting both bands together gives us the complete density of states for a semiconductor:



9.7 Effective Density of States

For many practical calculations in semiconductor devices, we do not need the full energy-dependent density of states. Instead, we use a useful simplification: the **effective density of states** at the band edges.

Definition 13 (Effective Density of States)**.** The **effective density of states** N_c and N_v are defined as:

$$N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2} \quad \text{(conduction band)}$$

$$N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2} \quad \text{(valence band)}$$

These represent the **total number of available states** (per unit volume) within a few $k_B T$ of the band edges, treated as if they were all concentrated at the band edges.

Typical Values at Room Temperature (300 K)

Material	N_c (cm^{-3})	N_v (cm^{-3})
Silicon (Si)	2.8×10^{19}	1.04×10^{19}
Germanium (Ge)	1.04×10^{19}	6.0×10^{18}
Gallium Arsenide (GaAs)	4.7×10^{17}	7.0×10^{18}

Notice that N_c and N_v are typically on the order of 10^{19} cm^{-3}, much larger than typical doping concentrations ($10^{14} - 10^{18}$ cm^{-3}). This means that even heavily doped semiconductors have plenty of available states—the limitation on carrier concentration comes from the occupation probability, not the availability of states.

9.8 What's Next: The Occupation Probability

Now that we know **how many states are available** at each energy ($g(E)$), we need to know **how likely each state is to be occupied by an electron**. This is given by the **Fermi-Dirac distribution function** $f(E)$, which depends on:

- The energy of the state E
- The Fermi energy E_F
- The temperature T

The actual number of electrons in the conduction band is then:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE$$

Similarly, the number of holes in the valence band is:

$$p = \int_{-\infty}^{E_v} g_v(E)\, [1 - f(E)]\, dE$$

In the next section, we will introduce the Fermi-Dirac distribution and use it together with the density of states to calculate carrier concentrations in semiconductors.

Key Takeaways

- The density of states $g(E)$ tells us how many quantum states are available at each energy.
- For the conduction band: $g_c(E) \propto \sqrt{E - E_c}$ with effective mass m_e^*.
- For the valence band (holes): $g_v(E) \propto \sqrt{E_v - E}$ with effective mass m_h^*.
- The effective densities of states N_c and N_v are convenient totals that incorporate temperature dependence.
- The actual carrier concentration depends on both the available states and the probability that those states are occupied.

10 Fermi-Dirac Distribution: The Probability of Occupation

We now have a way to count available quantum states through the density of states $g(E)$. The next question is: given these available states, what is the probability that each one is actually occupied by an electron? This is determined by quantum statistics, which governs how particles distribute themselves among available energy states in thermal equilibrium.

10.1 Quantum Statistics: Fermions and Bosons

In classical physics, particles are distinguishable and can be treated as independent entities. In quantum mechanics, however, identical particles are fundamentally **indistinguishable**. This leads to two fundamentally different classes of particles.

Definition 14 (Fermions)**. Fermions** are particles with **half-integer spin** $(\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots)$. They obey the **Pauli exclusion principle**: no two fermions can occupy the same quantum state simultaneously.

Examples: electrons, protons, neutrons, quarks.

Definition 15 (Bosons)**. Bosons** are particles with **integer spin** $(0, 1, 2, \ldots)$. They do **not** obey the Pauli exclusion principle; multiple bosons can occupy the same quantum state.

Examples: photons, phonons, gluons, helium-4 atoms.

The Spin-Statistics Theorem

The connection between spin and statistics is a fundamental result of quantum field theory known as the **spin-statistics theorem** (Pauli, 1940):

Particles with half-integer spin are fermions and obey Fermi-Dirac statistics. Particles with integer spin are bosons and obey Bose-Einstein statistics.

This explains why matter is stable (electrons cannot all collapse into the lowest energy state) and why lasers work (photons are bosons and can occupy the same state).

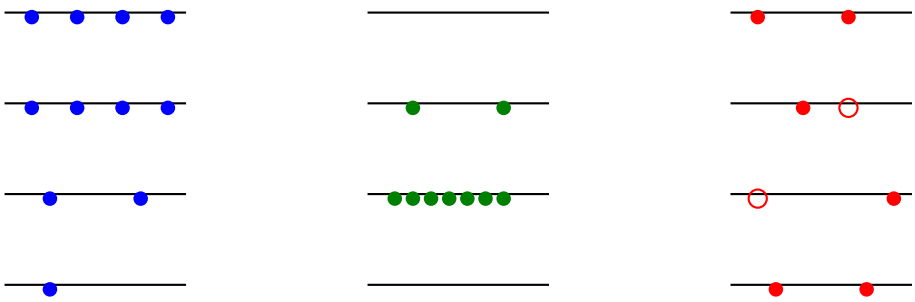
10.2 The Three Fundamental Distribution Functions

Depending on the type of particles and the conditions, three different statistical distributions describe how particles occupy energy states.

1. **Maxwell-Boltzmann distribution**: Classical statistics for distinguishable particles at low density and high temperature.

2. **Bose-Einstein distribution**: Quantum statistics for bosons (integer spin). Allows multiple particles in the same state.

3. **Fermi-Dirac distribution**: Quantum statistics for fermions (half-integer spin). Enforces the Pauli exclusion principle.



Maxwell-Boltzmann	**Bose-Einstein**	**Fermi-Dirac**
Classical, distinguishable	Bosons, integer spin	Fermions, half-integer spin
Multiple particles per state	Multiple particles per state	Pauli exclusion
	Bose-Einstein condensation	Maximum one particle per state

10.3 Mathematical Forms

Maxwell-Boltzmann Distribution

$$\boxed{f_{MB}(E) = e^{-(E-\mu)/k_BT}}$$

Valid for classical distinguishable particles or quantum particles in the low-density, high-temperature limit ($e^{-(E-\mu)/k_BT} \ll 1$). Can exceed 1 (not a true probability).

Bose-Einstein Distribution

$$\boxed{f_{BE}(E) = \frac{1}{e^{(E-\mu)/k_BT} - 1}}$$

Applies to bosons (integer spin). Can exceed 1. Diverges as $E \to \mu$ (leads to Bose-Einstein condensation). *Note: Not used in this course.*

Fermi-Dirac Distribution

$$\boxed{f_{FD}(E) = \frac{1}{e^{(E-E_F)/k_BT} + 1}}$$

Applies to fermions (half-integer spin). Always $0 \leq f_{FD} \leq 1$ (true probability). This is the distribution we use for electrons in semiconductors.

When to Use Which Distribution

Condition	Distribution	Particle Type	Example
Classical limit (low density, high T)	Maxwell-Boltzmann	Distinguishable	Ideal gas
Bosons, any energy	Bose-Einstein	Integer spin	Photons, phonons
Fermions, any energy	Fermi-Dirac	Half-integer spin	Electrons, protons
Semiconductors (non-degenerate)	MB approximation	Electrons	Most devices
Semiconductors (degenerate)	Fermi-Dirac (exact)	Electrons	Heavily doped

10.4 The Fermi-Dirac Distribution Function

Since electrons are fermions with spin 1/2, they obey Fermi-Dirac statistics. The probability that a quantum state with energy E is occupied by an electron in thermal equilibrium is:

The Fermi-Dirac Distribution

$$\boxed{f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}}$$

where:

- E is the energy of the state
- E_F is the **Fermi energy** (chemical potential)
- $k_B = 8.617 \times 10^{-5}$ eV/K (Boltzmann's constant)
- T is the absolute temperature in Kelvin

Definition 16 (Fermi Energy)**.** The **Fermi energy** E_F is the energy at which the occupation probability is exactly 1/2:

$$f(E = E_F) = \frac{1}{2}$$

States well below E_F are almost always occupied; states well above E_F are almost always empty.

10.5 Key Properties

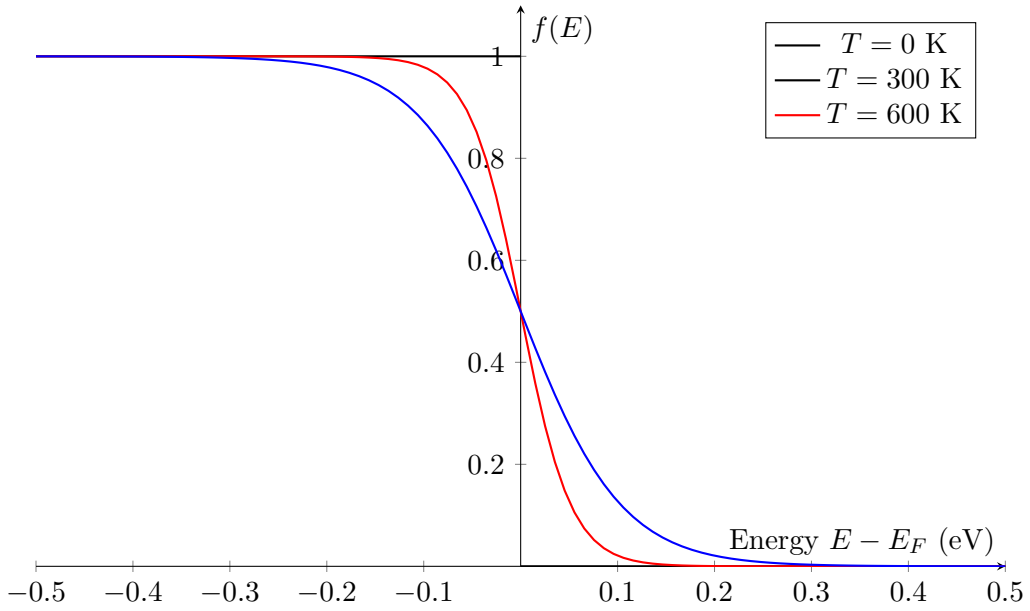
1. **Range**: $0 \leq f(E) \leq 1$ for all energies (true probability).
2. **At $T = 0$ K**: Becomes a step function:

$$f(E) = \begin{cases} 1 & \text{for } E < E_F \\ 0 & \text{for } E > E_F \end{cases}$$

3. **At $E = E_F$**: $f(E_F) = 1/2$ at any temperature.

4. **Symmetry**: $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$
5. **Temperature dependence**: The sharp step at E_F "softens" over $\sim$ a few $k_B T$.

Fermi-Dirac Distribution at Different Temperatures



10.6 Physical Interpretation

The Fermi-Dirac distribution captures two competing effects:

- **Pauli exclusion principle**: Forces electrons to occupy higher energies even at low temperatures, creating the "Fermi sea."
- **Thermal energy**: At finite temperatures, some electrons gain enough energy to be excited above E_F, leaving holes below.

Understanding the Distribution

- At $T = 0$ K: Electrons fill states up to E_F like water in a bucket.
- As T increases: The surface becomes "fuzzy" over $\sim \pm$ a few $k_B T$ around E_F.
- At room temperature: $k_B T \approx 0.026$ eV, so only electrons within ~ 0.1 eV of E_F are thermally affected.
- Electrons deep below E_F remain "frozen" by the Pauli principle.

10.7 The Maxwell-Boltzmann Approximation

For semiconductors, the important region is the **high-energy tail** where $E - E_F \gg k_B T$. In this regime:

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}} \approx e^{-(E-E_F)/k_B T} \quad \text{(Maxwell-Boltzmann approximation)}$$

This applies when the Fermi level is at least a few $k_B T$ away from the band edge (non-degenerate doping).

10.8 Summary: Computing Carrier Concentrations

We now have the two essential ingredients for calculating carrier concentrations:

Density of States $g(E)$	**Fermi-Dirac Distribution** $f(E)$
How many states exist at each energy	Probability each state is occupied
Depends on effective mass and band structure	Depends on E_F and temperature
$g_c(E) \propto \sqrt{E - E_c}$	$f(E) = 1/[1 + e^{(E-E_F)/k_BT}]$

Electron concentration in the conduction band:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE$$

Hole concentration in the valence band:

$$p = \int_{-\infty}^{E_v} g_v(E)\, [1 - f(E)]\, dE$$

In the next section, we will evaluate these integrals to obtain explicit formulas for n and p.

Key Takeaways

- Electrons are **fermions** and obey the Pauli exclusion principle.
- **Fermi-Dirac distribution** $f(E)$ gives occupation probability.
- At $T = 0$ K: step function (all states below E_F filled, above empty).
- As T increases: step "softens" over $\sim$ a few k_BT.
- For $E - E_F \gg k_BT$: reduces to exponential (Maxwell-Boltzmann approximation).
- This approximation is valid for most semiconductor conditions (non-degenerate).

11 Equilibrium Carrier Concentrations

We now have the two essential ingredients for calculating how many electrons and holes exist in a semiconductor at thermal equilibrium: the density of states $g(E)$, which tells us how many quantum states are available at each energy, and the Fermi-Dirac distribution $f(E)$, which tells us the probability that each state is occupied. In this section, we combine these to derive the fundamental expressions for electron and hole concentrations, the important $np = n_i^2$ relation, and the position of the Fermi level in intrinsic and doped semiconductors.

Notation Reference

n	Electron concentration (cm^{-3})
p	Hole concentration (cm^{-3})
N_D	Donor concentration (cm^{-3})
N_A	Acceptor concentration (cm^{-3})
N_D^+	Ionized donor concentration (cm^{-3})
N_A^-	Ionized acceptor concentration (cm^{-3})
n_i	Intrinsic carrier concentration (cm^{-3})
N_c	Effective density of states in conduction band (cm^{-3})
N_v	Effective density of states in valence band (cm^{-3})
E_c	Conduction band edge (eV)
E_v	Valence band edge (eV)
E_F	Fermi level (eV)
E_g	Bandgap energy (eV)

$$
\begin{gathered}
g(E) \times f(E) \\
\downarrow \\
n = \int g_c(E) f(E) dE \\
\downarrow \\
n = N_c e^{-(E_c - E_F)/k_B T} \\
\downarrow \\
np = n_i^2 \\
\downarrow \\
\text{Charge Neutrality: } p + N_D = n + N_A \\
\downarrow \\
\text{Exact } n, p \text{ from quadratic equation}
\end{gathered}
$$

11.1 Electron Concentration in the Conduction Band

The electron concentration n (number of electrons per unit volume) in the conduction band is obtained by integrating the product of the density of states in the conduction band and the probability that those states are occupied:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE$$

where:

- $g_c(E)$ is the density of states in the conduction band
- $f(E)$ is the Fermi-Dirac distribution function
- E_c is the conduction band edge energy

The Density of States for the Conduction Band

From our earlier derivation, the density of states in the conduction band for a parabolic band structure is:

$$g_c(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c} \quad \text{for } E \geq E_c$$

This expression tells us that the density of states increases as the square root of the energy above the band edge.

The Fermi-Dirac Distribution in the Non-Degenerate Limit

The Fermi-Dirac distribution is:

$$f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

For most semiconductors under typical operating conditions, the Fermi level E_F lies within the bandgap, at least several $k_B T$ away from both band edges. In the conduction band, $E \geq E_c$, so $E - E_F \gg k_B T$. In this case, the exponential term in the denominator dominates, and we can make the **Maxwell-Boltzmann approximation**:

$$f(E) \approx \exp\left(-\frac{E - E_F}{k_B T}\right) \quad \text{for } E - E_F \gg k_B T$$

This approximation is excellent when $E_c - E_F > 3k_B T$, which is true for non-degenerate semiconductors.

Evaluating the Integral

Substituting both expressions into our integral:

$$n = \int_{E_c}^{\infty} \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\sqrt{E - E_c}\,\exp\left(-\frac{E - E_F}{k_B T}\right) dE$$

We can simplify by making a change of variables. Let:

$$x = \frac{E - E_c}{k_B T} \quad \Rightarrow \quad E = E_c + x k_B T$$

Then:

$$dE = k_B T\, dx$$

$$\sqrt{E - E_c} = \sqrt{x k_B T}$$

$$\exp\left(-\frac{E - E_F}{k_B T}\right) = \exp\left(-\frac{E_c + x k_B T - E_F}{k_B T}\right) = \exp\left(-\frac{E_c - E_F}{k_B T}\right)\exp(-x)$$

Substituting everything:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}\int_0^{\infty}\sqrt{x k_B T}\,\exp\left(-\frac{E_c - E_F}{k_B T}\right)\exp(-x)\,k_B T\, dx$$

Factor out the constants from the integral:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(k_B T)^{3/2}\exp\left(-\frac{E_c - E_F}{k_B T}\right)\int_0^{\infty} x^{1/2} e^{-x}\, dx$$

The integral $\int_0^{\infty} x^{1/2} e^{-x}\, dx$ is a standard form known as the Gamma function:

$$\int_0^\infty x^{1/2} e^{-x}\, dx = \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

Therefore:

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2} (k_B T)^{3/2} \frac{\sqrt{\pi}}{2} \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

Introducing the Effective Density of States

We can simplify this expression by defining the **effective density of states in the conduction band**, N_c:

$$\boxed{N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}}$$

Let's verify that this matches our expression. Note that $\hbar = h/2\pi$, so:

$$\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2} = \left(\frac{2m_e^* \cdot 4\pi^2}{h^2}\right)^{3/2} = \left(\frac{8\pi^2 m_e^*}{h^2}\right)^{3/2}$$

Also, $(k_B T)^{3/2} \cdot \frac{\sqrt{\pi}}{2} \cdot \frac{1}{2\pi^2} = \frac{\sqrt{\pi}}{4\pi^2}(k_B T)^{3/2}$

Combining these carefully yields exactly $2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}$.

Thus, the electron concentration simplifies beautifully to:

$$\boxed{n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)}$$

This is one of the most important equations in semiconductor physics. It tells us that the electron concentration in the conduction band depends on:

- The effective density of states N_c (which depends on effective mass and temperature)
- The distance between the conduction band edge E_c and the Fermi level E_F
- Temperature through both N_c and the exponential factor

11.2 Hole Concentration in the Valence Band

For holes, we need the probability that a state in the valence band is *empty*. Since $f(E)$ is the probability a state is occupied by an electron, the probability it is empty (i.e., occupied by a hole) is $1 - f(E)$.

The hole concentration p is:

$$p = \int_{-\infty}^{E_v} g_v(E)\,[1 - f(E)]\, dE$$

where $g_v(E)$ is the density of states in the valence band.

Density of States for the Valence Band

For the valence band, with hole effective mass m_h^*, the density of states is:

$$g_v(E) = \frac{1}{2\pi^2}\left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \sqrt{E_v - E} \quad \text{for } E \le E_v$$

The Hole Probability Function

For $E \leq E_v$, and with E_F well above the valence band edge in non-degenerate materials, $E_F - E \gg k_B T$. We can simplify $1 - f(E)$:

$$1 - f(E) = 1 - \frac{1}{1 + \exp\left(\frac{E-E_F}{k_B T}\right)} = \frac{\exp\left(\frac{E-E_F}{k_B T}\right)}{1 + \exp\left(\frac{E-E_F}{k_B T}\right)}$$

For $E_F - E \gg k_B T$, the denominator is approximately 1, so:

$$1 - f(E) \approx \exp\left(\frac{E - E_F}{k_B T}\right) = \exp\left(-\frac{E_F - E}{k_B T}\right)$$

Evaluating the Hole Integral

Substituting:

$$p = \int_{-\infty}^{E_v} \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \sqrt{E_v - E}\, \exp\left(-\frac{E_F - E}{k_B T}\right) dE$$

Make a change of variables. Let:

$$x = \frac{E_v - E}{k_B T} \quad \Rightarrow \quad E = E_v - x k_B T$$

Then:

$$dE = -k_B T\, dx$$

Note that as $E \to -\infty$, $x \to \infty$, and as $E \to E_v$, $x \to 0$. The negative sign flips the integration limits:

$$p = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \int_{\infty}^{0} \sqrt{x k_B T}\, \exp\left(-\frac{E_F - (E_v - x k_B T)}{k_B T}\right) (-k_B T)\, dx$$

$$p = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} \int_{0}^{\infty} \sqrt{x k_B T}\, \exp\left(-\frac{E_F - E_v}{k_B T}\right) \exp(-x)\, k_B T\, dx$$

This is exactly the same form as the electron integral, but with m_e^* replaced by m_h^* and with the exponential factor $\exp\left(-\frac{E_F - E_v}{k_B T}\right)$.

Evaluating the same Gamma function integral gives:

$$p = \frac{1}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2}\right)^{3/2} (k_B T)^{3/2} \frac{\sqrt{\pi}}{2} \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Effective Density of States for the Valence Band

Define the **effective density of states in the valence band**, N_v:

$$\boxed{N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2}}$$

Then the hole concentration becomes:

$$\boxed{p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)}$$

This is the companion equation to our electron concentration expression. It shows that the hole concentration depends on:

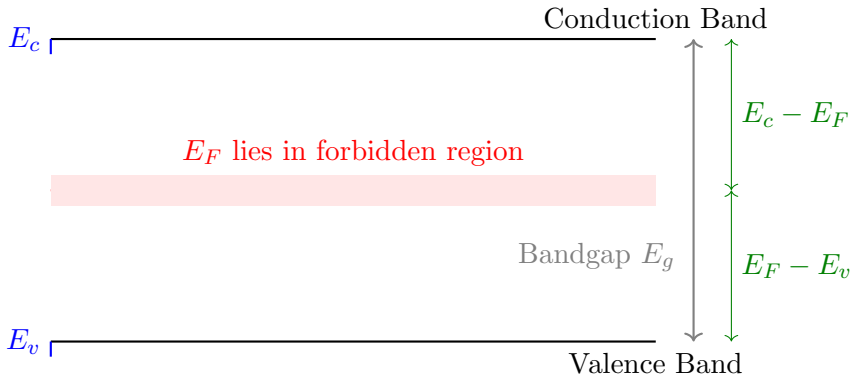
- The effective density of states N_v (determined by hole effective mass and temperature)
- The distance between the Fermi level E_F and the valence band edge E_v

11.3 Intuitive Picture: Where Do the Carriers Reside?

We now have the mathematical expressions for electron and hole concentrations. But what do these equations tell us physically? Where are the electrons and holes located in energy space? Understanding this intuitively is crucial for developing a feel for semiconductor behavior.

The Key Physical Assumption

The entire argument rests on one crucial assumption: **the Fermi level E_F lies inside the forbidden bandgap**. This is true for non-degenerate semiconductors — the materials we work with in most practical devices. Because E_F is in the bandgap, it is several k_BT away from both band edges.



The Product $g_c(E) \cdot f(E)$: Where Are the Electrons?

The number of electrons per unit energy interval at energy E in the conduction band is given by the product:

$$\text{Electron distribution} = g_c(E) \cdot f(E)$$

This product tells us how many electrons (per unit volume per unit energy) we expect to find at each energy level. Let's analyze its behavior:

1. **At $E = E_c$ (the band edge)**:

$$g_c(E_c) = 0 \quad \Rightarrow \quad \text{Product} = 0$$

   There are no states exactly at the band edge, so the electron density starts at zero.

2. **Just above E_c**:
   - $g_c(E)$ increases as $\sqrt{E - E_c}$ (opens up slowly)
   - $f(E) \approx \exp\left(-\frac{E-E_c}{k_BT}\right) \cdot \exp\left(-\frac{E_c-E_F}{k_BT}\right)$ (exponential decay)

   Initially, the square root increase dominates over the exponential decay, so the product rises from zero.

3. **Farther from E_c**:
   - Eventually, the exponential decay $\exp\left(-\frac{E-E_c}{k_BT}\right)$ becomes the dominant factor

- The product reaches a maximum and then decays toward zero

Similarly, the number of holes per unit energy interval is:

$$\text{Hole distribution} = g_v(E) \cdot [1 - f(E)]$$

For holes, the product starts at zero at $E = E_v$ (since $g_v(E_v) = 0$), rises to a maximum just below E_v, and then decays as we go deeper into the valence band.

Why This Matters: Carriers Concentrate Near Band Edges

The key insight is that **most electrons are concentrated in a narrow energy range just above** E_c, and **most holes are concentrated in a narrow energy range just below** E_v. This happens because:

- The exponential Boltzmann factor $\exp\left(-\frac{E-E_c}{k_BT}\right)$ decays very rapidly as we move away from the band edge
- At room temperature, $k_BT \approx 0.026$ eV, so the distribution decays by a factor of $e \approx 2.718$ for every 0.026 eV above E_c
- Even though $g_c(E)$ increases with energy, the exponential decay is much faster

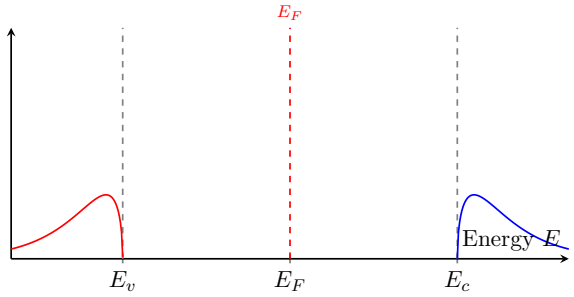
Physical Interpretation

The Fermi level acts like a "water level." Because it lies deep in the bandgap:

- Electrons in the conduction band are in the high-energy tail of the Fermi-Dirac distribution
- Their probability of occupation decreases exponentially with energy above E_c
- Consequently, almost all electrons cluster within a few k_BT of the conduction band edge
- Similarly, holes cluster within a few k_BT of the valence band edge

Visualizing the Distributions

The following figure shows how the electron and hole densities vary with energy. Note how both distributions start at zero at the band edges, rise to a peak just above/below the edges, and then decay rapidly.



The Mathematical Reason: Boltzmann Tail Dominance

Let's examine why the electron distribution peaks so close to E_c. Consider the product:

$$g_c(E) \cdot f(E) = A\sqrt{E - E_c} \cdot \exp\left(-\frac{E - E_F}{k_BT}\right)$$

where A is a constant. For $E \geq E_c$, we can write:

$$\text{Electron distribution} = A\sqrt{E - E_c} \cdot \exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot \exp\left(-\frac{E - E_c}{k_B T}\right)$$

The factor $\exp\left(-\frac{E_c - E_F}{k_B T}\right)$ is a constant that determines the overall magnitude. The energy-dependent part is:

$$h(E) = \sqrt{E - E_c} \cdot \exp\left(-\frac{E - E_c}{k_B T}\right)$$

We can find where this function peaks by taking its derivative and setting it to zero. Let $x = E - E_c \geq 0$. Then:

$$h(x) = \sqrt{x} \cdot e^{-x/k_B T}$$

Take the logarithm:

$$\ln h(x) = \frac{1}{2} \ln x - \frac{x}{k_B T}$$

Differentiate:

$$\frac{d}{dx} \ln h(x) = \frac{1}{2x} - \frac{1}{k_B T} = 0 \quad \Rightarrow \quad x = \frac{k_B T}{2}$$

Thus, the peak of the electron distribution occurs at:

$$\boxed{E_{\text{peak}} = E_c + \frac{k_B T}{2}}$$

That's only half of $k_B T$ above the band edge! At room temperature, $k_B T \approx 0.026$ eV, so the peak is a mere 0.013 eV above E_c. This is an incredibly narrow energy range.

Similarly, for holes, the peak occurs at:

$$E_{\text{peak}} = E_v - \frac{k_B T}{2}$$

just half of $k_B T$ below the valence band edge.

Key Insight: The "Narrow Band" Approximation

Because the peak of the carrier distribution lies only $\pm k_B T/2$ from the band edges, and because the exponential tail decays extremely rapidly, we can often assume that **all carriers reside exactly at the band edges** for purposes of rough estimates and device modeling.

This justifies using the expressions $n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$ and $p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$ — they effectively treat all states as if they were concentrated at the band edges with an effective density N_c or N_v.

For silicon at room temperature:

- $k_B T/2 \approx 0.013$ eV
- The thermal energy spread is less than 1% of the bandgap
- This is why the Boltzmann approximation works so well

11.4 The np Product and the Intrinsic Carrier Concentration

Multiplying n and p

One of the most powerful results in semiconductor physics emerges when we multiply the electron and hole concentrations:

$$np = \left[N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)\right]\left[N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)\right]$$

Notice that the exponential factors combine:

$$\exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot \exp\left(-\frac{E_F - E_v}{k_B T}\right) = \exp\left(-\frac{E_c - E_F + E_F - E_v}{k_B T}\right) = \exp\left(-\frac{E_c - E_v}{k_B T}\right)$$

But $E_c - E_v$ is exactly the bandgap energy E_g! Therefore:

$$\boxed{np = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)}$$

The Intrinsic Carrier Concentration

In an **intrinsic** (undoped) semiconductor, every electron in the conduction band comes from a thermally excited electron that left a hole behind in the valence band. Therefore:

$$n = p = n_i$$

where n_i is called the **intrinsic carrier concentration**.
Substituting into our product relation:

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)$$

Thus:

$$\boxed{n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)}$$

Physical Meaning of n_i

The intrinsic carrier concentration n_i is a fundamental material parameter that:

- Depends exponentially on bandgap E_g (larger bandgap $\Rightarrow$ smaller n_i)
- Increases rapidly with temperature
- Depends on effective masses through N_c and N_v
- At a given temperature, is constant for a specific material

For silicon at room temperature (300 K):

$$n_i \approx 1.5 \times 10^{10}\,\text{cm}^{-3}$$

For comparison:

- Germanium (Ge): $n_i \approx 2.4 \times 10^{13}\,\text{cm}^{-3}$
- Gallium Arsenide (GaAs): $n_i \approx 1.8 \times 10^{6}\,\text{cm}^{-3}$

The Mass Action Law

Returning to our product expression, we have the important **mass action law**:

$$\boxed{np = n_i^2}$$

This equation holds for any semiconductor in thermal equilibrium, whether intrinsic or doped. Its physical meaning is profound:

- In equilibrium, the product of electron and hole concentrations is constant at a given temperature
- If we increase n by doping, p must decrease to maintain the constant product
- Conversely, if we increase p, n must decrease

11.5 Position of the Fermi Level

We can now derive explicit expressions for the Fermi level position in intrinsic and doped semiconductors.

Fermi Level in Intrinsic Semiconductors

For an intrinsic semiconductor, $n = p = n_i$. Using our expressions:

$$N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Take natural logarithms of both sides:

$$\ln N_c - \frac{E_c - E_F}{k_B T} = \ln N_v - \frac{E_F - E_v}{k_B T}$$

Rearrange to collect terms with E_F:

$$-\frac{E_c}{k_B T} + \frac{E_F}{k_B T} + \frac{E_F}{k_B T} - \frac{E_v}{k_B T} = \ln N_v - \ln N_c$$

$$\frac{2E_F}{k_BT} - \frac{E_c + E_v}{k_BT} = \ln\left(\frac{N_v}{N_c}\right)$$

Solve for E_F:

$$\frac{2E_F}{k_BT} = \frac{E_c + E_v}{k_BT} + \ln\left(\frac{N_v}{N_c}\right)$$

$$E_F = \frac{E_c + E_v}{2} + \frac{k_BT}{2}\ln\left(\frac{N_v}{N_c}\right)$$

Recall that N_c and N_v depend on effective masses:

$$\frac{N_v}{N_c} = \left(\frac{m_h^*}{m_e^*}\right)^{3/2}$$

Therefore:

$$\boxed{E_F = \frac{E_c + E_v}{2} + \frac{3}{4}k_BT\ln\left(\frac{m_h^*}{m_e^*}\right)}$$

The first term is exactly the midpoint of the bandgap. The second term is a small correction that depends on the ratio of effective masses.

- If $m_e^* = m_h^*$, then E_F is exactly at midgap
- If $m_h^* > m_e^*$ (common in many semiconductors), the logarithm is positive, so E_F is slightly above midgap
- The correction is typically small — at room temperature, $k_BT \approx 0.026$ eV, so even a factor of 10 in mass ratio gives only about 0.045 eV shift

For most practical purposes, we can often say that the intrinsic Fermi level E_i is approximately at midgap.

Fermi Level in n-Type Semiconductors

For n-type material with donor concentration N_D, we have $n \approx N_D$. Using our expression for n:

$$N_c \exp\left(-\frac{E_c - E_F}{k_BT}\right) = N_D$$

Solve for E_F:

$$\exp\left(-\frac{E_c - E_F}{k_BT}\right) = \frac{N_D}{N_c}$$

$$-\frac{E_c - E_F}{k_BT} = \ln\left(\frac{N_D}{N_c}\right)$$

$$E_c - E_F = -k_BT\ln\left(\frac{N_D}{N_c}\right) = k_BT\ln\left(\frac{N_c}{N_D}\right)$$

Therefore:

$$\boxed{E_F = E_c - k_BT\ln\left(\frac{N_c}{N_D}\right)}$$

Since $N_D < N_c$ for non-degenerate doping, the logarithm is positive, so E_F lies below E_c. The heavier the doping, the closer E_F moves to the conduction band.

Example 4 (Fermi Level in n-Type Silicon)**.** For silicon at 300 K, $N_c \approx 2.8 \times 10^{19}\,\mathrm{cm}^{-3}$. For $N_D = 10^{16}\,\mathrm{cm}^{-3}$:

$$E_c - E_F = k_B T \ln\left(\frac{N_c}{N_D}\right) = 0.026\,\mathrm{eV} \times \ln\left(\frac{2.8 \times 10^{19}}{10^{16}}\right)$$

$$\frac{N_c}{N_D} = 2.8 \times 10^3, \quad \ln\big(2.8 \times 10^3\big) = \ln 2.8 + \ln 10^3 = 1.03 + 6.91 = 7.94$$

$$E_c - E_F = 0.026 \times 7.94 = 0.206\,\mathrm{eV}$$

So E_F is 0.206 eV below the conduction band edge. The intrinsic Fermi level E_i is approximately at midgap, which for silicon ($E_g = 1.12$ eV) is 0.56 eV below E_c. Thus doping has moved E_F significantly closer to the conduction band.

Fermi Level in p-Type Semiconductors

For p-type material with acceptor concentration N_A, we have $p \approx N_A$. Using our expression for p:

$$N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) = N_A$$

Solve for E_F:

$$\exp\left(-\frac{E_F - E_v}{k_B T}\right) = \frac{N_A}{N_v}$$

$$-\frac{E_F - E_v}{k_B T} = \ln\left(\frac{N_A}{N_v}\right)$$

$$E_F - E_v = -k_B T \ln\left(\frac{N_A}{N_v}\right) = k_B T \ln\left(\frac{N_v}{N_A}\right)$$

Therefore:

$$\boxed{E_F = E_v + k_B T \ln\left(\frac{N_v}{N_A}\right)}$$

Here E_F lies above E_v. Heavier doping moves E_F closer to the valence band.

Example 5 (Fermi Level in p-Type Silicon)**.** For silicon at 300 K, $N_v \approx 1.04 \times 10^{19}\,\mathrm{cm}^{-3}$. For $N_A = 10^{16}\,\mathrm{cm}^{-3}$:

$$E_F - E_v = k_B T \ln\left(\frac{N_v}{N_A}\right) = 0.026\,\mathrm{eV} \times \ln\left(\frac{1.04 \times 10^{19}}{10^{16}}\right)$$

$$\frac{N_v}{N_A} = 1.04 \times 10^3, \quad \ln\big(1.04 \times 10^3\big) = \ln 1.04 + \ln 10^3 = 0.039 + 6.91 = 6.949$$

$$E_F - E_v = 0.026 \times 6.949 = 0.181\,\mathrm{eV}$$

So E_F is 0.181 eV above the valence band edge. The intrinsic Fermi level is 0.56 eV below E_c, which is equivalent to 0.56 eV above E_v (since $E_g = 1.12$ eV). Thus doping has moved E_F significantly closer to the valence band.

11.6 Charge Neutrality and Exact Carrier Concentrations

In the previous sections, we derived the equilibrium carrier concentrations using the Boltzmann approximation:

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right), \qquad p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

We also introduced the mass action law $np = n_i^2$. However, these expressions alone do not determine n and p uniquely — they depend on the unknown Fermi level E_F. To find E_F, and hence the actual carrier concentrations, we must invoke a fundamental principle: **charge neutrality**.

The Charge Neutrality Condition

In a semiconductor at equilibrium, the total charge density must be zero. Any net charge would create an electric field that would cause carrier motion until neutrality is restored. This condition is expressed as:

$$\boxed{\rho = q(p - n + N_D^+ - N_A^-) = 0}$$

where:

- q is the elementary charge (1.602×10^{-19} C)
- n is the concentration of free electrons in the conduction band
- p is the concentration of free holes in the valence band
- N_D^+ is the concentration of **ionized donors** (positively charged)
- N_A^- is the concentration of **ionized acceptors** (negatively charged)

The Complete Ionization Approximation

At room temperature, for typical doping concentrations ($N_D, N_A \lesssim 10^{18}$ cm^{-3}), the donor and acceptor energy levels are very close to the band edges. Since the Fermi level in non-degenerate semiconductors lies within the bandgap, we have E_F well below E_D in n-type material and well above E_A in p-type material. Consequently, the exponentials in the denominators are extremely small, and we can make the **complete ionization approximation**:

$$N_D^+ \approx N_D, \qquad N_A^- \approx N_A$$

This approximation holds for shallow dopants at room temperature. It fails at very low temperatures (where thermal energy is insufficient to ionize all dopants) or in degenerate semiconductors (where the Fermi level approaches the band edges). For most practical room-temperature devices, complete ionization is an excellent assumption.

The Simplified Charge Neutrality Equation

Using the complete ionization approximation, the charge neutrality condition becomes:

$$\boxed{p - n + N_D - N_A = 0}$$

or equivalently:

$$\boxed{p + N_D = n + N_A}$$

This equation states that the total positive charge (holes plus ionized donors) equals the total negative charge (electrons plus ionized acceptors).

Exact Carrier Concentrations for Doped Semiconductors

We now combine charge neutrality with the mass action law to find n and p in terms of the doping concentrations N_D and N_A.

Step 1: Express p in terms of n
From the mass action law $np = n_i^2$, we have:

$$p = \frac{n_i^2}{n}$$

Step 2: Substitute into charge neutrality
The charge neutrality equation $p + N_D = n + N_A$ becomes:

$$\frac{n_i^2}{n} + N_D = n + N_A$$

Step 3: Rearrange into a quadratic equation
Multiply both sides by n:

$$n_i^2 + N_D n = n^2 + N_A n$$

Bring all terms to one side:

$$n^2 + N_A n - N_D n - n_i^2 = 0$$

$$\boxed{n^2 + (N_A - N_D)n - n_i^2 = 0}$$

Step 4: Solve for n
The quadratic formula gives:

$$n = \frac{-(N_A - N_D) \pm \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}$$

We take the positive root (since $n > 0$):

$$\boxed{n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}}$$

Step 5: Solve for p
Once n is known, we obtain p from the mass action law:

$$\boxed{p = \frac{n_i^2}{n}}$$

Alternatively, we could derive a similar quadratic for p by symmetry:

$$\boxed{p = \frac{-(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2}}$$

Understanding the Square Root Expression

The expressions above look complicated, but they have a simple interpretation. Let's define the **net doping concentration**:

$$N_{\text{net}} = |N_D - N_A|$$

There are two important regimes:

Regime 1: Non-Degenerate Doping ($N_{\text{net}} \gg n_i$)

When the net doping is much larger than the intrinsic carrier concentration, the term $4n_i^2$ under the square root is negligible compared to $(N_A - N_D)^2$. In this case:

For n-type ($N_D > N_A$):

$$n \approx \frac{-(N_A - N_D) + |N_A - N_D|}{2} = \frac{(N_D - N_A) + (N_D - N_A)}{2} = N_D - N_A$$

Thus:

$$n \approx N_D - N_A, \qquad p \approx \frac{n_i^2}{N_D - N_A}$$

For p-type ($N_A > N_D$):

$$p \approx N_A - N_D, \qquad n \approx \frac{n_i^2}{N_A - N_D}$$

Regime 2: Intrinsic or Lightly Doped ($N_{\text{net}} \ll n_i$)

When the doping is very light or the temperature is high such that n_i dominates, we can expand the square root:

$$\sqrt{(N_A - N_D)^2 + 4n_i^2} \approx 2n_i\sqrt{1 + \frac{(N_A - N_D)^2}{4n_i^2}} \approx 2n_i\left(1 + \frac{(N_A - N_D)^2}{8n_i^2}\right)$$

Then:

$$n \approx \frac{-(N_A - N_D) + 2n_i + \frac{(N_A - N_D)^2}{4n_i}}{2} \approx n_i + \frac{N_D - N_A}{2}$$

For $N_D = N_A = 0$ (intrinsic), we recover $n = n_i$.

Numerical Examples

Example 1: Standard n-Type Silicon (Complete Ionization Valid)

Consider silicon at $T = 300$ K where $n_i = 1.5 \times 10^{10}$ cm^{-3}. Suppose we dope with phosphorus at $N_D = 10^{16}$ cm^{-3}, with no acceptors ($N_A = 0$).

Step 1: Check if complete ionization is valid. Since N_D is moderate and room temperature, we can assume all donors are ionized.

Step 2: Use the exact formula.

$$n = \frac{-(0 - 10^{16}) + \sqrt{(0 - 10^{16})^2 + 4(1.5 \times 10^{10})^2}}{2}$$

First compute $(N_A - N_D)^2 = (-10^{16})^2 = 10^{32}$.

Compute $4n_i^2 = 4 \times (2.25 \times 10^{20}) = 9 \times 10^{20}$.

Since $10^{32} \gg 9 \times 10^{20}$, we have:

$$\sqrt{10^{32} + 9 \times 10^{20}} \approx 10^{16}\sqrt{1 + 9 \times 10^{-12}} \approx 10^{16}(1 + 4.5 \times 10^{-12})$$

Thus:

$$n \approx \frac{10^{16} + 10^{16}(1 + 4.5 \times 10^{-12})}{2} = \frac{10^{16}(2 + 4.5 \times 10^{-12})}{2} \approx 10^{16} + 2.25 \times 10^{4}\,\text{cm}^{-3}$$

The correction term 2.25×10^4 cm^{-3} is negligible compared to 10^{16} cm^{-3}.

Step 3: Approximation yields:

$$n \approx N_D = 10^{16}\,\text{cm}^{-3}$$

$$p = \frac{n_i^2}{n} = \frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^{4}\,\text{cm}^{-3}$$

Example 2: Compensated n-Type Semiconductor

Now consider a compensated semiconductor: silicon with $N_D = 10^{16}$ cm^{-3} and $N_A = 5 \times 10^{15}$ cm^{-3}.

Step 1: Compute net doping.

$$N_D - N_A = 10^{16} - 5 \times 10^{15} = 5 \times 10^{15}\,\text{cm}^{-3}$$

Step 2: Use the exact formula.

$$n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2} = \frac{5 \times 10^{15} + \sqrt{(5 \times 10^{15})^2 + 4n_i^2}}{2}$$

Compute $(N_A - N_D)^2 = (-5 \times 10^{15})^2 = 2.5 \times 10^{31}$.

$4n_i^2 = 9 \times 10^{20}$ is still negligible, so:

$$n \approx \frac{5 \times 10^{15} + 5 \times 10^{15}}{2} = 5 \times 10^{15}\,\text{cm}^{-3}$$

$$p = \frac{n_i^2}{n} = \frac{2.25 \times 10^{20}}{5 \times 10^{15}} = 4.5 \times 10^{4}\,\text{cm}^{-3}$$

Step 3: Verification of approximation. The net doping is still much larger than n_i, so the approximation $n \approx N_D - N_A$ is excellent.

Example 3: Lightly Doped n-Type (Approximation Begins to Fail)

Consider silicon at $T = 400$ K, where n_i increases significantly. For silicon at 400 K, $n_i \approx 7.7 \times 10^{12}$ cm^{-3} (rough estimate using $n_i^2 \propto T^3 e^{-E_g/k_B T}$). Take $N_D = 10^{12}$ cm^{-3} and $N_A = 0$.

Step 1: Use the exact formula.

$$n = \frac{10^{12} + \sqrt{(10^{12})^2 + 4(7.7 \times 10^{12})^2}}{2}$$

First compute $(N_A - N_D)^2 = 10^{24}$.

$4n_i^2 = 4 \times (5.93 \times 10^{25}) = 2.37 \times 10^{26}$.

Now the square root term:

$$\sqrt{10^{24} + 2.37 \times 10^{26}} = \sqrt{2.38 \times 10^{26}} = 1.543 \times 10^{13}$$

Thus:

$$n = \frac{10^{12} + 1.543 \times 10^{13}}{2} = \frac{1.643 \times 10^{13}}{2} = 8.215 \times 10^{12}\,\text{cm}^{-3}$$

Step 2: Compare with approximation $n \approx N_D$:

$$N_D = 10^{12}\,\text{cm}^{-3} \quad \text{(underestimates exact value by factor of } \sim 8.2\text{)}$$

Step 3: Compare with intrinsic value:

$$n_i = 7.7 \times 10^{12}\,\text{cm}^{-3}$$

Note that n lies between N_D and n_i, as expected. The exact value 8.22×10^{12} is slightly above n_i because the doping adds a small excess of electrons.

11.7 Practical Summary: When to Use Which Formula

Decision Tree for Carrier Concentration Calculations

1. **Identify the material and temperature** to find n_i and N_c, N_v (or use tabulated values).
2. **Check if complete ionization is valid**: For shallow dopants at room temperature with $N_D, N_A \lesssim 10^{18}$ cm^{-3}, complete ionization is an excellent approximation.
3. **Compute net doping**: $N_{\text{net}} = |N_D - N_A|$.
4. **Compare N_{net} with n_i**:
   - If $N_{\text{net}} \gg n_i$: Use the approximation

     $$n \approx N_D - N_A \quad \text{(n-type)}, \qquad p \approx \frac{n_i^2}{N_D - N_A}$$

     or

     $$p \approx N_A - N_D \quad \text{(p-type)}, \qquad n \approx \frac{n_i^2}{N_A - N_D}$$

   - If $N_{\text{net}} \lesssim n_i$: Use the exact quadratic formula:

     $$n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}$$

   - If $N_D = N_A$ (fully compensated): The material behaves intrinsically:

     $$n = p = n_i$$

11.8 Comprehensive Summary of Key Equations

Summary: Key Equations for Equilibrium Carrier Concentrations

Effective Density of States:

$$N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}, \qquad N_v = 2\left(\frac{2\pi m_h^* k_B T}{h^2}\right)^{3/2}$$

Carrier Concentrations (Boltzmann Approximation):

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right), \qquad p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Intrinsic Carrier Concentration and Mass Action Law:

$$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right), \qquad np = n_i^2$$

Fermi Level Positions:

$$\text{Intrinsic: } E_F = \frac{E_c + E_v}{2} + \frac{3}{4} k_B T \ln\left(\frac{m_h^*}{m_e^*}\right)$$

$$\text{n-Type: } E_F = E_c - k_B T \ln\left(\frac{N_c}{N_D}\right)$$

$$\text{p-Type: } E_F = E_v + k_B T \ln\left(\frac{N_v}{N_A}\right)$$

Charge Neutrality (Complete Ionization):

$$p + N_D = n + N_A$$

Exact Carrier Concentrations:

$$n = \frac{-(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2}, \qquad p = \frac{n_i^2}{n}$$

Approximation for $|N_D - N_A| \gg n_i$:

$$n \approx |N_D - N_A| \quad \text{(majority carriers)}, \qquad \text{minority carriers} \approx \frac{n_i^2}{|N_D - N_A|}$$

Key Takeaways

- The product $g_c(E) \cdot f(E)$ shows that electrons concentrate within a few $k_B T$ of E_c; similarly, holes concentrate near E_v.
- The mass action law $np = n_i^2$ is universal for all semiconductors in thermal equilibrium.
- **Charge neutrality** combined with $np = n_i^2$ yields exact carrier concentrations via a quadratic equation.
- For typical room-temperature devices with moderate doping ($|N_D - N_A| \gg n_i$), the simple approximations $n \approx N_D - N_A$ (n-type) or $p \approx N_A - N_D$ (p-type) are excellent.
- The Fermi level position is determined by doping: it rises toward E_c for n-type, falls toward E_v for p-type.

12 The General Drift-Diffusion-Poisson Model

Introduction: From Equilibrium to Nonequilibrium Dynamics

Thus far, we have established the equilibrium carrier concentrations and derived the fundamental laws governing drift and diffusion currents. The next critical step is to transition from static equilibrium to **nonequilibrium dynamics**, which lies at the heart of all active electrical transport devices—from basic resistors to diodes, transistors, and integrated circuits. The primary objective of this section is to derive the fundamental set of coupled partial differential equations that describe how the electron and hole concentrations evolve as functions of spatial position $\mathbf{x} \in \mathbb{R}^3$ and time $t \in \mathbb{R}$. Once $n(\mathbf{x}, t)$ and $p(\mathbf{x}, t)$ are determined in response to an applied electric field $\mathcal{E}(\mathbf{x}, t)$, subject to specific geometric boundary conditions and material doping profiles, we can compute the terminal currents and ultimately extract the complete I–V characteristics of the device.

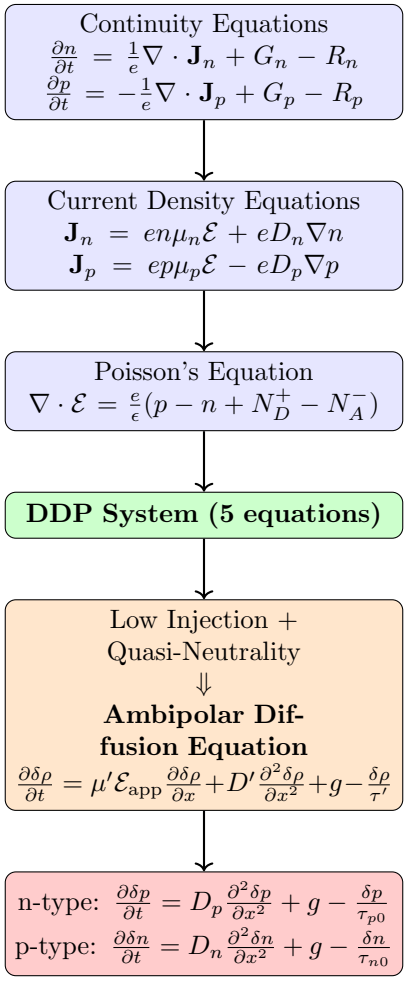


Figure 1: Roadmap from fundamental equations to the minority carrier diffusion equation.

12.1 Fundamental Equations Without Approximations

Definition of Physical Quantities

Below is a comprehensive list of the primary quantities used in the derivation. Emphasis is placed on their explicit space-time dependence where applicable.

Symbol	Physical Meaning	Units
$n(\mathbf{x},t)$	Electron concentration	cm^{-3}
$p(\mathbf{x},t)$	Hole concentration	cm^{-3}
$G_{\mathrm{n}}(\mathbf{x},t;n,p)$	Electron generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{n}}(\mathbf{x},t;n,p)$	Electron recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$G_{\mathrm{p}}(\mathbf{x},t;n,p)$	Hole generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{p}}(\mathbf{x},t;n,p)$	Hole recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$\mathbf{J}_{\mathrm{n}}(\mathbf{x},t)$	Total electron current density	$\mathrm{A\,cm}^{-2}$
$\mathbf{J}_{\mathrm{p}}(\mathbf{x},t)$	Total hole current density	$\mathrm{A\,cm}^{-2}$
e	Elementary charge (positive constant)	$1.602\times10^{-19}\,\mathrm{C}$
N_{D}^{+}	Ionized donor density (assumed fully ionized)	cm^{-3}
N_{A}^{-}	Ionized acceptor density (assumed fully ionized)	cm^{-3}
μ_{n}	Electron mobility	$\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
μ_{p}	Hole mobility	$\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}$
D_{n}	Electron diffusion coefficient	$\mathrm{cm}^2\,\mathrm{s}^{-1}$
D_{p}	Hole diffusion coefficient	$\mathrm{cm}^2\,\mathrm{s}^{-1}$
ϵ	Semiconductor permittivity	$\mathrm{F\,cm}^{-1}$

Recombination and Generation: Physical Context

We now consider the total recombination R and generation G rates, which encompass both equilibrium and nonequilibrium contributions. For the moment, we treat G as the total number of charge carriers generated per unit volume per unit time at location $\mathbf{x}$ and time t, given a local carrier content n and p. In general, G and R are highly nonlinear functionals of n and p, and may possess explicit space-time dependence $(\mathbf{x},t)$ beyond the implicit dependence carried by $n(\mathbf{x},t)$ and $p(\mathbf{x},t)$.

In semiconductor physics, we primarily focus on electron–hole pair recombination as the dominant loss mechanism in the bulk. Ohmic (Joule) dissipation is typically negligible in the semiconductor bulk and is instead accounted for at metal–semiconductor interfaces, ohmic contacts, or within metallic interconnects. Later in the course, we will introduce simplified phenomenological models (e.g., SRH, radiative, Auger) that effectively linearize or parameterize these relations for tractable analysis.

Derivation of the DDP Equations

The foundation of carrier transport is the **continuity equation**, which expresses local charge conservation. In its most general 3D form:

$$\frac{\partial\rho}{\partial t}+\nabla\cdot\mathbf{J}=\mathcal{G}-\mathcal{R}, \tag{6}$$

where ρ is the net charge density, $\mathbf{J}$ is the total current density, and $\mathcal{G}$, $\mathcal{R}$ are net generation and recombination source terms.

We now split this into two separate balance equations, one for electrons and one for holes. Using

$\rho_n = -en$ and $\rho_p = +ep$, and denoting the respective currents as $\mathbf{J}_\mathrm{n}$ and $\mathbf{J}_\mathrm{p}$, we obtain:

$$\frac{\partial n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{n} + G_\mathrm{n} - R_\mathrm{n}, \tag{7}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{p} + G_\mathrm{p} - R_\mathrm{p}. \tag{8}$$

Note the sign difference: electrons carry negative charge, so their current divergence enters with a positive prefactor, whereas holes carry positive charge.

The total current densities are composed of **drift** (response to electric field) and **diffusion** (response to concentration gradients) components. Using the standard sign conventions:

$$\mathbf{J}_\mathrm{n}(\mathbf{x},t) = en\mu_\mathrm{n}\mathcal{E}(\mathbf{x},t) + eD_\mathrm{n}\nabla n, \tag{9}$$

$$\mathbf{J}_\mathrm{p}(\mathbf{x},t) = ep\mu_\mathrm{p}\mathcal{E}(\mathbf{x},t) - eD_\mathrm{p}\nabla p. \tag{10}$$

The final equation closes the system via **Gauss's Law** (Poisson equation), relating the electric field to the net charge density:

$$\nabla \cdot \mathcal{E}(\mathbf{x},t) = \frac{1}{\epsilon}\rho_\mathrm{net} = \frac{e}{\epsilon}\left[p(\mathbf{x},t) - n(\mathbf{x},t) + N_\mathrm{D}^+ - N_\mathrm{A}^-\right]. \tag{11}$$

The 3D Drift–Diffusion–Poisson (DDP) System

$$\boxed{\frac{\partial n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{n} + G_\mathrm{n} - R_\mathrm{n}} \tag{12}$$

$$\boxed{\frac{\partial p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_\mathrm{p} + G_\mathrm{p} - R_\mathrm{p}} \tag{13}$$

$$\boxed{\mathbf{J}_\mathrm{n} = en\mu_\mathrm{n}\mathcal{E} + eD_\mathrm{n}\nabla n} \tag{14}$$

$$\boxed{\mathbf{J}_\mathrm{p} = ep\mu_\mathrm{p}\mathcal{E} - eD_\mathrm{p}\nabla p} \tag{15}$$

$$\boxed{\nabla \cdot \mathcal{E} = \frac{e}{\epsilon}\left(p - n + N_\mathrm{D}^+ - N_\mathrm{A}^-\right)} \tag{16}$$

One-Dimensional Reduction

For planar device structures or pedagogical analysis, we often assume all quantities vary only along a single spatial coordinate x. Replacing $\nabla \to \frac{\partial}{\partial x}\hat{\mathbf{x}}$ and $\mathcal{E}(\mathbf{x},t) \to \mathcal{E}(x,t)\,\hat{\mathbf{x}}$, the system reduces to:

The 1D Drift–Diffusion–Poisson System

$$\boxed{\frac{\partial n}{\partial t} = \frac{1}{e}\frac{\partial J_\mathrm{n}}{\partial x} + G_\mathrm{n} - R_\mathrm{n}} \tag{17}$$

$$\boxed{\frac{\partial p}{\partial t} = -\frac{1}{e}\frac{\partial J_\mathrm{p}}{\partial x} + G_\mathrm{p} - R_\mathrm{p}} \tag{18}$$

$$\boxed{J_\mathrm{n} = en\mu_\mathrm{n}\mathcal{E} + eD_\mathrm{n}\frac{\partial n}{\partial x}} \tag{19}$$

$$\boxed{J_\mathrm{p} = ep\mu_\mathrm{p}\mathcal{E} - eD_\mathrm{p}\frac{\partial p}{\partial x}} \tag{20}$$

$$\boxed{\frac{\partial \mathcal{E}}{\partial x} = \frac{e}{\epsilon}\left(p - n + N_\mathrm{D}^+ - N_\mathrm{A}^-\right)} \tag{21}$$

General Comments and Numerical Considerations

The five coupled partial differential equations above constitute the **Drift–Diffusion–Poisson (DDP) model**. They are highly nonlinear, strongly coupled, and generally lack closed-form analytical solutions. Even numerical resolution requires advanced computational techniques such as the Finite Element Method (FEM), Finite Difference Method (FDM), or specialized Semiconductor Device Simulators (TCAD tools like Sentaurus, Silvaco, or COMSOL).

The nonlinearity arises from:

- Products like $n\mathcal{E}$ and $p\mathcal{E}$ in the drift terms
- Spatial gradients ∇n, ∇p coupled to time derivatives
- Nonlinear, carrier-dependent $G(n,p)$ and $R(n,p)$ functions

Key Takeaways

- The DDP model provides a complete, self-consistent description of carrier transport and electrostatics in semiconductors.
- It is inherently **nonlinear** and **coupled**; analytical solutions require strategic approximations (depletion approximation, quasi-neutrality, steady-state, etc.).
- Modern device simulation relies on **FEM/FDM** discretization of these exact equations.
- In subsequent lectures, we will systematically simplify this system to derive tractable analytical models for diodes, BJTs, and MOSFETs while preserving the essential physics.

12.2 The Concept of Carrier Lifetime and Recombination Dynamics

In this subsection, we begin to simplify the highly coupled five-equation Drift–Diffusion–Poisson (DDP) system by introducing a series of well-justified physical assumptions. The first step in the analysis is to decompose the fundamental quantities into equilibrium (thermal) and nonequilibrium (excess) components.

Symbol	Physical Meaning	Units
n_0	Equilibrium electron concentration	cm^{-3}
p_0	Equilibrium hole concentration	cm^{-3}
$\delta n(\mathbf{x},t)$	Nonequilibrium (excess) electron concentration	cm^{-3}
$\delta p(\mathbf{x},t)$	Nonequilibrium (excess) hole concentration	cm^{-3}
$G_{\mathrm{th}}^{\mathrm{n}}$	Equilibrium electron generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$G_{\mathrm{th}}^{\mathrm{p}}$	Equilibrium hole generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{th}}^{\mathrm{n}}$	Equilibrium electron recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$R_{\mathrm{th}}^{\mathrm{p}}$	Equilibrium hole recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$g_{\mathrm{n}}(\mathbf{x},t)$	Excess electron generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$g_{\mathrm{p}}(\mathbf{x},t)$	Excess hole generation rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$r_{\mathrm{n}}(\mathbf{x},t)$	Excess electron recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$
$r_{\mathrm{p}}(\mathbf{x},t)$	Excess hole recombination rate	$\mathrm{cm}^{-3}\mathrm{s}^{-1}$

The total generation and recombination rates are decomposed as:

$$G_{\rm n} = G_{\rm th}^{\rm n} + g_{\rm n}, \quad G_{\rm p} = G_{\rm th}^{\rm p} + g_{\rm p}, \tag{22}$$
$$R_{\rm n} = R_{\rm th}^{\rm n} + r_{\rm n}, \quad R_{\rm p} = R_{\rm th}^{\rm p} + r_{\rm p}. \tag{23}$$

At thermal equilibrium, we assume a homogeneous, steady-state distribution. Consequently, all spatial gradients and time derivatives of the equilibrium quantities vanish:

$$\frac{\partial n_0}{\partial t} = 0, \quad \frac{\partial p_0}{\partial t} = 0, \quad \nabla n_0 = 0, \quad \nabla p_0 = 0.$$

Furthermore, detailed balance requires that at equilibrium, every generation event is exactly compensated by a recombination event:

$$G_{\rm th}^{\rm n} = R_{\rm th}^{\rm n}, \qquad G_{\rm th}^{\rm p} = R_{\rm th}^{\rm p}.$$

This equality holds fundamentally because electron–hole (EH) pair creation via direct interband transitions (or thermal generation) occurs continuously. In steady state, the rate of pair creation must exactly match the rate of pair annihilation, otherwise carrier densities would grow or decay indefinitely.

Charge Neutrality: Global vs. Local It is critical to distinguish two distinct notions of charge neutrality.

- **Global Charge Neutrality**: The net total charge within a macroscopic, isolated semiconductor domain D is zero. Mathematically:

$$\int_D \rho(\mathbf{x}, t)\, {\rm d}V = 0 \quad \Rightarrow \quad \langle p - n + N_{\rm D}^+ - N_{\rm A}^- \rangle_D = 0.$$

  This condition ensures the device remains electrically neutral overall but does *not* imply $n(\mathbf{x}, t) = p(\mathbf{x}, t)$ at every point.

- **Local Charge Neutrality**: The net charge density vanishes at every point in the domain:

$$\rho(\mathbf{x}, t) = e\big[p(\mathbf{x}, t) - n(\mathbf{x}, t) + N_{\rm D}^+ - N_{\rm A}^-\big] = 0 \quad \forall \mathbf{x} \in D.$$

In equilibrium, we already know $p_0 - n_0 + N_{\rm D}^+ - N_{\rm A}^- = 0$. If we now impose local charge neutrality on the total nonequilibrium densities, we substitute $n = n_0 + \delta n$ and $p = p_0 + \delta p$:

$$(p_0 + \delta p) - (n_0 + \delta n) + N_{\rm D}^+ - N_{\rm A}^- = 0.$$

Canceling the equilibrium terms immediately yields a fundamental relation:

$$\boxed{\delta n(\mathbf{x}, t) = \delta p(\mathbf{x}, t)} \tag{24}$$

This local charge neutrality condition states that excess electrons and holes are generated (or removed) strictly in pairs. However, it is **crucial** to note that $\delta n = \delta p$ does *not* imply $n = p$. In n-type material, for example, $n_0 \gg p_0$, so even with equal excess densities, $n \gg p$ remains true.

Reduction of the Continuity Equations Assuming spatial homogeneity ($\nabla \cdot \mathbf{J}_{\rm n,p} = 0$) and applying the decomposition above, the continuity equations reduce to ordinary time derivatives. Since n_0 and p_0 are constant in time (${\rm d}n_0/{\rm d}t = {\rm d}p_0/{\rm d}t = 0$), we obtain:

$$\frac{{\rm d}\delta n}{{\rm d}t} = G_{\rm n} - R_{\rm n}, \tag{25}$$
$$\frac{{\rm d}\delta p}{{\rm d}t} = G_{\rm p} - R_{\rm p}. \tag{26}$$

These equations govern the temporal evolution of excess carriers but require a model for how $r_{\rm n,p}$ and $g_{\rm n,p}$ depend on carrier densities.

Modeling Recombination Dynamics The physics of recombination involves the random encounter of two distinct species: electrons and holes. This is a universal phenomenon observed in chemical kinetics, population dynamics, and laser physics. When modeling collision-driven interaction rates $R(n,p)$, several functional dependencies might be proposed: $R(n)$, $R(p)$, $R(n+p)$, or $R(np)$.

Physical counting arguments rule out the first three: a rate depending only on n or only on p contradicts the requirement that both species must participate in an encounter. A sum $n+p$ fails because the probability of pairwise collisions scales with the product of the population densities, not their sum. Thus, the only logically consistent leading-order dependence is:

$$R = \alpha np,$$

where α is a recombination coefficient. To account for possible asymmetries, we introduce distinct coefficients α_{n} and α_{p}. Expanding $R(n,p)$ in a Taylor series around equilibrium and retaining only the linear term (consistent with low-injection regimes) yields the standard bimolecular form.

At thermal equilibrium, $n_0 p_0 = n_i^2$, so $R_{\mathrm{th}} = \alpha_{\mathrm{n,p}} n_i^2$. Since $G_{\mathrm{th}}^{\mathrm{n,p}} = R_{\mathrm{th}}^{\mathrm{n,p}}$, we substitute into the rate equations:

$$\frac{\mathrm{d}n}{\mathrm{d}t} = \alpha_{\mathrm{n}} n_i^2 + g_{\mathrm{n}} - \alpha_{\mathrm{n}} np, \tag{27}$$

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \alpha_{\mathrm{p}} n_i^2 + g_{\mathrm{p}} - \alpha_{\mathrm{p}} np. \tag{28}$$

Why Focus on Minority Carriers?

In practical device analysis, we typically derive dynamic equations for **minority carriers** because:

- Minority carrier concentrations are orders of magnitude smaller than majority carriers, making their *relative* change much larger and more sensitive to external perturbations.
- Diffusion currents in p-n junctions, BJTs, and solar cells are dominated by minority carrier gradients.
- Majority carriers rapidly screen external fields and equilibrate locally, while minority carriers dictate the temporal response speed (switching, modulation, photoconductive decay).

Thus, tracking minority carrier dynamics captures the essential transient and steady-state behavior of active semiconductor devices.

Let us focus on an **n-type** material and derive the equation for minority carriers (holes). Using $n = n_0 + \delta n$, $p = p_0 + \delta p$, and the local neutrality condition $\delta n = \delta p$, the excess recombination rate for holes becomes:

$$r_{\mathrm{p}} = R_{\mathrm{p}} - R_{\mathrm{th}}^{\mathrm{p}} = \alpha_{\mathrm{p}}(n_0 + \delta p)(p_0 + \delta p) - \alpha_{\mathrm{p}} n_i^2.$$

Since $n_0 p_0 = n_i^2$, the equilibrium terms cancel exactly, leaving:

$$r_{\mathrm{p}} = \alpha_{\mathrm{p}}\left[n_0 \delta p + p_0 \delta p + (\delta p)^2\right] = \alpha_{\mathrm{p}}\left[(n_0 + p_0)\delta p + (\delta p)^2\right].$$

The exact continuity equation for excess holes is therefore:

$$\frac{\mathrm{d}\delta p}{\mathrm{d}t} = g_{\mathrm{p}} - \alpha_{\mathrm{p}}(n_0 + p_0)\delta p - \alpha_{\mathrm{p}}(\delta p)^2.$$

This equation is nonlinear due to the $(\delta p)^2$ term and generally requires numerical solution. To obtain an analytical model, we invoke the **low-injection approximation**:

$$\boxed{\delta n = \delta p \ll n_0 + p_0.}$$

Under this condition, the quadratic term is negligible compared to the linear term. The equation linearizes to:

$$\frac{\mathrm{d}\delta p}{\mathrm{d}t} = g_{\mathrm{p}} - \frac{\delta p}{\tau_{\mathrm{p0}}},$$

where

$$\boxed{\tau_{\mathrm{p0}} = \frac{1}{\alpha_{\mathrm{p}}(n_0 + p_0)} \approx \frac{1}{\alpha_{\mathrm{p}} n_0}}$$

The parameter τ_{p0} is called the **minority carrier lifetime**. To see its physical meaning, consider a pulse excitation where $g_{\mathrm{p}} = 0$ for $t > 0$ and an initial excess density $\delta p(0) = \delta p_0$. The solution is:

$$\delta p(t) = \delta p_0 \, e^{-t/\tau_{\mathrm{p0}}}.$$

When $t \gg \tau_{\mathrm{p0}}$, the excess pulse effectively disappears. Thus, τ_{p0} represents the average time an excess hole survives in the semiconductor before recombining with an electron.

A completely symmetric derivation for **p-type** material yields the electron lifetime:

$$\boxed{\tau_{\mathrm{n0}} = \frac{1}{\alpha_{\mathrm{n}}(n_0 + p_0)} \approx \frac{1}{\alpha_{\mathrm{n}} p_0}}$$

Lifetime vs. Momentum Relaxation Time

It is vital **not** to confuse the carrier lifetimes $\tau_{\mathrm{n0}}, \tau_{\mathrm{p0}}$ with the momentum relaxation (scattering) times $\tau_{\mathrm{n}}, \tau_{\mathrm{p}}$ introduced in the drift analysis. They describe fundamentally different physical processes:

- $\tau_{\mathrm{n}}, \tau_{\mathrm{p}}$: Average time between momentum-randomizing collisions with the lattice or impurities (picosecond scale).
- $\tau_{\mathrm{n0}}, \tau_{\mathrm{p0}}$: Average time before an electron and hole meet and annihilate (microsecond to millisecond scale).

Typical values at room temperature are summarized below:

Material	τ_{scat} (s)	τ_{n0} (s)	τ_{p0} (s)	$\tau_{\mathrm{life}}/\tau_{\mathrm{scat}}$
Silicon (Si)	$\sim 10^{-13}$	10^{-5}–10^{-3}	10^{-5}–10^{-3}	$\sim 10^{8}$–10^{10}
Gallium Arsenide (GaAs)	$\sim 10^{-13}$	10^{-9}–10^{-7}	10^{-9}–10^{-7}	$\sim 10^{4}$–10^{6}
Germanium (Ge)	$\sim 10^{-13}$	10^{-4}–10^{-2}	10^{-4}–10^{-2}	$\sim 10^{9}$–10^{11}

The carrier lifetime is typically *orders of magnitude larger* than the relaxation time. This is physically necessary: a carrier must scatter off the lattice or impurities many thousands of times before it randomly encounters a carrier of the opposite sign to annihilate in an EH recombination event. If $\tau_{\mathrm{life}} \approx \tau_{\mathrm{scat}}$, conduction would be impossible because all carriers would recombine immediately upon generation.

Finally, we express the nonequilibrium recombination rates in the standard linear form:

$$\boxed{r_{\mathrm{n}} = \frac{\delta n}{\tau_{\mathrm{n0}}}, \qquad r_{\mathrm{p}} = \frac{\delta p}{\tau_{\mathrm{p0}}}.}$$

These represent the canonical recombination models used throughout this course. They close the DDP system under the low-injection, local charge neutrality, and homogeneous assumptions, enabling tractable analytical solutions for diode, transistor, and optoelectronic device characteristics.

12.3 Derivation of Coupled Carrier Density Equations

In the previous section, we assumed spatial homogeneity to reduce the transport problem to ordinary differential equations. We now relax this assumption to treat the general case where carrier concentrations vary in both space and time. The objective is to derive two coupled partial differential equations governing the dynamics of the excess carrier densities $\delta n(\mathbf{x}, t)$ and $\delta p(\mathbf{x}, t)$.

We begin by decomposing the total generation and recombination rates into equilibrium and excess components:

$$G_{\mathrm{n}} - R_{\mathrm{n}} = g_{\mathrm{n}} - r_{\mathrm{n}}, \qquad G_{\mathrm{p}} - R_{\mathrm{p}} = g_{\mathrm{p}} - r_{\mathrm{p}}.$$

Since the equilibrium concentrations n_0 and p_0 are spatially uniform and time-independent, the partial derivatives satisfy

$$\frac{\partial n}{\partial t} = \frac{\partial \delta n}{\partial t}, \quad \frac{\partial p}{\partial t} = \frac{\partial \delta p}{\partial t}, \quad \nabla n = \nabla \delta n, \quad \nabla p = \nabla \delta p.$$

Substituting these into the general continuity equations yields:

$$\frac{\partial \delta n}{\partial t} = \frac{1}{e}\nabla \cdot \mathbf{J}_{\mathrm{n}} + g_{\mathrm{n}} - r_{\mathrm{n}}, \tag{29}$$

$$\frac{\partial \delta p}{\partial t} = -\frac{1}{e}\nabla \cdot \mathbf{J}_{\mathrm{p}} + g_{\mathrm{p}} - r_{\mathrm{p}}. \tag{30}$$

Next, we substitute the Drift–Diffusion–Poisson (DDP) expressions for the current densities:

$$\mathbf{J}_{\mathrm{n}} = en\mu_{\mathrm{n}}\mathcal{E} + eD_{\mathrm{n}}\nabla n, \qquad \mathbf{J}_{\mathrm{p}} = ep\mu_{\mathrm{p}}\mathcal{E} - eD_{\mathrm{p}}\nabla p.$$

Inserting these into (29) and (30) gives:

$$\frac{\partial \delta n}{\partial t} = \frac{1}{e}\nabla \cdot (en\mu_{\mathrm{n}}\mathcal{E} + eD_{\mathrm{n}}\nabla n) + g_{\mathrm{n}} - r_{\mathrm{n}}, \tag{31}$$

$$\frac{\partial \delta p}{\partial t} = -\frac{1}{e}\nabla \cdot (ep\mu_{\mathrm{p}}\mathcal{E} - eD_{\mathrm{p}}\nabla p) + g_{\mathrm{p}} - r_{\mathrm{p}}. \tag{32}$$

Assuming constant mobility and diffusion coefficients, we extract them from the divergence operator:

$$\frac{\partial \delta n}{\partial t} = \mu_{\mathrm{n}}\nabla \cdot (n\mathcal{E}) + D_{\mathrm{n}}\nabla^2 n + g_{\mathrm{n}} - r_{\mathrm{n}}, \tag{33}$$

$$\frac{\partial \delta p}{\partial t} = -\mu_{\mathrm{p}}\nabla \cdot (p\mathcal{E}) + D_{\mathrm{p}}\nabla^2 p + g_{\mathrm{p}} - r_{\mathrm{p}}. \tag{34}$$

We now apply the **Leibniz product rule** for the divergence of a scalar field ψ multiplied by a vector field $\mathbf{A}$:

$$\nabla \cdot (\psi \mathbf{A}) = \psi\, \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla \psi.$$

Applying this rule term-by-term and substituting $n = n_0 + \delta n$, $p = p_0 + \delta p$ (with $\nabla n_0 = \nabla p_0 = 0$), we obtain the step-by-step expansions:

$$\begin{aligned}
\frac{\partial \delta n}{\partial t} &= \mu_{\mathrm{n}}\nabla \cdot (n\mathcal{E}) + D_{\mathrm{n}}\nabla^2 n + g_{\mathrm{n}} - r_{\mathrm{n}} \\
&= \mu_{\mathrm{n}}\left[n\nabla \cdot \mathcal{E} + \mathcal{E} \cdot \nabla n\right] + D_{\mathrm{n}}\nabla^2 n + g_{\mathrm{n}} - r_{\mathrm{n}} \\
&= \mu_{\mathrm{n}}(n_0 + \delta n)\nabla \cdot \mathcal{E} + \mu_{\mathrm{n}}\mathcal{E} \cdot \nabla \delta n + D_{\mathrm{n}}\nabla^2 \delta n + g_{\mathrm{n}} - r_{\mathrm{n}},
\end{aligned} \tag{35}$$

$$\begin{aligned}
\frac{\partial \delta p}{\partial t} &= -\mu_{\mathrm{p}}\nabla \cdot (p\mathcal{E}) + D_{\mathrm{p}}\nabla^2 p + g_{\mathrm{p}} - r_{\mathrm{p}} \\
&= -\mu_{\mathrm{p}}\left[p\nabla \cdot \mathcal{E} + \mathcal{E} \cdot \nabla p\right] + D_{\mathrm{p}}\nabla^2 p + g_{\mathrm{p}} - r_{\mathrm{p}} \\
&= -\mu_{\mathrm{p}}(p_0 + \delta p)\nabla \cdot \mathcal{E} - \mu_{\mathrm{p}}\mathcal{E} \cdot \nabla \delta p + D_{\mathrm{p}}\nabla^2 \delta p + g_{\mathrm{p}} - r_{\mathrm{p}}.
\end{aligned} \tag{36}$$

Equations (35) and (36) are second-order partial differential equations in space and explicitly involve $\nabla \cdot \mathcal{E}$. To close the system, we invoke Gauss's Law:

$$\nabla \cdot \mathcal{E} = \frac{e}{\epsilon}\left(p - n + N_{\mathrm{D}}^{+} - N_{\mathrm{A}}^{-}\right).$$

Using the equilibrium charge neutrality condition $p_0 - n_0 + N_{\mathrm{D}}^{+} - N_{\mathrm{A}}^{-} = 0$, we substitute $p = p_0 + \delta p$ and $n = n_0 + \delta n$ to obtain:

$$\nabla \cdot \mathcal{E} = \frac{e}{\epsilon}\left(\delta p - \delta n\right). \tag{37}$$

Note that we have **not** assumed local charge neutrality ($\delta n = \delta p$) at this stage. Retaining $\nabla \cdot \mathcal{E}$ explicitly allows us to treat the most general situation rigorously. (The consequences of imposing local charge neutrality will be examined in detail when we discuss the ambipolar approximation in subsequent lectures.)

Self-Consistent Coupling

The system (35)–(37) is inherently **nonlinear and strongly coupled**. Physically:

- Spatial gradients or temporal changes in δn and δp alter the net charge density.
- This modified charge density generates an electric field $\mathcal{E}$ via (37).
- The resulting $\mathcal{E}$ feeds back into the drift terms $\mu\mathcal{E} \cdot \nabla \delta n$ and $\mu\mathcal{E} \cdot \nabla \delta p$, which in turn modify the carrier evolution.

Consequently, these equations must be solved **self-consistently**. In analytical treatments, simplifying assumptions are introduced. In computational device simulation, iterative numerical schemes (e.g., Gummel iteration or Newton–Raphson coupled with FEM/FDM) are employed to converge to the self-consistent $\delta n(\mathbf{x}, t)$, $\delta p(\mathbf{x}, t)$, and $\mathcal{E}(\mathbf{x}, t)$ fields at each time step.

For planar device geometries, we reduce the system to one spatial dimension (x). Replacing $\nabla \to \frac{\partial}{\partial x}$, $\nabla^2 \to \frac{\partial^2}{\partial x^2}$, and $\mathcal{E} \cdot \nabla \to \mathcal{E}\frac{\partial}{\partial x}$, we obtain:

$$\boxed{\frac{\partial \delta n}{\partial t} = \mu_{\mathrm{n}}\mathcal{E}\frac{\partial \delta n}{\partial x} + \mu_{\mathrm{n}}(n_0 + \delta n)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{n}}\frac{\partial^2 \delta n}{\partial x^2} + g_{\mathrm{n}} - r_{\mathrm{n}}} \tag{38}$$

$$\boxed{\frac{\partial \delta p}{\partial t} = -\mu_{\mathrm{p}}\mathcal{E}\frac{\partial \delta p}{\partial x} - \mu_{\mathrm{p}}(p_0 + \delta p)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{p}}\frac{\partial^2 \delta p}{\partial x^2} + g_{\mathrm{p}} - r_{\mathrm{p}}} \tag{39}$$

Finally, under the **low-injection regime** ($\delta n = \delta p \ll n_0 + p_0$), the recombination rates linearize to:

$$r_{\mathrm{n}} = \frac{\delta n}{\tau_{\mathrm{n0}}}, \qquad r_{\mathrm{p}} = \frac{\delta p}{\tau_{\mathrm{p0}}},$$

where τ_{n0} and τ_{p0} are the minority carrier lifetimes derived previously. Substituting these into the 1D equations yields the standard coupled transport equations used in most analytical device models:

$$\frac{\partial \delta n}{\partial t} = \mu_{\mathrm{n}}\mathcal{E}\frac{\partial \delta n}{\partial x} + \mu_{\mathrm{n}}(n_0 + \delta n)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{n}}\frac{\partial^2 \delta n}{\partial x^2} + g_{\mathrm{n}} - \frac{\delta n}{\tau_{\mathrm{n0}}}, \tag{40}$$

$$\frac{\partial \delta p}{\partial t} = -\mu_{\mathrm{p}}\mathcal{E}\frac{\partial \delta p}{\partial x} - \mu_{\mathrm{p}}(p_0 + \delta p)\frac{e}{\epsilon}(\delta p - \delta n) + D_{\mathrm{p}}\frac{\partial^2 \delta p}{\partial x^2} + g_{\mathrm{p}} - \frac{\delta p}{\tau_{\mathrm{p0}}}. \tag{41}$$

These equations form the complete dynamical foundation for analyzing carrier injection, extraction, and diffusion in semiconductor devices under realistic spatially varying conditions.

Key Features of the Coupled Transport System

- **Second-Order Spatial Dependence**: The Laplacian terms $D_{\mathrm{n}}\partial^2\delta n/\partial x^2$ and $D_{\mathrm{p}}\partial^2\delta p/\partial x^2$ capture diffusion-driven carrier spreading from regions of high concentration to low concentration.
- **Field-Carrier Coupling**: The terms proportional to $\mathcal{E}\,\partial\delta n/\partial x$ and the explicit $(\delta p - \delta n)$ factors demonstrate how spatial charge imbalances instantly modify the local electric field, which in turn feeds back into the drift velocities.
- **Linearized Recombination Dynamics**: Under low injection, the recombination channels reduce to simple first-order decay terms $-\delta n/\tau_{\mathrm{n0}}$ and $-\delta p/\tau_{\mathrm{p0}}$, introducing characteristic relaxation timescales that compete with transport.
- **Boundary Conditions**: To obtain unique physical solutions, these PDEs must be supplemented with appropriate boundary conditions at device interfaces (e.g., ohmic contacts, surface recombination velocities, or reflective boundaries).

The coupled system above constitutes the most general description of nonequilibrium carrier transport within the drift-diffusion framework. While analytically formidable due to the explicit nonlinear coupling between δn, δp, and $\mathcal{E}$, this formulation rigorously captures the competition between injection, diffusion, drift, and recombination. In practical device simulation, these equations are discretized and solved iteratively using self-consistent numerical solvers (e.g., Gummel or Newton–Raphson methods) to resolve the electrostatic and transport fields at each spatiotemporal node.

The Inconsistency of Exact Local Charge Neutrality

Motivational Scenario: Uniform Optical Excitation Consider a semiconductor sample uniformly illuminated by light with photon energy $h\nu > E_g$. Each absorbed photon generates one electron-hole pair. Therefore, in the bulk of the semiconductor (away from surfaces and contacts), the generation rates for electrons and holes are intrinsically equal:

$$g_n(\mathbf{x},t) = g_p(\mathbf{x},t) \equiv g(\mathbf{x},t)$$

This is a fundamental physical fact: photons create electrons and holes in pairs. There is no mechanism in the bulk that generates an electron without simultaneously generating a hole, or vice versa.

Now, in the low-injection regime, it is tempting to also enforce **exact local charge neutrality** everywhere by setting $\delta n(x,t) = \delta p(x,t) \equiv \delta\rho(x,t)$. This assumption is often motivated by the observation that bulk semiconductors cannot sustain large space-charge regions over macroscopic distances — any local imbalance would be quickly neutralized by the high conductivity of the majority carriers.

Under these two assumptions — equal generation rates and equal excess carrier densities — the system appears to simplify dramatically. Substituting these conditions into the general 1D coupled carrier density equations, the Poisson space-charge term vanishes identically:

$$\frac{e}{\epsilon}(\delta p - \delta n) = 0.$$

The two evolution equations then decouple from the electric field and reduce to:

$$\frac{\partial\delta\rho}{\partial t} = \mu_{\mathrm{n}}\mathcal{E}\frac{\partial\delta\rho}{\partial x} + D_{\mathrm{n}}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{n0}}}, \tag{42}$$

$$\frac{\partial\delta\rho}{\partial t} = -\mu_{\mathrm{p}}\mathcal{E}\frac{\partial\delta\rho}{\partial x} + D_{\mathrm{p}}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{p0}}}. \tag{43}$$

The Inconsistency Since both equations describe the same physical quantity $\delta\rho(x,t)$, their left-hand sides are identical. For consistency, the right-hand sides must also be equal at every point (x,t). Equating (42) and (43) yields the constraint:

$$(\mu_{\rm n}+\mu_{\rm p})\mathcal{E}\frac{\partial\delta\rho}{\partial x}+(D_{\rm n}-D_{\rm p})\frac{\partial^2\delta\rho}{\partial x^2}+\delta\rho\left(\frac{1}{\tau_{\rm p0}}-\frac{1}{\tau_{\rm n0}}\right)=0.$$

For this equation to hold **identically** for all possible $\delta\rho(x,t)$ and $\mathcal{E}(x,t)$ that might arise in a device, the coefficients would need to satisfy:

- $\mu_{\rm n}+\mu_{\rm p}=0 \quad\Rightarrow\quad \mu_{\rm n}=-\mu_{\rm p}$ (impossible, since both mobilities are strictly positive),
- $D_{\rm n}-D_{\rm p}=0 \quad\Rightarrow\quad D_{\rm n}=D_{\rm p}$ (generally false, as $D=\mu k_BT/e$ and $\mu_{\rm n}\neq\mu_{\rm p}$),
- $\frac{1}{\tau_{\rm p0}}-\frac{1}{\tau_{\rm n0}}=0 \quad\Rightarrow\quad \tau_{\rm n0}=\tau_{\rm p0}$ (only true in highly symmetric or compensated materials).

Since these equalities are not satisfied in real semiconductors, the system (42)–(43) is **mathematically inconsistent**.

Physical Interpretation: Why Exact Neutrality Fails

The contradiction reveals a fundamental dynamical fact: electrons and holes possess different mobilities and diffusion coefficients. If they were forced to move exactly together ($\delta n=\delta p$ everywhere), their natural drift and diffusion velocities would differ. This mismatch would instantly generate a local separation of charge, which in turn produces an internal electric field.
Therefore, **exact local charge neutrality cannot be strictly enforced** in the presence of spatial gradients or external fields. The semiconductor maintains a *quasi-neutral* state where $\delta n\approx\delta p$, but a tiny, self-consistent deviation $\delta p-\delta n\neq 0$ is precisely what generates the internal field required to couple the electron and hole flows.

Assumption	Physical Motivation	Why It Fails
$g_n=g_p$	Optical generation creates e-h pairs	✓ Correct in bulk, not the problem
$\delta n=\delta p$	Bulk cannot sustain large space charge	× Fails because $\mu_n\neq\mu_p$, $D_n\neq D_p$

How Do We Proceed? The Quasi-Neutrality and Ambipolar Approximations The resolution lies in relaxing the strict equality $\delta n=\delta p$ while retaining the physical insight that bulk semiconductors cannot sustain large space-charge regions outside depletion zones. We adopt the **quasi-neutrality assumption**:

$$\boxed{\delta n(x,t)\approx\delta p(x,t),\quad \text{but}\quad \nabla\cdot\mathcal{E}\neq 0\ \text{(small but finite)}.}$$

Instead of discarding the Poisson term, we use it to eliminate the unknown electric field $\mathcal{E}$ from the coupled system. By combining the two continuity equations in a way that cancels the drift-diffusion mismatch, we can derive a single transport equation for the excess carrier density. This procedure yields the celebrated **ambipolar diffusion equation**, which introduces effective transport parameters (ambipolar mobility μ' and ambipolar diffusion coefficient D') that describe the coupled motion of electrons and holes as a single quasi-particle ensemble.

In the following subsection, we will rigorously derive this ambipolar transport equation, show how it resolves the inconsistency highlighted above, and demonstrate its power in analyzing bipolar devices such as p-n junctions, solar cells, and photodetectors.

12.4 The Ambipolar Approximation

The strong electrostatic coupling between electrons and holes often enforces quasi-neutrality in the bulk of optoelectronic and bipolar devices. In this subsection, we will exploit this property to combine the two continuity equations into a single *ambipolar diffusion equation*, dramatically simplifying analytical treatments of p-n junctions, photodetectors, and bipolar transistors while preserving the essential physics of coupled carrier transport.

Key Physical Idea: The Internal Electric Field

When excess electrons and holes are injected into a semiconductor, they tend to move under the influence of:

- Any **applied electric field** $\mathcal{E}_{\text{app}}$ (from external bias)
- **Diffusion** due to concentration gradients

However, electrons and holes have different mobilities ($\mu_{\text{n}} \neq \mu_{\text{p}}$) and different diffusion coefficients ($D_{\text{n}} \neq D_{\text{p}}$). If they were to move independently, their natural drift and diffusion velocities would differ. This would instantly create a local separation of charge, generating an **internal electric field** $\mathcal{E}_{\text{int}}$ via Poisson's equation.

The total electric field is therefore:

$$\boxed{\mathcal{E}_{\text{total}} = \mathcal{E}_{\text{app}} + \mathcal{E}_{\text{int}}}$$

This internal field acts as a **restoring force** that binds the excess electron and hole populations together. Physically:

- If electrons (negative) try to move faster than holes (positive), the resulting charge imbalance creates an internal field that **slows down** the faster species and **accelerates** the slower species
- This Coulomb coupling ensures that the excess electron and hole pulses travel together as a single *ambipolar* packet

Crucial Note: Binding Only Applies to Excess Carriers

The internal field binds the **excess** carriers (δn and δp) to each other. It does **not** bind the total carrier populations ($n = n_0 + \delta n$, $p = p_0 + \delta p$).
For example, in n-type material:

- The majority carrier concentration n_0 is much larger than δn
- The internal field does not prevent the majority electrons from flowing independently to contacts
- This is why we can have large terminal currents while maintaining $\delta n \approx \delta p$ in the bulk

Derivation of the Ambipolar Transport Equation

We now present the complete derivation of the ambipolar transport equation. Starting from the 1D continuity equations for excess carriers under low injection ($\delta n = \delta p = \delta\rho$) and assuming quasi-neutrality, we combine the equations to eliminate the electric field.

The 1D continuity equations for electrons and holes under low injection, including drift, diffusion, generation, and recombination, are:

$$\frac{\partial \delta n}{\partial t} = \mu_{\text{n}} \mathcal{E}_{\text{total}} \frac{\partial \delta n}{\partial x} + D_{\text{n}} \frac{\partial^2 \delta n}{\partial x^2} + g - \frac{\delta n}{\tau_{\text{n0}}}, \tag{44}$$

$$\frac{\partial \delta p}{\partial t} = -\mu_{\text{p}} \mathcal{E}_{\text{total}} \frac{\partial \delta p}{\partial x} + D_{\text{p}} \frac{\partial^2 \delta p}{\partial x^2} + g - \frac{\delta p}{\tau_{\text{p0}}}. \tag{45}$$

Under quasi-neutrality, we set $\delta n = \delta p = \delta\rho$. Also, the total electric field $\mathcal{E}_{\text{total}}$ is the sum of applied and internal fields. However, for the purpose of deriving the ambipolar equation, we treat $\mathcal{E}_{\text{total}}$ as the field that appears in both equations.

To eliminate $\mathcal{E}_{\text{total}}$, we multiply Equation (44) by $\mu_{\text{p}}p_0$ and Equation (45) by $\mu_{\text{n}}n_0$, then add them. This weighting choice combines the drift terms into a single effective drift term characterized by the ambipolar mobility μ'.

First, compute the weighted equations:

$$\mu_{\text{p}}p_0\frac{\partial\delta\rho}{\partial t} = \mu_{\text{n}}\mu_{\text{p}}p_0\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + \mu_{\text{p}}p_0D_{\text{n}}\frac{\partial^2\delta\rho}{\partial x^2} + \mu_{\text{p}}p_0g - \mu_{\text{p}}p_0\frac{\delta\rho}{\tau_{\text{n0}}}, \tag{46}$$

$$\mu_{\text{n}}n_0\frac{\partial\delta\rho}{\partial t} = -\mu_{\text{n}}\mu_{\text{p}}n_0\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + \mu_{\text{n}}n_0D_{\text{p}}\frac{\partial^2\delta\rho}{\partial x^2} + \mu_{\text{n}}n_0g - \mu_{\text{n}}n_0\frac{\delta\rho}{\tau_{\text{p0}}}. \tag{47}$$

Now add Equations (46) and (47):

$$(\mu_{\text{p}}p_0+\mu_{\text{n}}n_0)\frac{\partial\delta\rho}{\partial t} = \mu_{\text{n}}\mu_{\text{p}}(p_0-n_0)\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x}+(\mu_{\text{p}}p_0D_{\text{n}}+\mu_{\text{n}}n_0D_{\text{p}})\frac{\partial^2\delta\rho}{\partial x^2}+(\mu_{\text{p}}p_0+\mu_{\text{n}}n_0)g-\left(\frac{\mu_{\text{p}}p_0}{\tau_{\text{n0}}}+\frac{\mu_{\text{n}}n_0}{\tau_{\text{p0}}}\right)\delta\rho.$$

The drift term simplifies to:

$$-\mu_{\text{n}}\mu_{\text{p}}(n_0-p_0)\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x}.$$

Divide both sides by $(\mu_{\text{p}}p_0+\mu_{\text{n}}n_0)$:

$$\frac{\partial\delta\rho}{\partial t} = -\frac{\mu_{\text{n}}\mu_{\text{p}}(n_0-p_0)}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + \frac{\mu_{\text{p}}p_0D_{\text{n}}+\mu_{\text{n}}n_0D_{\text{p}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\mu_{\text{p}}p_0/\tau_{\text{n0}}+\mu_{\text{n}}n_0/\tau_{\text{p0}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}\delta\rho.$$

Thus, we identify the **ambipolar mobility** μ' and **ambipolar diffusion coefficient** D':

$$\boxed{\mu' = \frac{\mu_{\text{n}}\mu_{\text{p}}(n_0-p_0)}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}}$$

$$\boxed{D' = \frac{\mu_{\text{p}}p_0D_{\text{n}}+\mu_{\text{n}}n_0D_{\text{p}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}}$$

Using the Einstein relation $D = \mu k_BT/e$, we substitute $\mu_{\text{n}} = \frac{e}{k_BT}D_{\text{n}}$ and $\mu_{\text{p}} = \frac{e}{k_BT}D_{\text{p}}$ into the expression for D'. The constant factors $\frac{e}{k_BT}$ cancel, yielding the alternative form:

$$\boxed{D' = \frac{D_{\text{n}}D_{\text{p}}(n_0+p_0)}{D_{\text{n}}n_0+D_{\text{p}}p_0}}$$

The effective lifetime term τ' is given by:

$$\frac{1}{\tau'} = \frac{\mu_{\text{p}}p_0/\tau_{\text{n0}}+\mu_{\text{n}}n_0/\tau_{\text{p0}}}{\mu_{\text{n}}n_0+\mu_{\text{p}}p_0}.$$

Under low injection, τ' reduces to the minority carrier lifetime, as shown below.

The final **ambipolar transport equation** is:

$$\boxed{\frac{\partial\delta\rho}{\partial t} = -\mu'\mathcal{E}_{\text{total}}\frac{\partial\delta\rho}{\partial x} + D'\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau'}}$$

Effective Ambipolar Parameters

For the case where both electrons and holes contribute significantly to conduction (i.e., in intrinsic or lightly doped material, or under high injection), the **ambipolar mobility** and **ambipolar diffusion coefficient** are given by:

$$\boxed{\mu' = \frac{(n_0 - p_0)\mu_\mathrm{n}\mu_\mathrm{p}}{n_0\mu_\mathrm{n} + p_0\mu_\mathrm{p}}}$$

$$\boxed{D' = \frac{D_\mathrm{n}D_\mathrm{p}(n_0 + p_0)}{D_\mathrm{n}n_0 + D_\mathrm{p}p_0}}$$

(Note: The sign in μ' indicates the drift direction of the ambipolar packet. For $n_0 > p_0$, $\mu' > 0$ and the packet drifts in the direction of the field, analogous to hole motion. The expression for D' is obtained by substituting the Einstein relation into the mobility-weighted form.)

The effective lifetime τ' depends on the recombination mechanism. Under low injection, it reduces to the minority carrier lifetime.

Important Simplifications for Practical Cases

Case 1: n-Type Material ($n_0 \gg p_0$, low injection $\delta\rho \ll n_0$)
For an n-type semiconductor, $n_0 \gg p_0$ and $\delta\rho \ll n_0$. Then:

$$\mu' \approx \frac{n_0\mu_\mathrm{n}\mu_\mathrm{p}}{n_0\mu_\mathrm{n}} = \mu_\mathrm{p}$$

$$D' \approx \frac{n_0 D_\mathrm{p}}{n_0} = D_\mathrm{p}$$

$$\frac{1}{\tau'} \approx \frac{\mu_\mathrm{n}n_0/\tau_{\mathrm{p}0}}{\mu_\mathrm{n}n_0} = \frac{1}{\tau_{\mathrm{p}0}}$$

Thus, the ambipolar equation reduces to the **hole diffusion equation**:

$$\boxed{\frac{\partial\delta\rho}{\partial t} = D_\mathrm{p}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{p}0}}}$$

Case 2: p-Type Material ($p_0 \gg n_0$, low injection $\delta\rho \ll p_0$)
Similarly, for p-type material:

$$\mu' \approx -\mu_\mathrm{n}, \qquad D' \approx D_\mathrm{n}, \qquad \tau' \approx \tau_{\mathrm{n}0}$$

The ambipolar equation reduces to the **electron diffusion equation**:

$$\boxed{\frac{\partial\delta\rho}{\partial t} = D_\mathrm{n}\frac{\partial^2\delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{n}0}}}$$

Physical Interpretation of the Ambipolar Parameters

- **Ambipolar mobility** μ': In intrinsic material ($n_0 = p_0$), $\mu' = 0$ — there is no net drift of the excess packet in an applied field because electrons and holes drift in opposite directions, canceling each other's motion.
- **Ambipolar diffusion coefficient** D': This is a weighted average of D_n and D_p. The slower species (usually holes) limits the diffusion of the coupled pair.

- **Why $D' \approx D_{\mathrm{p}}$ in n-type**: The excess holes (minority carriers) control the diffusion rate because they are the limiting species. The abundant majority electrons quickly adjust to maintain neutrality.

Summary: The Ambipolar Approximation in a Nutshell

Key Takeaways

- The **internal electric field** $\mathcal{E}_{\mathrm{int}}$ couples electron and hole motion, forcing excess carriers to travel together
- This binding applies only to **excess carriers** (δn, δp), not total carriers
- The result is a single **ambipolar diffusion equation** for $\delta\rho = \delta n = \delta p$
- Under low injection in n-type material: $D' = D_{\mathrm{p}}$, $\mu' = \mu_{\mathrm{p}}$ (hole diffusion controls)
- Under low injection in p-type material: $D' = D_{\mathrm{n}}$, $\mu' = -\mu_{\mathrm{n}}$ (electron diffusion controls)
- In intrinsic material, the ambipolar mobility $\mu' = 0$ — the excess packet does not drift

Why Not Simply Set $\delta n = \delta p$ Exactly?

Recall our earlier inconsistency: setting $\delta n = \delta p$ exactly led to contradictory equations because $\mu_{\mathrm{n}} \neq \mu_{\mathrm{p}}$. The ambipolar approximation resolves this by:

1. Acknowledging that $\delta n \approx \delta p$ (quasi-neutrality), but not exactly equal
2. The small difference $\delta p - \delta n$ generates $\mathcal{E}_{\mathrm{int}}$ via Poisson's equation
3. This $\mathcal{E}_{\mathrm{int}}$ couples the equations and allows a single equation to emerge
4. The factor e/ϵ can amplify tiny charge imbalances into significant fields

Thus, the ambipolar approximation is **self-consistent**: the internal field adjusts to make the two species move together, and we describe their coupled motion with effective parameters.

Final Ambipolar Equation for Minority Carrier Diffusion

For most practical device analysis (p-n junctions, BJTs, solar cells), we work in the low-injection regime where the minority carrier diffusion equation applies:

$$\boxed{\frac{\partial \delta\rho}{\partial t} = D_{\mathrm{minority}} \frac{\partial^2 \delta\rho}{\partial x^2} + g - \frac{\delta\rho}{\tau_{\mathrm{minority}}}}$$

where:

- For n-type: $D_{\mathrm{minority}} = D_{\mathrm{p}}$, $\tau_{\mathrm{minority}} = \tau_{\mathrm{p0}}$, $\delta\rho =$ excess holes
- For p-type: $D_{\mathrm{minority}} = D_{\mathrm{n}}$, $\tau_{\mathrm{minority}} = \tau_{\mathrm{n0}}$, $\delta\rho =$ excess electrons

This equation forms the foundation for analyzing all bipolar devices, as we will see in subsequent lectures.

Conceptual Clarification: How Can Electrons and Holes Travel Together Without Annihilating?

A common conceptual hurdle arises when students compare two seemingly contradictory statements:

1. In the ambipolar transport model, excess electrons and holes travel together as a coupled pair, maintaining approximate local charge neutrality ($\delta n \approx \delta p$) at every point in space and time.
2. In the recombination dynamics analysis, we derived that the excess carrier recombination rate is $r = \delta n/\tau_{n0}$ (or $r = \delta p/\tau_{p0}$), meaning that when an electron and a hole meet, they annihilate.

The natural question arises: **If electrons and holes are traveling together, why don't they instantly recombine and vanish?**

The Resolution: Two Vastly Different Time Scales The key to resolving this apparent paradox lies in recognizing that there are **two completely distinct time scales** at play in semiconductor physics:

Parameter	Symbol	Typical Magnitude	Physical Meaning
Momentum relaxation time	τ_n, τ_p	10^{-13} – 10^{-12} s	Time between scattering events (collisions with lattice, impurities)
Carrier lifetime	τ_{n0}, τ_{p0}	10^{-9} – 10^{-3} s	Time before an electron and hole recombine

The critical observation is that:

$$\boxed{\tau_{n0}, \tau_{p0} \gg \tau_n, \tau_p} \tag{48}$$

Typically, the carrier lifetime is **orders of magnitude larger** than the momentum relaxation time. For silicon at room temperature:

- $\tau_n \approx \tau_p \approx 10^{-13}$ s (picosecond scale)
- $\tau_{n0} \approx 10^{-6} - 10^{-3}$ s (microsecond to millisecond scale)

The ratio τ_{n0}/τ_n can be as large as 10^7 – 10^{10}!

What This Means Physically

Analogy: Two People Walking Together in a Crowded Room

Imagine two people (an electron and a hole) trying to find each other in a vast, crowded stadium filled with thousands of other people (lattice atoms, impurities, other carriers).

- **Momentum relaxation (scattering)**: Each person constantly bumps into other people in the crowd, changing direction every fraction of a second. This is like τ_n and τ_p — frequent, random collisions that randomize velocity.
- **Recombination (annihilation)**: The two specific people (electron and hole) are searching for each other to "annihilate" (recombine). Even if they are walking in the same direction, the stadium is so crowded and large that it takes a **very long time** (millions of collisions) before they actually meet.
- The **lifetime** τ_{n0} is the average time before this meeting happens.

Why the Internal Field Binds Without Causing Recombination The internal electric field $\mathcal{E}_{\text{int}}$ that couples the excess electron and hole pulses arises from **electrostatic attraction** — it is a long-range Coulomb force that acts continuously over large distances. This force ensures that the *centers* of the excess carrier pulses move together.

However, recombination requires a **close encounter** — the electron and hole must come within a very short distance (typically a few nanometers, comparable to the exciton Bohr radius) to recombine. The probability of such a close encounter at any given moment is extremely small.

Thus:

- The **internal field** ensures the pulses drift and diffuse together on average
- **Recombination** is a rare event that occurs when, by chance, a specific electron and hole come very close
- Because $\tau_{\text{n0}} \gg \tau_{\text{n}}$, the carriers undergo **billions of scattering events** before finally recombining

Mathematical Justification We can make this argument more quantitative. Consider an n-type semiconductor under low injection. The excess holes (minority carriers) have:

- Mean time between scattering events: $\tau_{\text{p}} \approx 10^{-13}$ s
- Mean time before recombination: $\tau_{\text{p0}} \approx 10^{-6}$ s

The number of scattering events a typical hole experiences before recombining is:

$$N_{\text{scat}} = \frac{\tau_{\text{p0}}}{\tau_{\text{p}}} \approx \frac{10^{-6}}{10^{-13}} = 10^{7}$$

The hole undergoes **ten million collisions** with the lattice and impurities before finally encountering an electron to recombine. During this time, it can travel a significant distance (the diffusion length $L_p = \sqrt{D_p \tau_{\text{p0}}}$, typically tens of micrometers).

The Role of Charge Neutrality Approximate local charge neutrality ($\delta n \approx \delta p$) does **not** mean that every electron is physically coincident with a hole. Rather, it means that in any small volume (large enough to contain many carriers but small compared to device dimensions), the *total number* of excess electrons approximately equals the *total number* of excess holes.

Within that volume, individual electrons and holes are moving randomly, scattering frequently. The probability that any specific electron-hole pair comes close enough to recombine during a given short time interval is tiny. Only after many scattering events (on the time scale of τ_{n0} or τ_{p0}) does recombination eventually occur.

Key Takeaways

- The **internal electric field** couples the *centers of mass* of the excess electron and hole distributions via long-range Coulomb forces
- **Recombination** requires a short-range encounter (within nanometers) and is therefore a much rarer event
- The vast separation of time scales ($\tau_{\text{n0}}, \tau_{\text{p0}} \gg \tau_{\text{n}}, \tau_{\text{p}}$) allows the ambipolar transport approximation to be valid: excess carriers can travel together for long distances and times before eventually recombining
- Approximate local charge neutrality ($\delta n \approx \delta p$) refers to *average* concentrations over volumes containing many carriers, not to individual electron-hole pairs

Connection to the Drift-Diffusion-Poisson Model In the DDP model, the continuity equations track *concentrations* (averages over many carriers), not individual particles. The recombination terms $-\delta n/\tau_{\mathrm{n0}}$ and $-\delta p/\tau_{\mathrm{p0}}$ represent the *average rate* at which carriers disappear from the ensemble. This average rate is small precisely because τ_{n0} and τ_{p0} are large compared to the scattering time.

Thus, there is no contradiction:

- At the **macroscopic (continuum)** level, we describe coupled drift-diffusion of excess electron and hole *populations*
- At the **microscopic (particle)** level, individual electrons and holes undergo frequent scattering but rarely recombine
- The internal field ensures the *average* motions are correlated, while recombination represents the slow, statistical annihilation of the populations

This separation of time scales is what makes semiconductor devices possible: carriers can travel significant distances (across a p-n junction, through a transistor base) before recombining, allowing for current flow and amplification.

12.5 Applications of the Ambipolar Transport Equation

We now solve the ambipolar transport equation for several important problems. These examples illustrate the behavior of excess carriers in semiconductor materials, and the results will be used later in the discussion of p-n junctions and other semiconductor devices.

The following examples use several common simplifications in the solution of the ambipolar transport equation. Table 1 summarizes these simplifications and their effects.

Specification	**Effect**
Steady state	$\dfrac{\partial(\delta n)}{\partial t} = 0, \quad \dfrac{\partial(\delta p)}{\partial t} = 0$
Uniform distribution of excess carriers (uniform generation rate)	$\dfrac{\partial^2(\delta n)}{\partial x^2} = 0, \quad \dfrac{\partial^2(\delta p)}{\partial x^2} = 0$
Zero electric field	$\mathcal{E} = 0, \quad \mathcal{E}\dfrac{\partial(\delta n)}{\partial x} = 0, \quad \mathcal{E}\dfrac{\partial(\delta p)}{\partial x} = 0$
No excess carrier generation	$g = 0$
No excess carrier recombination (infinite lifetime)	$\dfrac{\delta n}{\tau_{n0}} = 0, \quad \dfrac{\delta p}{\tau_{p0}} = 0$

Example 6.2: Time Decay of Excess Carriers (Recombination)

Objective: Determine the time behavior of excess carriers as a semiconductor returns to thermal equilibrium.

Consider an infinitely large, homogeneous n-type semiconductor with zero applied electric field. Assume that at time $t = 0$, a uniform concentration of excess carriers exists in the crystal, but assume that $g = 0$ for

$t > 0$. If we assume that the concentration of excess carriers is much smaller than the thermal-equilibrium electron concentration, then the low-injection condition applies. Calculate the excess carrier concentration as a function of time for $t \geq 0$.

Solution:

For the n-type semiconductor, we need to consider the ambipolar transport equation for the minority carrier holes, which is:

$$D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p \mathcal{E} \frac{\partial(\delta p)}{\partial x} + g - \frac{\delta p}{\tau_{p0}} = \frac{\partial(\delta p)}{\partial t}$$

We are assuming a uniform concentration of excess holes so that:

$$\frac{\partial^2(\delta p)}{\partial x^2} = \frac{\partial(\delta p)}{\partial x} = 0$$

For $t > 0$ we are also assuming that $g = 0$. The equation reduces to:

$$\frac{d(\delta p)}{dt} = -\frac{\delta p}{\tau_{p0}}$$

Since there is no spatial variation, the total time derivative may be used. At low injection, the minority carrier hole lifetime τ_{p0} is a constant. The left-hand side is the time rate of change of δp and the right-hand side is the recombination rate. The solution is:

$$\delta p(t) = \delta p(0)\, e^{-t/\tau_{p0}}$$

where $\delta p(0)$ is the uniform concentration of excess carriers that exists at time $t = 0$. The concentration of excess holes decays exponentially with time, with a time constant equal to the minority carrier hole lifetime.

From the charge neutrality condition, we have $\delta n = \delta p$, so the excess electron concentration is given by:

$$\delta n(t) = \delta p(t) = \delta p(0)\, e^{-t/\tau_{p0}}$$

Comment: The excess electrons and holes recombine at the rate determined by the excess minority carrier hole lifetime in the n-type semiconductor.

Example 6.3: Time-Dependent Buildup to Steady State

Objective: Determine the time dependence of excess carriers in reaching a steady-state condition.

Again consider an infinitely large, homogeneous n-type semiconductor with zero applied electric field. Assume that for $t < 0$, the semiconductor is in thermal equilibrium and that for $t \geq 0$, a uniform generation rate exists in the crystal. Calculate the excess carrier concentration as a function of time assuming the condition of low injection.

Solution:

The condition of a uniform generation rate and a homogeneous semiconductor again implies that:

$$\frac{\partial^2(\delta p)}{\partial x^2} = \frac{\partial(\delta p)}{\partial x} = 0$$

The ambipolar transport equation for minority carrier holes reduces to:

$$g - \frac{\delta p}{\tau_{p0}} = \frac{d(\delta p)}{dt}$$

This is a first-order differential equation. To solve it, rearrange:

$$\frac{d(\delta p)}{dt} + \frac{\delta p}{\tau_{p0}} = g$$

The solution consists of a homogeneous part plus a particular solution. The homogeneous solution is $\delta p_h = Ae^{-t/\tau_{p0}}$. The particular solution (steady state) is found by setting $d(\delta p)/dt = 0$, giving $\delta p_{ss} = g\tau_{p0}$. The complete solution is:

$$\delta p(t) = g\tau_{p0} + Ae^{-t/\tau_{p0}}$$

The constant A is determined from the initial condition. At $t = 0$, $\delta p(0) = 0$ (thermal equilibrium). Therefore:

$$0 = g\tau_{p0} + A \quad \Rightarrow \quad A = -g\tau_{p0}$$

Thus:

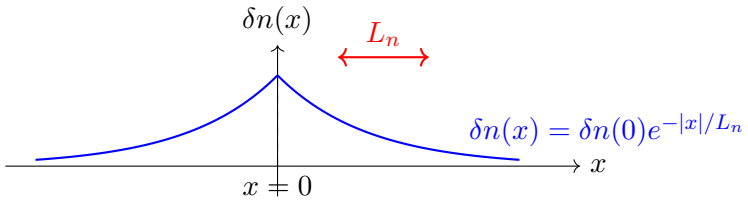
$$\boxed{\delta p(t) = g\tau_{p0}\left(1 - e^{-t/\tau_{p0}}\right)}$$

Comment: We may note that as $t \to \infty$, a steady-state excess hole and electron concentration of $g\tau_{p0}$ is reached. The excess minority carrier hole concentration increases with time with the same time constant τ_{p0}, which is the excess minority carrier lifetime. The excess carrier concentration reaches a steady-state value as time goes to infinity, even though a steady-state generation of excess electrons and holes exists. This steady-state effect can be seen by setting $d(\delta p)/dt = 0$ in the original equation, which yields $g = \delta p/\tau_{p0}$, i.e., generation rate equals recombination rate.

Example 6.4: Steady-State Spatial Decay (Diffusion Length)

Objective: Determine the steady-state spatial dependence of the excess carrier concentration.

Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only. The excess carriers being generated at $x = 0$ will begin diffusing in both the $+x$ and $-x$ directions. Calculate the steady-state excess carrier concentration as a function of x.



Solution:

The ambipolar transport equation for excess minority carrier electrons in p-type material is:

$$D_n\frac{\partial^2(\delta n)}{\partial x^2} + \mu_n\mathcal{E}\frac{\partial(\delta n)}{\partial x} + g - \frac{\delta n}{\tau_{n0}} = \frac{\partial(\delta n)}{\partial t}$$

From our assumptions:

- $\mathcal{E} = 0$ (zero applied field)
- $g = 0$ for $x \neq 0$ (generation only at $x = 0$)
- $\partial(\delta n)/\partial t = 0$ (steady state)

Assuming a one-dimensional crystal, the equation reduces to:

$$D_n\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{\tau_{n0}} = 0$$

Dividing by the diffusion coefficient:

$$\frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{D_n \tau_{n0}} = \frac{d^2(\delta n)}{dx^2} - \frac{\delta n}{L_n^2} = 0$$

where we have defined $L_n^2 = D_n \tau_{n0}$. The parameter L_n has units of length and is called the **minority carrier electron diffusion length**.

The general solution to this differential equation is:

$$\delta n(x) = Ae^{-x/L_n} + Be^{x/L_n}$$

As the minority carrier electrons diffuse away from $x = 0$, they recombine with the majority carrier holes. The minority carrier electron concentration will then decay toward zero at both $x = +\infty$ and $x = -\infty$. These boundary conditions mean that $B = 0$ for $x > 0$ and $A = 0$ for $x < 0$.

The solution may then be written as:

$$\delta n(x) = \delta n(0)\, e^{-x/L_n} \quad \text{for } x \geq 0$$
$$\delta n(x) = \delta n(0)\, e^{+x/L_n} \quad \text{for } x \leq 0$$

or more compactly:

$$\boxed{\delta n(x) = \delta n(0)\, e^{-|x|/L_n}}$$

where $\delta n(0)$ is the value of the excess electron concentration at $x = 0$.

Comment: The steady-state excess concentration decays to $1/e$ of its value at $x = \pm L_n$. The diffusion length $L_n = \sqrt{D_n \tau_{n0}}$ represents the average distance a minority carrier diffuses before recombining.

From charge neutrality, $\delta p(x) = \delta n(x)$, so the steady-state excess majority carrier hole concentration also decays exponentially with distance with the same characteristic minority carrier electron diffusion length L_n.

Under low injection ($\delta n(0) \ll p_0$), the total concentration of majority carrier holes barely changes. However, the minority carrier concentration may change by many orders of magnitude.

Typical values for silicon at 300 K:

- For p-type Si with $N_a = 10^{16}$ cm^{-3}: $D_n \approx 25$ cm^2/s, $\tau_{n0} \approx 10^{-6}$ s $\Rightarrow L_n \approx 50$ μm
- For n-type Si with $N_d = 10^{16}$ cm^{-3}: $D_p \approx 10$ cm^2/s, $\tau_{p0} \approx 10^{-6}$ s $\Rightarrow L_p \approx 32$ μm

13 The PN Junction Diode

13.1 Basic Structure of the PN Junction

A PN junction is formed when a p-type and an n-type semiconductor are brought into contact to form a single-crystal structure. Figure 1 shows a simplified geometry of the PN junction, and Figure 2 illustrates the space charge region and the electric field that forms at the junction.

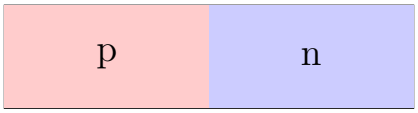


Figure 1: Simplified geometry of a Metallurgical PN junction

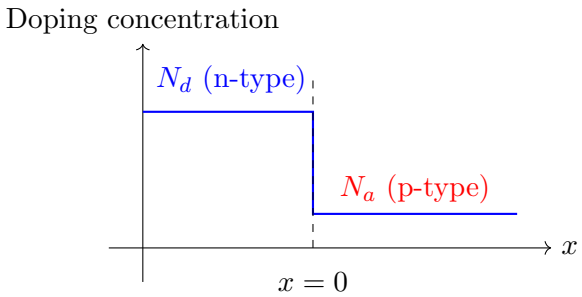


Figure 2: Doping profile of an ideal uniformly doped PN junction

13.2 Zero Applied Bias (Thermal Equilibrium)

When a PN junction is formed, majority carrier electrons in the n region diffuse into the p region, and majority carrier holes in the p region diffuse into the n region. This diffusion leaves behind ionized donors in the n region (positive charge) and ionized acceptors in the p region (negative charge), creating a **space charge region** (also called the **depletion region**) near the metallurgical junction. The electric field created by these fixed charges opposes further diffusion, establishing thermal equilibrium.

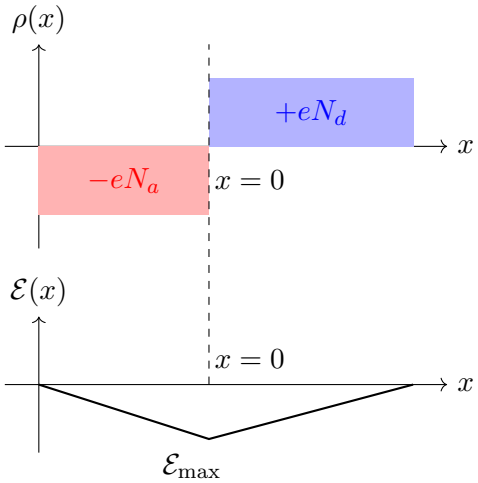


Figure 3: Space charge density and electric field in a uniformly doped PN junction

Energy Band Diagram and Built-in Potential

Figure 4 shows the energy band diagram of a PN junction in thermal equilibrium. The Fermi level is constant throughout the structure, and the conduction and valence bands bend through the space charge region. The built-in potential barrier V_{bi} is the potential difference across the junction that maintains equilibrium.

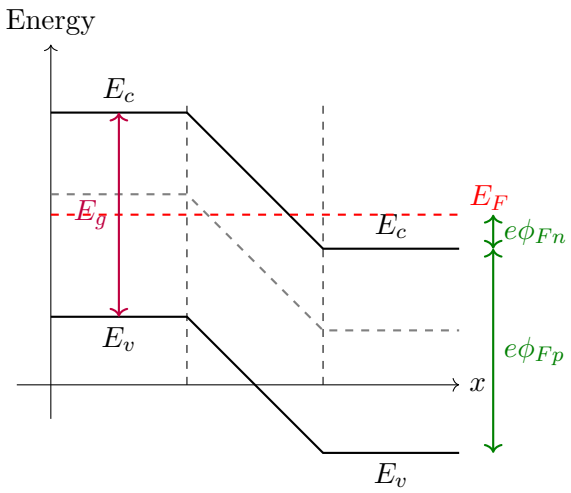


Figure 4: Energy band diagram of a PN junction in thermal equilibrium

Derivation of the Built-in Potential

The built-in potential barrier V_{bi} can be determined from the difference between the intrinsic Fermi levels in the p and n regions. From earlier chapters, we have:

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right), \qquad p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

For the n region, with $n_0 = N_d$, we define:

$$e\phi_{Fn} = E_{Fi} - E_F \quad \Rightarrow \quad N_d = n_i \exp\left(-\frac{e\phi_{Fn}}{kT}\right)$$

Solving for ϕ_{Fn}:

$$\phi_{Fn} = -\frac{kT}{e} \ln\left(\frac{N_d}{n_i}\right)$$

For the p region, with $p_0 = N_a$, we define:

$$e\phi_{Fp} = E_{Fi} - E_F \quad \Rightarrow \quad N_a = n_i \exp\left(\frac{e\phi_{Fp}}{kT}\right)$$

Solving for ϕ_{Fp}:

$$\phi_{Fp} = \frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right)$$

The built-in potential is the sum of the magnitudes of these two potentials:

$$\boxed{V_{bi} = |\phi_{Fn}| + |\phi_{Fp}| = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)}$$

where $V_t = kT/e$ is the thermal voltage (approximately 0.0259 V at 300 K).

Example 6 (Built-in Potential Calculation)**.** Consider a silicon PN junction at $T = 300$ K with doping concentrations $N_a = 2 \times 10^{17}$ cm^{-3} and $N_d = 10^{15}$ cm^{-3}. The intrinsic carrier concentration for silicon is $n_i = 1.5 \times 10^{10}$ cm^{-3}. Calculate the built-in potential.

Solution:

$$V_{bi} = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right) = (0.0259) \ln\left[\frac{(2 \times 10^{17})(10^{15})}{(1.5 \times 10^{10})^2}\right] = 0.713 \text{ V}$$

If the doping in the p region is reduced to $N_a = 10^{16}$ cm^{-3}, then:

$$V_{bi} = 0.0259 \ln \left[\frac{(10^{16})(10^{15})}{(1.5 \times 10^{10})^2} \right] = 0.635 \text{ V}$$

Comment: The built-in potential changes only slightly as doping concentrations change by orders of magnitude due to the logarithmic dependence.

13.3 Electrostatics of the Space Charge Region

The electric field and potential in the space charge region are determined by solving Poisson's equation. For a uniformly doped abrupt junction, the charge density is:

$$\rho(x) = -eN_a \quad (-x_p < x < 0), \qquad \rho(x) = +eN_d \quad (0 < x < x_n)$$

Integrating Poisson's equation yields the electric field distribution (linear through the junction) and the potential distribution (quadratic). The key results are:

- **Charge neutrality condition:** $N_a x_p = N_d x_n$
- **Maximum electric field (at $x = 0$):** $|\mathcal{E}_{\max}| = \dfrac{eN_a x_p}{\epsilon_s} = \dfrac{eN_d x_n}{\epsilon_s}$
- **Built-in potential:** $V_{bi} = \dfrac{e}{2\epsilon_s}(N_a x_p^2 + N_d x_n^2)$

The space charge widths extending into each region are:

$$\boxed{x_n = \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_a}{N_d} \right) \left(\frac{1}{N_a + N_d} \right) \right]^{1/2}}, \qquad \boxed{x_p = \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_d}{N_a} \right) \left(\frac{1}{N_a + N_d} \right) \right]^{1/2}}$$

The total space charge width is:

$$\boxed{W = x_n + x_p = \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}}$$

where:

- ϵ_s = semiconductor permittivity ($\epsilon_s = \epsilon_r \epsilon_0$)
- e = elementary charge (1.602×10^{-19} C)
- N_a, N_d = acceptor and donor concentrations (cm^{-3})
- V_{bi} = built-in potential (V)

For a detailed derivation, students are referred to Neamen's *Semiconductor Physics and Devices*, Chapter 7.

13.4 Reverse Bias

When a positive voltage V_R is applied to the n region with respect to the p region (reverse bias), the potential barrier increases. The total barrier becomes $V_{bi} + V_R$. The space charge width increases as:

$$\boxed{W = \left[\frac{2\epsilon_s (V_{bi} + V_R)}{e} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}}$$

The space charge widths in each region increase by the same factor:

$$x_n(V_R) = x_n(0)\sqrt{1 + \frac{V_R}{V_{bi}}}, \qquad x_p(V_R) = x_p(0)\sqrt{1 + \frac{V_R}{V_{bi}}}$$

The maximum electric field also increases:

$$|\mathcal{E}_{\max}| = \frac{2(V_{bi} + V_R)}{W}$$

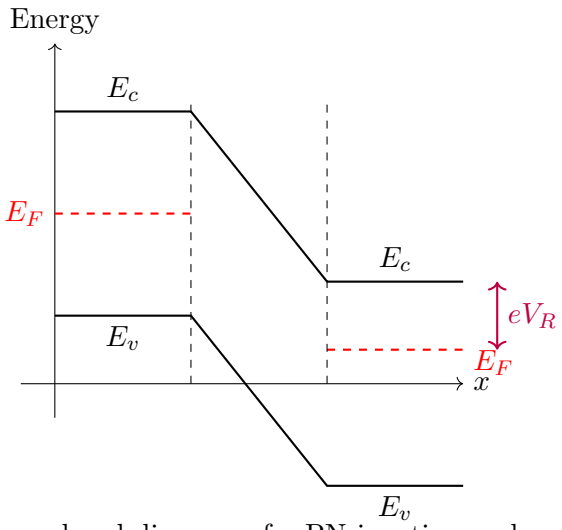


Figure 5: Energy band diagram of a PN junction under reverse bias

13.5 Junction Capacitance

The separation of fixed charges in the depletion region creates a capacitance. When the reverse bias voltage changes by dV_R, the charge stored in the depletion region changes by dQ', giving rise to the **junction capacitance** (also called the **depletion layer capacitance**):

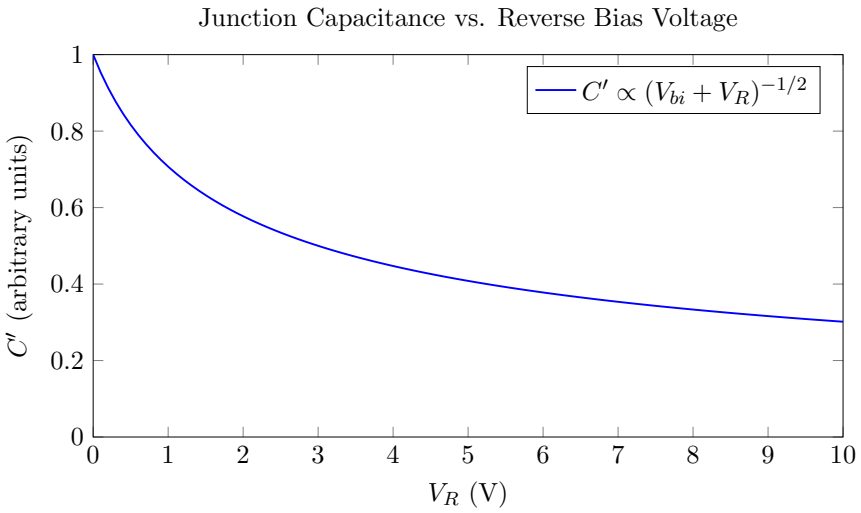
$$\boxed{C' = \frac{dQ'}{dV_R} = \left[\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right]^{1/2}}$$

A useful alternative expression shows that the junction capacitance behaves like a parallel plate capacitor:

$$\boxed{C' = \frac{\epsilon_s}{W}}$$

For a one-sided junction (e.g., p^+n where $N_a \gg N_d$), the capacitance simplifies to:

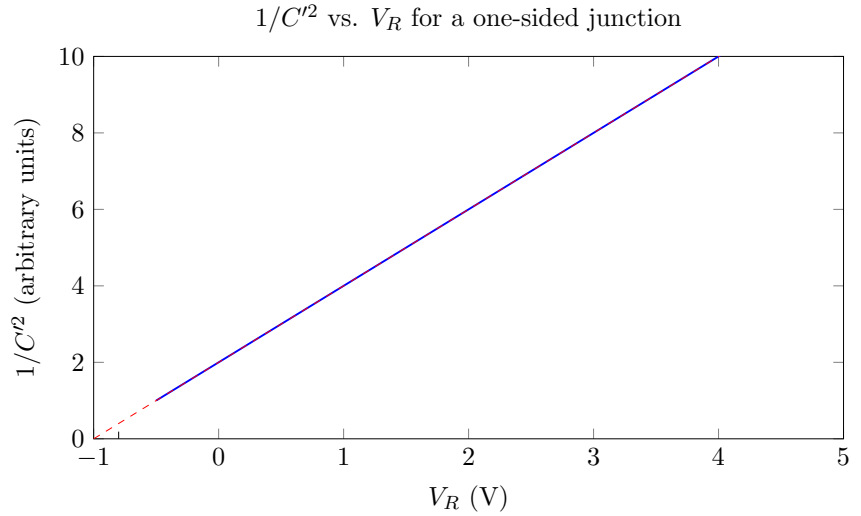
$$\boxed{C' \approx \left[\frac{e\epsilon_s N_d}{2(V_{bi} + V_R)}\right]^{1/2}}$$



Importance of Junction Capacitance

The junction capacitance has several important practical implications:

- **High-frequency operation:** In PN junction diodes (e.g., switching diodes, rectifiers), the junction capacitance limits the maximum operating frequency. When the diode is reverse-biased, the depletion region acts as a capacitor that must be charged and discharged during switching.
- **Voltage-controlled capacitors:** The strong dependence of C' on V_R is exploited in **varactor diodes** (varicap diodes). By varying the reverse bias voltage, the capacitance can be tuned. This property is used in:
  - Voltage-controlled oscillators (VCOs) for frequency modulation
  - Electronic tuning circuits (e.g., radio tuners)
  - Phase-locked loops (PLLs) and frequency synthesizers
- **Device modeling:** The junction capacitance is an essential parameter in SPICE models for diodes and transistors, affecting transient response and AC performance.
- **Characterization of doping profiles:** Measuring C' as a function of V_R allows experimental determination of doping concentrations. A plot of $1/C'^2$ vs. V_R yields a straight line whose slope gives N_d (for a one-sided junction) and whose intercept gives V_{bi}.



Summary of Key Parameters

Parameter	**Expression**
Built-in potential (V_{bi})	$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$
Space charge width (zero bias)	$W = \left[\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$
Space charge width (reverse bias)	$W = \left[\frac{2\epsilon_s (V_{bi} + V_R)}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$
Junction capacitance	$C' = \frac{\epsilon_s}{W} = \left[\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right]^{1/2}$
One-sided junction (p^+n)	$C' \approx \left[\frac{e\epsilon_s N_d}{2(V_{bi} + V_R)}\right]^{1/2}$

The material presented here establishes the electrostatic foundation of the PN junction. In the next chapter, we will extend this analysis to consider current flow under forward bias and derive the diode current-voltage characteristics.

14 Bipolar Junction Transistor (BJT)

14.1 Motivation: From Two-Terminal to Three-Terminal Amplification

In our study of the PN junction, we discovered something remarkable. When we forward-bias the junction, the minority carrier concentration at the edge of the depletion region becomes:

$$\delta p(x_n) = p_{n0}\left(e^{qV/kT} - 1\right)$$

This is an **exponential function** of the applied voltage. A small change in voltage produces a huge change in carrier concentration—and therefore a huge change in current. This is, in essence, an **amplification mechanism**: a small input signal (voltage) controls a much larger output signal (current).

The Problem: No Third Terminal

If a single PN junction can amplify, why do we need a more complex device? The answer lies in **circuit practicality**:

- A two-terminal device (diode) has the same two terminals for both input and output
- You cannot independently control the input and monitor the output
- There is no **ground** or reference terminal to separate input from output
- The input signal and output signal are intrinsically mixed together

What we need is a **three-terminal device** where:

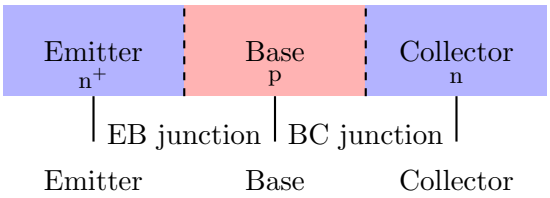
- One terminal serves as the input (signal applied here)
- Another terminal serves as the output (amplified signal taken from here)
- The third terminal serves as a common reference (ground)

This allows the input and output circuits to be separate while still allowing the input to control the output. This is exactly what the **Bipolar Junction Transistor (BJT)** provides—a **controlled PN junction**.

14.2 Device Structure: Three Layers, Two Junctions

The BJT consists of three alternating layers of doped semiconductor, forming two back-to-back PN junctions. There are two types:

- **npn transistor**: emitter and collector are n-type, base is p-type
- **pnp transistor**: emitter and collector are p-type, base is n-type



npn BJT Structure

14.3 The Four Modes of Operation

With two junctions (emitter-base and base-collector), each can be either forward-biased (FB) or reverse-biased (RB). This gives four distinct operating modes:

Mode	EB Junction	BC Junction	Application
Active (Forward)	Forward bias	Reverse bias	**Amplification**
Cutoff	Reverse bias	Reverse bias	Switch (OFF)
Saturation	Forward bias	Forward bias	Switch (ON)
Reverse Active	Reverse bias	Forward bias	Rarely used

14.4 Understanding the Base Current: Recombination and Charge Neutrality

A critical question: **Where does the base current come from?**

In the active mode, the emitter injects electrons into the base (for npn). These electrons are minority carriers in the p-type base. As they diffuse across the base toward the collector, some of them recombine with holes (the majority carriers in the base).

Charge Neutrality and Base Current Components

When an electron recombines with a hole in the base, that hole is **annihilated**. To maintain charge neutrality in the base, the missing hole must be replenished. This is accomplished by the **base current** I_B.

The base current in a modern BJT has two physical components:

1. **Recombination component:** Holes supplied to replace those lost to recombination with injected electrons.
2. **Back-injection component:** Holes injected from the base into the emitter (dominant in modern, high-efficiency transistors).

In well-designed BJTs, the back-injection component often dominates I_B, though the recombination model is pedagogically useful and captures the essential physics. The key point remains: I_B is small because both loss mechanisms are small relative to the injected current.

The base current I_B is therefore **proportional to the net rate of hole loss** from the base:

$$I_B = \frac{Q_b}{\tau_b} = \frac{\text{stored charge in base}}{\text{effective lifetime}}$$

Since I_B is typically much smaller than I_C, this means **most injected electrons reach the collector without being lost**.

14.5 Direction of Base Current: Why Holes, Not Electrons?

A common source of confusion is understanding why the base current consists of **holes entering the base** (for npn) rather than electrons. The answer lies in understanding **which terminal supplies which carrier**.

14.5.1 What Carriers Are Lost in the Base?

In an npn BJT in active mode, electrons are injected from the emitter into the base, and holes are the majority carriers in the base. When an electron recombines with a hole, **both carriers are lost** from the

base. Each lost carrier must be replenished by the terminal that is designed to supply that type of carrier.

14.5.2 Who Replaces the Lost Carriers?

The lost electron originally came from the emitter. The emitter is n-type and its ohmic contact is designed to supply electrons. Therefore, the emitter current I_E continuously supplies new electrons to the base, replacing both those that recombine and those that are collected. The lost hole originally came from the base itself. The base is p-type and its ohmic contact is designed to supply holes. Therefore, the base current I_B supplies holes to replenish those lost to recombination.

Why No Electron Current Enters the Base

The base contact is an **ohmic contact to p-type material**. Such a contact:

- Supplies holes easily (majority carriers)
- Does not supply electrons efficiently (electrons are minority carriers in the base)

Therefore, the base current is carried by **holes entering the base**, not by electrons. The electrons lost to recombination are replaced by the **emitter**, not by the base.

14.5.3 Resulting Current Directions for NPN in Active Mode

For an npn transistor operating in the forward active mode, the terminal currents flow as follows:

- **Emitter current** I_E flows **out of** the emitter terminal. This outward flow represents electrons leaving the emitter and entering the base (conventional current opposite to electron flow).
- **Base current** I_B flows **into** the base terminal. This inward flow represents holes entering the base from the external circuit to replenish those lost to recombination.
- **Collector current** I_C flows **into** the collector terminal. This inward flow represents electrons being collected from the base and entering the collector.

These directions satisfy Kirchhoff's current law:

$$\boxed{I_E = I_B + I_C}$$

with all three currents taken as positive quantities.

Important: Sign Convention and Base Current Direction

The analysis above uses the **conventional current direction** (positive charge flow).

- In an **npn BJT**: The base is p-type. Forward bias of the EB junction **enables hole injection from base to emitter** (back-injection) alongside dominant electron injection from emitter to base. Therefore, conventional current **enters** the base terminal (I_B into base) to maintain charge neutrality.
- In a **pnp BJT**: The base is n-type. Forward bias of the EB junction **enables electron injection from base to emitter** alongside dominant hole injection from emitter to base. Conventional current (positive charge flow) is opposite to electron flow. Therefore, conventional current **leaves** the base terminal (I_B out of base).

Physical carriers:

- npn: Holes enter the base terminal (same direction as conventional current).
- pnp: Electrons enter the base terminal (opposite to conventional current direction).

When using other textbooks or circuit simulators, always verify the sign convention being used. The equations $I_E = I_B + I_C$ and $I_C = \beta I_B$ hold with **all currents taken as positive** using the directions defined in these notes.

14.6 Physical Mechanism of Amplification

The amplification mechanism proceeds as follows:

1. **Emitter injection:** The forward-biased EB junction injects electrons from emitter into base. The injected electron concentration at the edge of the base is:

$$\delta n_b(0) = n_{b0}\left(e^{qV_{BE}/kT} - 1\right) \approx n_{b0}e^{qV_{BE}/kT}$$

   This is the same exponential relationship we saw in the PN junction.

2. **Diffusion across the base:** The base is extremely thin ($W_b \ll L_n$). Most injected electrons diffuse to the base-collector junction without recombining.

3. **Collection:** The reverse-biased BC junction creates a strong electric field that sweeps the diffusing electrons into the collector. The collector current is:

$$I_C = \alpha I_E \quad \text{where } \alpha = \gamma \cdot \alpha_T \approx 0.95 - 0.99$$

   Here γ is the emitter injection efficiency and α_T is the base transport factor.

4. **Base current:** The small fraction of electrons that recombine in the base, plus holes back-injected into the emitter, must be replaced. This constitutes the base current:

$$I_B = I_E - I_C$$

5. **Current gain:** Because $I_C = \alpha I_E$ and $I_B = (1-\alpha)I_E$:

$$\boxed{\beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha}}$$

   where $\alpha = I_C/I_E$ is the **common-base current gain** (typically $0.95 - 0.99$).

6. **Amplification in action (for small signals):** A small change in base current produces a large change in collector current:

$$\boxed{i_c = \beta\, i_b}$$

This is the fundamental amplifier equation of the BJT, valid for small-signal variations around a DC operating point.

Decomposition of α into γ and α_T

The common-base current gain α can be decomposed as:

$$\alpha = \gamma \cdot \alpha_T$$

where:

- $\gamma = \dfrac{I_{E,n}}{I_E}$ is the **emitter injection efficiency**: the fraction of emitter current carried by electrons injected into the base (desired component).
- $\alpha_T \approx 1 - \dfrac{W_b^2}{2L_n^2}$ is the **base transport factor**: the fraction of injected electrons that reach the collector without recombining.

This decomposition helps in understanding the physical factors limiting current gain:

- To maximize β, we need $\gamma \to 1$ (emitter much more heavily doped than base) and $\alpha_T \to 1$ (base very thin compared to diffusion length).

The Role of Base Width

The transport factor α_T depends on base width:

$$\alpha_T \approx 1 - \frac{W_b^2}{2L_n^2}$$

where $L_n = \sqrt{D_n \tau_n}$ is the diffusion length. This shows:

- **Thinner base** (W_b small) $\to$ less recombination $\to$ higher $\alpha \to$ higher β
- Typical base widths are $0.1 - 1\mu$m, much smaller than L_n (typically $10 - 100\mu$m)
- This is why the base must be extremely thin for good amplification

14.7 Understanding Saturation: The ON State of the Switch

What is $V_{CE,sat}$? $V_{CE,sat}$ is the **collector-emitter saturation voltage**—the voltage across the transistor when it is fully turned on. It is not a fixed constant but depends on operating conditions such as collector current and base current.

Physical explanation of saturation: When the base-collector junction becomes **forward-biased** (as well as the EB junction), the electrostatic conditions change fundamentally:

1. In the active mode, the reverse-biased BC junction creates a depletion region that sweeps carriers from base to collector.
2. Under saturation, the forward bias of the BC junction reduces this electric field. The depletion region shrinks, and the built-in potential barrier at the BC junction is partially overcome.

3. Without a strong collecting field, carriers that diffuse to the base-collector boundary are no longer efficiently swept into the collector. Some may even be injected backward from collector into base.

4. The collector current becomes limited by the external circuit rather than by the base current. The collector-emitter voltage drops to a small value:

$$V_{CE,sat} = V_{BE,sat} - V_{BC,sat}$$

For silicon, typical values are $V_{CE,sat} \approx 0.1 - 0.3$ V, depending on current level and device geometry.

The Electrostatics of Saturation

In saturation:

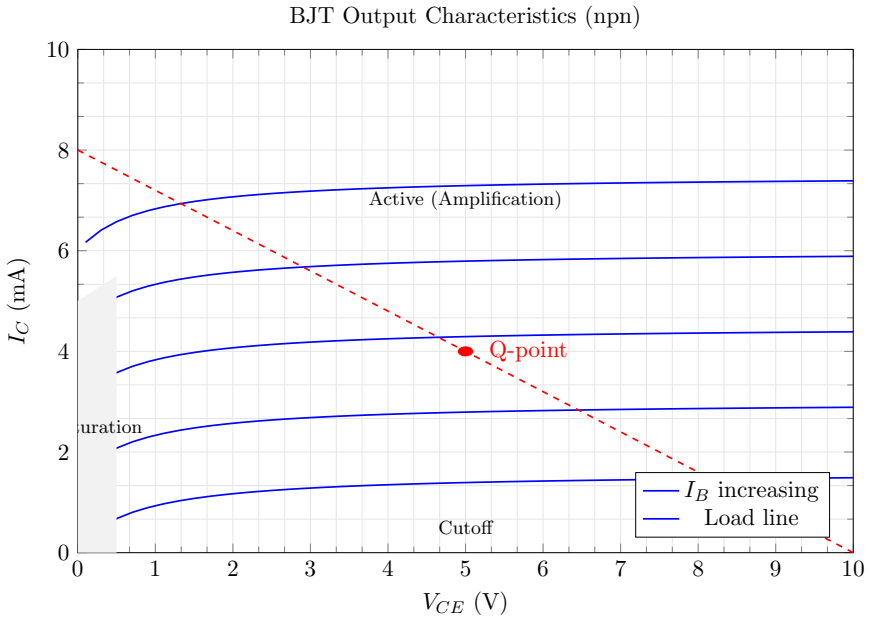
- Both junctions are forward-biased
- The base-collector depletion region is greatly reduced
- Carriers can flow from emitter to collector, but also from collector to base (reverse injection)
- The collector current is determined primarily by the external circuit (V_{CC} and R_C), not by I_B
- V_{CE} is the difference between two forward-biased diode voltages:

$$V_{CE} = V_{BE} - V_{BC}$$

- The transistor behaves as a closed switch: small voltage drop (typically $0.1 - 0.3$ V), large current determined by the external circuit

14.8 BJT Output Characteristics

The family of curves shows I_C as a function of V_{CE} for different base currents I_B.



Three distinct regions are visible:

- **Cutoff region ($V_{BE} < 0.6$ V)**: Both junctions reverse-biased. $I_C \approx 0$, transistor is OFF. This is the **open switch** state.

- **Active region ($V_{CE} > V_{CE,sat}$)**: EB forward, BC reverse. Here $I_C = \beta I_B$, with a weak dependence on V_{CE} due to the **Early effect** (base-width modulation). This is the **amplification region**.
- **Saturation region ($V_{CE} < V_{CE,sat}$)**: Both junctions forward-biased. I_C is no longer controlled by I_B; the transistor behaves like a **closed switch**.

14.9 The Load Line: Finding the Operating Point

When a BJT is used in an amplifier circuit, it is biased with resistors and power supplies. The **load line** represents all possible combinations of I_C and V_{CE} allowed by the external circuit:

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

Choosing the Q-Point for Amplification

The intersection of the load line with the transistor's output characteristic for a given I_B is the **Q-point** (quiescent point). For linear amplification:

- The Q-point should be centered in the active region
- Far enough from cutoff to avoid distortion of negative signal swings
- Far enough from saturation to avoid distortion of positive signal swings
- The signal should not drive the transistor into cutoff or saturation

For small-signal variations around the DC bias point:

- The base current varies around its DC value ($I_B \pm i_b$)
- The collector current varies around its DC value ($I_C \pm i_c$)
- The voltage v_{CE} varies according to the load line
- The small-signal current gain is $i_c/i_b = \beta$

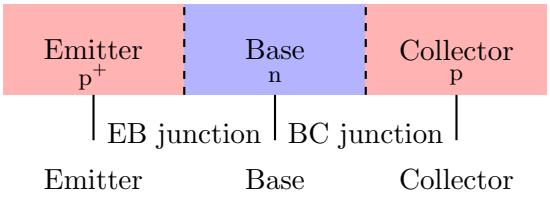
14.10 The pnp Bipolar Junction Transistor

The pnp BJT is the complementary counterpart to the npn transistor. All physical principles remain the same, but the doping types and therefore the polarity of voltages and directions of currents are reversed.

Device Structure

In a pnp transistor:

- Emitter: **p-type** (heavily doped, p^+)
- Base: **n-type** (lightly doped)
- Collector: **p-type**



pnp BJT Structure

Active Mode Biasing (Amplification)

For the pnp transistor to operate in the forward active mode (amplification region), the biasing conditions are opposite to those of the npn:

- **Emitter-Base junction: Forward-biased**. This requires the emitter to be at a higher potential than the base: $V_{EB} > 0$ (or $V_{BE} < 0$). For a silicon pnp, $V_{EB} \approx 0.7$ V.
- **Base-Collector junction: Reverse-biased**. This requires the collector to be at a lower potential than the base: $V_{BC} < 0$ (or $V_{CB} > 0$).

Carrier Flow in the pnp Transistor (Active Mode)

The physical mechanism is the same as for the npn, but with holes replacing electrons as the injected carriers:

1. **Emitter injection:** Holes injected from emitter into base.
2. **Diffusion across the base:** Most injected holes reach the BC junction.
3. **Collection:** Reverse-biased BC junction sweeps holes into collector.
4. **Base current:** Lost electrons (from recombination with injected holes) are replenished by electrons entering the base from the external circuit.

Base Current Direction in pnp

In a pnp transistor:

- The base is n-type. Its ohmic contact supplies electrons (majority carriers).
- When a hole recombines with an electron in the base, an electron is lost.
- To replenish the lost electron, the external base current supplies **electrons entering the base**.

By conventional current convention (positive charge flow), electrons entering the base correspond to conventional current **leaving** the base. Therefore, for a pnp in active mode, the base current I_B flows **out of** the base terminal.

Current Directions for pnp in Active Mode

Summarizing the terminal current directions for a pnp in active mode:

- **Emitter current** I_E flows **into** the emitter terminal.
- **Base current** I_B flows **out of** the base terminal.
- **Collector current** I_C flows **out of** the collector terminal.

These directions satisfy Kirchhoff's current law $I_E = I_B + I_C$ with all currents taken as positive.

Key Equations for pnp (Active Mode)

The mathematical relationships for the pnp are identical in form to those of the npn, provided the voltages are defined with the correct polarity.

Key pnp BJT Equations for Active Mode

Current relationships:

$$\boxed{I_E = I_B + I_C}, \qquad \boxed{I_C = \beta I_B}, \qquad \boxed{I_C = \alpha I_E}$$

Current gain relationships:

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}, \qquad \boxed{\alpha = \frac{\beta}{\beta+1}}$$

Shockley equation for active region (using V_{EB}):

$$\boxed{I_C = I_S\left(e^{V_{EB}/V_T} - 1\right)} \approx I_S e^{V_{EB}/V_T} \quad (e^{V_{EB}/V_T} \gg 1)$$

Emitter-base diode equation:

$$\boxed{I_B = \frac{I_S}{\beta}\left(e^{V_{EB}/V_T} - 1\right)} \approx \frac{I_S}{\beta} e^{V_{EB}/V_T}$$

Note: $V_T = kT/q = 0.0259$ V at 300 K. For a pnp, V_{EB} (emitter-to-base voltage) is positive in active mode. $V_{EB} \approx 0.7$ V for silicon.

14.11 Summary

Key BJT Equations for Active Mode

Current relationships (same form for npn and pnp, with appropriate sign conventions):

$$\boxed{I_E = I_B + I_C}, \qquad \boxed{I_C = \beta I_B}, \qquad \boxed{I_C = \alpha I_E}$$

Current gain relationships:

$$\boxed{\beta = \frac{\alpha}{1-\alpha}}, \qquad \boxed{\alpha = \frac{\beta}{\beta+1}}$$

Shockley equation (npn: V_{BE}, pnp: V_{EB}):

$$\boxed{I_C = I_S\left(e^{V_{BE}/V_T} - 1\right)} \approx I_S e^{V_{BE}/V_T} \quad \text{(npn)}$$

$$\boxed{I_C = I_S\left(e^{V_{EB}/V_T} - 1\right)} \approx I_S e^{V_{EB}/V_T} \quad \text{(pnp)}$$

Emitter-base diode equation:

$$\boxed{I_B = \frac{I_S}{\beta}\left(e^{V_{BE}/V_T} - 1\right)} \approx \frac{I_S}{\beta} e^{V_{BE}/V_T} \quad \text{(npn)}$$

$$\boxed{I_B = \frac{I_S}{\beta}\left(e^{V_{EB}/V_T} - 1\right)} \approx \frac{I_S}{\beta} e^{V_{EB}/V_T} \quad \text{(pnp)}$$

Thermal voltage: $V_T = kT/q = 0.0259$ V at 300 K.

Current Directions in Active Mode (All Currents Positive)

For an npn transistor:

- Emitter current I_E **leaves** the emitter terminal.
- Base current I_B **enters** the base terminal.
- Collector current I_C **enters** the collector terminal.

For a pnp transistor:

- Emitter current I_E **enters** the emitter terminal.
- Base current I_B **leaves** the base terminal.
- Collector current I_C **leaves** the collector terminal.

Kirchhoff's current law $I_E = I_B + I_C$ holds for both.

Physical Insights

In an npn transistor, the emitter injects electrons into the base. The base current consists of holes entering the base to replenish holes lost to recombination (plus holes back-injected into the emitter). In a pnp transistor, the emitter injects holes into the base. The base current consists of electrons entering the base to replenish electrons lost to recombination (plus electrons back-injected into the emitter). In both devices, the base current is small because only a small fraction of injected carriers are lost. Most injected carriers reach the collector, giving $\beta \gg 1$.

Operating Regions

Region	EB Junction	BC Junction	Application
Cutoff	Reverse	Reverse	Switch (OFF)
Active (Forward)	Forward	Reverse	Amplification
Saturation	Forward	Forward	Switch (ON)
Reverse Active	Reverse	Forward	Rarely used

Circuit Symbol Convention

The arrow on the emitter points in the direction of conventional current when the EB junction is forward-biased. For an npn transistor, the arrow points outward from the emitter. For a pnp transistor, the arrow points inward toward the emitter.

The BJT solves the two-terminal amplification problem by providing a third terminal (the base) that controls the current between emitter and collector. The base current is small because only a small fraction of injected carriers are lost. This small control current I_B regulates a much larger output current I_C, giving β typically between 50 and 500. The transistor's operating regions (cutoff, active, saturation, and reverse active) enable both linear amplifiers and digital switches, making the BJT one of the most versatile devices in electronics.

15 The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

15.1 Motivation: Voltage-Controlled Switching and Amplification

The BJT is a powerful amplifier, but it has inherent limitations. The base current, while small, still flows into the device. This input current loads the preceding stage, consumes power, and limits input impedance. Furthermore, BJTs require a continuous base current to remain in the ON state, which is problematic for low-power digital logic where millions of transistors must switch efficiently.

What if we could build a three-terminal device where the input terminal draws **no DC current**? Such a device would have infinite input impedance, consume negligible input power, and be ideal for both high-impedance sensing and low-power digital switching. This is precisely what the **Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)** provides.

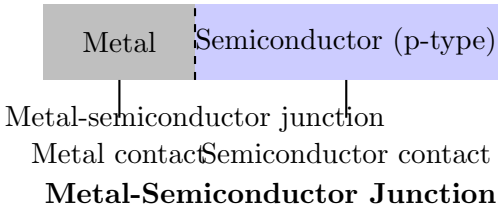
The Key Insight: Voltage Control, Not Current Control

The BJT is a **current-controlled current source**: $I_C = \beta I_B$.
The MOSFET is a **voltage-controlled current source**: $I_D = f(V_{GS})$.
This distinction is profound:

- In a BJT, the input terminal (base) draws current. You must supply power to control the device.
- In a MOSFET, the input terminal (gate) is electrically isolated by an insulating oxide layer. It draws **no DC current** (only a tiny transient current to charge the gate capacitance).
- This makes MOSFETs ideal for ultra-high-input-impedance amplifiers and for CMOS logic, where power is consumed only during switching.

15.2 The Metal-Semiconductor Junction: A Necessary Prelude

Before we can understand the MOSFET, we must understand the **metal-semiconductor junction**—the interface between a metal contact and the semiconductor that forms the gate terminal. This junction is fundamentally different from a PN junction.



Metal-Semiconductor Junction

15.2.1 Schottky Junction vs. Ohmic Contact

Depending on the metal work function and semiconductor doping, the metal-semiconductor junction can be one of two types:

- **Schottky junction (rectifying)**: Forms a potential barrier at the interface. Current flows easily in one direction (forward bias) but is strongly blocked in the opposite direction (reverse bias). This behaves like a diode, but with a lower forward voltage drop (≈ 0.3 V for silicon) than a PN junction (≈ 0.7 V).
- **Ohmic contact (non-rectifying)**: The barrier is so thin that carriers can tunnel through it regardless of bias direction. The junction exhibits linear current-voltage behavior: $I = V/R$. This is essential for making electrical connections to semiconductor devices without introducing unwanted diodes.

The MOS Structure: Metal-Oxide-Semiconductor

In a MOSFET, the gate terminal is a metal (or heavily doped polysilicon) layer separated from the semiconductor by a thin insulating layer of **silicon dioxide (SiO_2)**. This is not a metal-semiconductor junction—it is a **metal-insulator-semiconductor (MIS)** capacitor.
The key difference: The oxide layer is an **insulator**. No DC current can flow from the gate into the semiconductor. The gate voltage creates an **electric field** that penetrates through the oxide and modulates the carrier concentration in the semiconductor **without any gate current**.
This is the fundamental principle of the MOSFET: voltage control with zero DC input current.

15.3 The MOS Capacitor and Gate Oxide Capacitance

Before analyzing the complete MOSFET, we must understand the **MOS capacitor**—the gate structure itself. The gate, oxide, and semiconductor form a parallel-plate capacitor where the semiconductor acts as one plate whose charge density can be modulated.

15.3.1 Gate Oxide Capacitance per Unit Area

The gate oxide capacitance per unit area is defined as:

$$\boxed{C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}}$$

where:

- $\epsilon_{ox} = \epsilon_r \epsilon_0$ is the permittivity of the oxide
  - $\epsilon_0 = 8.854 \times 10^{-14}$ F/cm (permittivity of free space)
  - $\epsilon_r \approx 3.9$ for SiO_2 (relative permittivity)
  - Therefore $\epsilon_{ox} \approx 3.45 \times 10^{-13}$ F/cm for SiO_2
- t_{ox} is the oxide thickness (typically 1–10 nm in modern devices)

Example: Calculating C_{ox}

For a typical MOSFET with $t_{ox} = 2$ nm $= 2 \times 10^{-7}$ cm:

$$C_{ox} = \frac{3.45 \times 10^{-13} \text{ F/cm}}{2 \times 10^{-7} \text{ cm}} \approx 1.73 \times 10^{-6} \text{ F/cm}^2 = 1.73 \ \mu\text{F/cm}^2$$

In more convenient units: $C_{ox} \approx 17.3$ fF/μm^2.
Key insight: Thinner oxide → higher C_{ox} → stronger gate control → higher drive current. This is why modern transistors use extremely thin oxides (or high-κ dielectrics).

15.3.2 Total Gate Capacitance

For a MOSFET with channel width W and length L, the total gate capacitance is:

$$C_G = C_{ox} \cdot W \cdot L$$

This capacitance must be charged and discharged during switching, determining the dynamic power consumption and switching speed.

15.4 Work Function and Flat-Band Voltage

To understand the threshold voltage, we must first define the work function and flat-band conditions.

15.4.1 Definition of Work Function

The **work function** ϕ of a material is the minimum energy required to remove an electron from the Fermi level to a point immediately outside the material (the vacuum level).

$$\boxed{\phi = E_{\text{vac}} - E_F}$$

- **Metals**: The work function is a well-defined material constant:
  - Aluminum: $\phi_m \approx 4.1$ eV
  - Gold: $\phi_m \approx 5.1$ eV
  - Polysilicon (n^+): $\phi_m \approx 4.0$ eV
  - Polysilicon (p^+): $\phi_m \approx 5.2$ eV
- **Semiconductors**: The work function is **doping-dependent** because the Fermi level position changes with doping:

  $$\phi_s = \chi + \frac{E_g}{2} \pm \phi_F$$

  where χ is the electron affinity, E_g is the bandgap, and ϕ_F is the Fermi potential (positive for p-type, negative for n-type).

15.4.2 Work Function Difference

The work function difference between the gate metal and semiconductor is:

$$\phi_{ms} = \phi_m - \phi_s$$

This difference creates a built-in potential that must be compensated by the gate voltage even before any external bias is applied.

15.4.3 Fermi Potential

The **Fermi potential** ϕ_F is the potential difference between the intrinsic Fermi level E_i and the actual Fermi level E_F:

$$\boxed{\phi_F = \frac{kT}{q} \ln\left(\frac{N_a}{n_i}\right)} \quad \text{(for p-type)}$$

$$\boxed{\phi_F = -\frac{kT}{q} \ln\left(\frac{N_d}{n_i}\right)} \quad \text{(for n-type)}$$

where:

- N_a, N_d = acceptor/donor doping concentration
- n_i = intrinsic carrier concentration ($\approx 1.5 \times 10^{10}$ cm^{-3} for Si at 300 K)
- $kT/q \approx 0.0259$ V at 300 K

For typical p-type substrate with $N_a = 10^{16}$ cm^{-3}:

$$\phi_F \approx 0.0259 \ln\left(\frac{10^{16}}{1.5 \times 10^{10}}\right) \approx 0.35 \text{ V}$$

15.4.4 Flat-Band Voltage

The **flat-band voltage** V_{FB} is the gate voltage required to make the energy bands in the semiconductor flat (no band bending), compensating for the work function difference and any oxide charge:

$$\boxed{V_{FB} = \phi_{ms} - \frac{Q_{ox}}{C_{ox}}}$$

where Q_{ox} is the fixed oxide charge per unit area (typically positive for SiO_2).
In ideal conditions ($Q_{ox} = 0$): $V_{FB} = \phi_{ms}$.

15.5 The Threshold Voltage

The threshold voltage V_{TH} is the gate-to-source voltage at which the semiconductor surface **inverts** and a conducting channel forms. At threshold, the surface potential $\psi_s = 2\phi_F$.

15.5.1 Threshold Voltage Formula

$$\boxed{V_{TH} = V_{FB} + 2\phi_F + \frac{\sqrt{4\epsilon_s q N_a \phi_F}}{C_{ox}}}$$

where:

- V_{FB} = flat-band voltage (defined above)
- ϕ_F = Fermi potential of the substrate
- ϵ_s = permittivity of semiconductor ($\approx 1.04 \times 10^{-12}$ F/cm for Si)
- q = electron charge (1.6×10^{-19} C)
- N_a = substrate doping concentration (for nMOS)
- C_{ox} = gate oxide capacitance per unit area (defined earlier)

The three terms represent:

1. V_{FB}: Voltage to compensate work function difference
2. $2\phi_F$: Voltage to bend bands to inversion condition
3. $\frac{\sqrt{4\epsilon_s q N_a \phi_F}}{C_{ox}}$: Voltage to create depletion charge

Typical Threshold Voltages

- **nMOS** (n-channel on p-substrate): $V_{TH} \approx +0.4$ to $+0.7$ V
- **pMOS** (p-channel on n-substrate): $V_{TH} \approx -0.4$ to -0.7 V

The sign convention: V_{TH} is positive for nMOS (need positive V_{GS} to turn on) and negative for pMOS (need negative V_{GS} to turn on).

15.5.2 Body Effect (Substrate Bias Effect)

When the source-to-body voltage $V_{SB} \neq 0$, the threshold voltage changes:

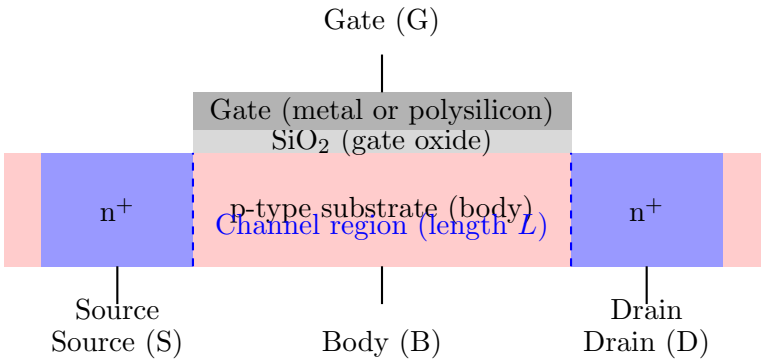
$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F} \right)$$

where:

- V_{TH0} = threshold voltage at $V_{SB} = 0$
- $\gamma = \frac{\sqrt{2q\epsilon_s N_a}}{C_{ox}}$ = body-effect coefficient

Physical insight: Reverse-biasing the source-body junction increases the depletion width, requiring more gate voltage to achieve inversion.

15.6 MOSFET Structure and Operation



n-Channel MOSFET (nMOS) Structure

The MOSFET has four terminals:

- **Gate (G)**: Controls the transistor. Voltage applied here creates an electric field that modulates the channel conductivity.
- **Source (S)**: The terminal through which carriers enter the channel.
- **Drain (D)**: The terminal through which carriers exit the channel.
- **Body (B)**: The substrate (usually connected to the source for simplicity).

15.7 Physical Intuition: Channel Inversion

The term "inversion" captures the essence of MOSFET operation. In a p-type substrate, the majority carriers are holes. When a positive voltage is applied to the gate, the electric field penetrates the semiconductor and repels these holes away from the surface.

As holes are pushed away, a depletion region forms. When the gate voltage exceeds the threshold voltage V_{TH}, the electric field becomes strong enough to attract **minority carriers** (electrons) to the surface. The surface region **inverts** from p-type to n-type, creating a thin, continuous layer of mobile electrons. This layer is the **channel** that connects source to drain.

15.8 Physical Intuition: Pinch-Off and Current Saturation

Once the channel is formed, applying a drain voltage V_{DS} causes current to flow from drain to source. However, the channel is not uniform along its length.

Why does the channel thin out toward the drain? The voltage along the channel increases from 0 V at the source to V_{DS} at the drain. The effective gate-to-channel voltage is $V_{GS} - V(x)$, where $V(x)$ is the voltage at position x. Near the drain, $V(x)$ is largest, so the local overdrive voltage $V_{GS} - V(x) - V_{TH}$ is smallest, resulting in a thinner inversion layer.

When V_{DS} reaches $V_{GS} - V_{TH}$, the channel at the drain end is **pinched off**—the inversion charge goes to zero. For $V_{DS} > V_{GS} - V_{TH}$, the pinch-off point moves slightly toward the source, but the voltage drop across the remaining channel remains approximately constant at $V_{GS} - V_{TH}$. Electrons that reach the pinch-off point are swept into the drain by the strong electric field in the high-field region. The drain current saturates because further increases in V_{DS} do not increase the voltage drop across the channel.

15.9 MOSFET Operating Regions: A Practical Guide

Students often struggle to determine which formula to apply. Here is a simple decision tree:

How to Determine the MOSFET Operating Region

1. **Step 1: Check if the transistor is ON or OFF.**
   - If $V_{GS} < V_{TH}$: Transistor is in **cutoff**. $I_D = 0$.
   - If $V_{GS} \geq V_{TH}$: Transistor is ON. Proceed to Step 2.
2. **Step 2: Compare V_{DS} with $V_{GS} - V_{TH}$ (the overdrive voltage).**
   - If $V_{DS} < V_{GS} - V_{TH}$: Transistor is in **linear (triode) region**. Use:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

   - If $V_{DS} \geq V_{GS} - V_{TH}$: Transistor is in **saturation region**. Use:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

Quick Reference

$V_{GS} < V_{TH} \rightarrow$ Cutoff ($I_D = 0$)
$V_{GS} \geq V_{TH}$ and $V_{DS} < V_{GS} - V_{TH} \rightarrow$ Linear region
$V_{GS} \geq V_{TH}$ and $V_{DS} \geq V_{GS} - V_{TH} \rightarrow$ Saturation region

15.10 p-Channel MOSFET (pMOS)

The p-channel MOSFET is the complementary counterpart to the nMOS transistor. All physical principles remain the same, but the doping types, carrier types, and voltage polarities are reversed.

15.10.1 Structure

- **Substrate**: n-type (instead of p-type)
- **Source and Drain**: p^+ regions (instead of n^+)
- **Channel carriers**: Holes (instead of electrons)
- **Gate oxide and gate**: Same structure as nMOS

15.10.2 Operating Principles

- **To turn ON**: Apply $V_{GS} < V_{TH}$ (negative gate voltage relative to source)
- **Threshold voltage**: $V_{TH} < 0$ (typically -0.4 to -0.7 V)
- **Current flow**: Holes flow from source to drain (conventional current from source to drain)
- **Mobility**: Hole mobility $\mu_p \approx 2-3\times$ lower than electron mobility μ_n

15.10.3 Current-Voltage Equations for pMOS

Using the convention that all currents and voltages are positive quantities:

Cutoff ($V_{SG} < |V_{TH}|$):

$$I_D = 0$$

Linear region ($V_{SG} \geq |V_{TH}|$ and $V_{SD} < V_{SG} - |V_{TH}|$):

$$I_D = \mu_p C_{ox} \frac{W}{L}\left[(V_{SG} - |V_{TH}|)V_{SD} - \frac{V_{SD}^2}{2}\right]$$

Saturation ($V_{SG} \geq |V_{TH}|$ and $V_{SD} \geq V_{SG} - |V_{TH}|$):

$$I_D = \frac{1}{2}\mu_p C_{ox} \frac{W}{L}(V_{SG} - |V_{TH}|)^2$$

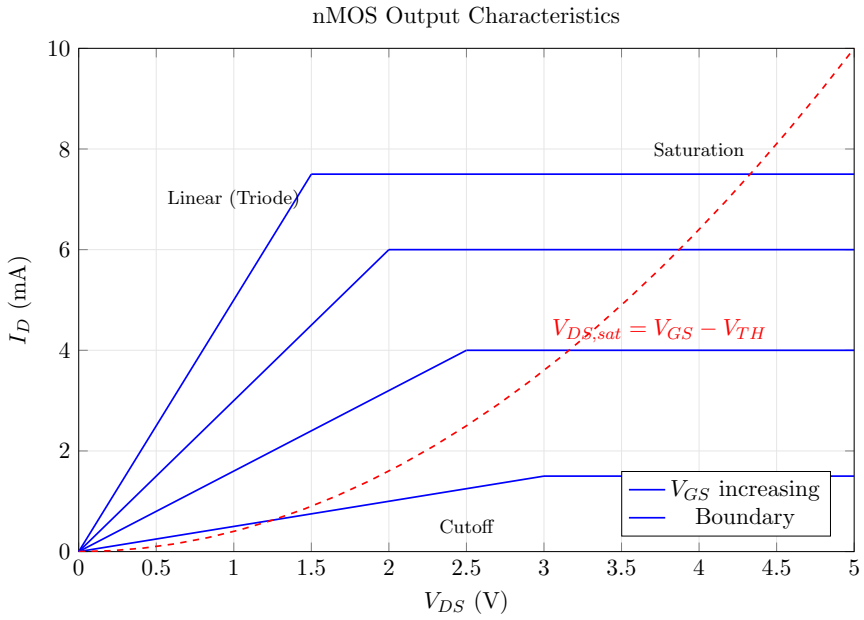
where $V_{SG} = V_S - V_G$ and $V_{SD} = V_S - V_D$ (source-referenced voltages).

nMOS vs. pMOS: Key Differences

Property	nMOS	pMOS
Substrate type	p-type	n-type
Source/Drain doping	n^+	p^+
Channel carriers	Electrons	Holes
Threshold voltage V_{TH}	Positive ($+0.4$ to $+0.7$ V)	Negative (-0.4 to -0.7 V)
To turn ON	$V_{GS} > V_{TH}$	$V_{GS} < V_{TH}$ (or $V_{SG} > \|V_{TH}\|$)
Carrier mobility	$\mu_n \approx 600$ cm^2/V·s	$\mu_p \approx 200$ cm^2/V·s
Relative drive strength	$1\times$ (reference)	$\approx 1/2$ to $1/3$ of nMOS

Design implication: To achieve equal drive strength, pMOS transistors are typically made 2–3× wider than nMOS transistors ($W_p \approx 2-3W_n$).

15.11 MOSFET Output Characteristics



15.12 Key MOSFET Equations Summary

MOSFET Equations (n-channel)

Parameters:

- $C_{ox} = \epsilon_{ox}/t_{ox}$ (gate oxide capacitance per unit area)
- μ_n = electron mobility in channel
- W/L = channel width-to-length ratio
- V_{TH} = threshold voltage

Cutoff region ($V_{GS} < V_{TH}$)**:**

$$\boxed{I_D = 0}$$

Linear/Triode region ($V_{GS} \geq V_{TH}$, $V_{DS} < V_{GS} - V_{TH}$)**:**

$$\boxed{I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \frac{V_{DS}^2}{2} \right]}$$

Saturation region ($V_{GS} \geq V_{TH}$, $V_{DS} \geq V_{GS} - V_{TH}$)**:**

$$\boxed{I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2}$$

Transconductance (saturation):

$$\boxed{g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}}$$

Key Insight

The BJT and MOSFET are complements, not competitors:

- **BJTs** offer higher transconductance for the same bias current and are preferred in high-speed analog, RF, and low-noise applications.
- **MOSFETs** offer zero gate current, excellent scalability, and are the foundation of digital logic (CMOS) and low-power analog circuits.
- **BiCMOS** technology combines both on the same chip for optimal performance.

15.13 Summary: The MOSFET as a Voltage-Controlled Switch and Amplifier

The MOSFET fundamentally changed electronics by offering a voltage-controlled switch with zero DC input current. This property enabled:

- Ultra-high-input-impedance amplifiers (ideal for sensors)
- CMOS digital logic with near-zero static power dissipation
- Continuous scaling (Moore's Law) for over five decades
- Integration of billions of transistors on a single chip

While BJTs remain superior for certain high-speed and high-current applications, the MOSFET is the undisputed workhorse of modern electronics, from microprocessors to memory chips to power management.